

Reconstruction-independent moment tests for quantum-computed dynamical spectra

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Sample-based quantum diagonalization now reconstructs dynamical spectral functions $A(\omega)$ classically in subspaces sampled on a quantum processor. We develop and calibrate a screen comparing the moments read off a reconstruction with ground-state operator expectations that never use $A(\omega)$. A mismatch beyond a preset threshold rejects it; a pass does not certify it. In a hash-sealed, blinded simulation of 300 doped Hubbard-ring instances, the pre-registered primary endpoint was a detection rate of 53.7% (101/188; Wilson 95% CI 46.6–60.7%) at a false-positive rate of 0.9% (1/112), with an unbiased simulated estimator. The screen caught all 56 determinant truncations, 74% of spurious features and none of 71 under-converged Krylov reconstructions; two sealed secondary predictions failed. Classical moment brackets state what a pass implies; at the deployed order (m_0 – m_2) guaranteed power against fixed-pole redistributions is zero. On IBM Heron only the sampled support comes from the device. The missed weight Δ_0 grows from 0.004 to 0.37 as coverage falls from 94% to 35%, but it rewards support spreading: a uniform in-sector sampler reaches lower Δ_0 , and the reconstruction with $\Delta_0 = 0.004$ has a relative L_1 error of 0.35. The discriminating moments m_1, m_2 remain simulation-only; a sealed noise gate for an on-device momentum-distribution test failed.

I. INTRODUCTION

Dynamical response functions connect many-body theory to experiment. The single-particle spectral function $A(k, \omega)$ is measured by angle-resolved photoemission [1], the dynamical structure factors $S(q, \omega)$ and $S^{zz}(q, \omega)$ by inelastic neutron and resonant x-ray scattering, and the optical conductivity by infrared spectroscopy [2–4]. Quantum processors, first used for static ground-state energies [5–9], now return such functions as well. Trapped-ion devices have produced dynamical structure factors and single-particle spectral functions [10, 11], and Green’s functions built from measured Hamiltonian moments [12]. On superconducting processors, a DFT+DMFT workflow has computed its impurity Green’s function with up to 14 qubits [13], and structure factors computed with up to 50 qubits have been compared quantitatively with neutron-scattering data [14]. Sample-based diagonalization and quantum Krylov methods, developed for ground states [15–19], have been extended to spectra [20, 21]. In this quantum-centric pattern the device samples a subspace, and classical resources diagonalize the Hamiltonian in it and assemble $A(\omega)$ [22]. The hardware run of our companion study is exact by coverage and establishes only that this pipeline executes; its post-selection read the bitstrings in reversed bit order, so all of its retained determinants came from device errors [20].

Such results matter most where no trusted classical cross-check exists, which is the regime that motivates the quantum resource. The edge of that regime moves as classical methods improve and earlier demonstrations are dequantized [23–28]; for sample-based diagonaliza-

tion it is surveyed in Ref. [29]. That regime is the eventual target, not the setting of this paper. A check intended for it must first be specified and calibrated where the ground truth is known [30–34], and that is what we do here. Physics supplies the conditions. Every $T = 0$ Lehmann spectrum must satisfy moment sum rules [35–37] and positivity [38], and the state behind it must satisfy its conserved-charge identities [39], however it was computed.

Current practice validates a quantum-computed spectrum against a reference. Lee *et al.* compare with neutron data [14]. Buchs *et al.* build a validation framework with experimental and classical anchors, whose metrics include windowed weights, low-order moments and sum-rule residuals [40]; for the regime without full classical benchmarks they outline an asymmetric mode that would lean more on experiment, sum rules and internal consistency, and leave it to future work. Other verification tools certify a number rather than a function. Self-verifying variational simulation [41] prepares a state variationally [6, 42, 43] and bounds its distance to an eigenstate through a measured energy variance. State certification from single-qubit measurements [44], error mitigation [45–49], randomized-measurement estimation [50, 51], observable estimation without classical verification [52], and bounded-error analog simulation from learned Hamiltonians and Lindbladians [53] all act on states or expectation values. None of them checks a reconstructed $A(\omega)$ against the Hamiltonian that produced it without a classical or experimental reference. The failure we target is spectral weight placed at the wrong frequencies while integrated quantities stay within tolerance. Critical studies of quantum-selected configuration interaction and subspace diagonalization document the limitations of these methods for ground-state energies [54–58]; that the same mechanisms misassign spectral weight is our extrapolation. The same concern

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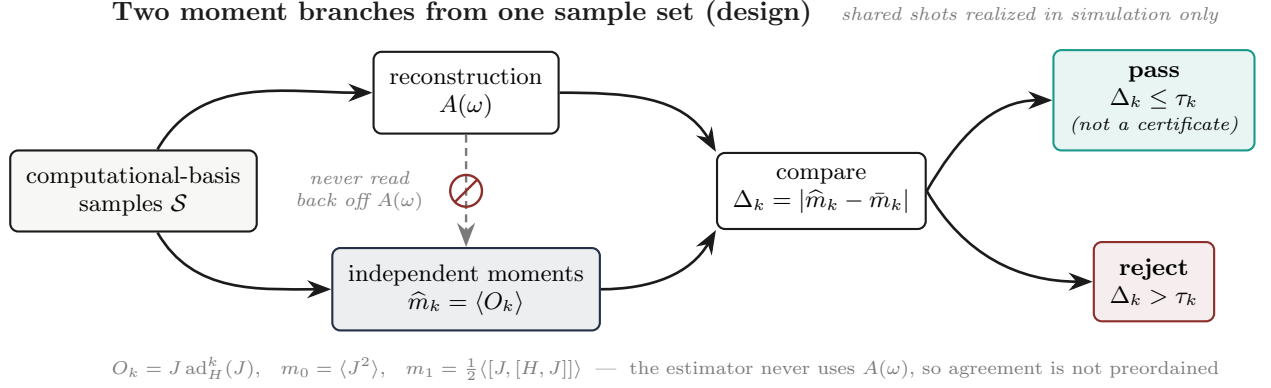


FIG. 1. **Two moment branches from one sample set (design).** Computational-basis samples fix a subspace $\mathcal{S}$. The reconstruction branch diagonalizes H in $\mathcal{S}$, assembles $A(\omega)$ and reads off $\bar{m}_k = \int \omega^k A(\omega) d\omega$. The estimator branch evaluates the ground-state operator moment $\hat{m}_k = \langle J(H - e_0)^k J \rangle$ with $e_0 = \langle H \rangle$ on the prepared state; on an exact eigenstate this equals $\langle J \text{ad}_H^k(J) \rangle$, the form printed in the drawing (Appendix A). It never uses $A(\omega)$, so it can disagree with the reconstruction. The reconstruction is rejected when $\Delta_k = |\hat{m}_k - \bar{m}_k|$ exceeds a threshold $\tau_k = z \delta_k$ fixed in advance from shot noise and a bias budget; a pass does not certify it. In this paper the two branches share one set of shots only in simulation; on hardware the device supplies the support $\mathcal{S}$ alone (Sec. VI).

motivates response-targeted determinant selection on the classical side [59]. Figure 3 illustrates the target with an injected example: a corrupted $A(k, \omega)$ that passes visual inspection and a total-weight check but carries the wrong first moment.

The screen compares two sets of moments of the spectral measure (Fig. 1). For a zero-temperature Lehmann function of a probe J , the moments $m_k = \int \omega^k A(\omega) d\omega$ are ground-state expectations, $m_0 = \langle J^2 \rangle$, $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$ and in general $m_k = \langle J(H - E_0)^k J \rangle$; they follow from operator algebra, not from the dynamics [35, 38, 60]. The reconstruction supplies its own moments $\bar{m}_k$. The estimate $\hat{m}_k$ of the ground-state expectation is never obtained by integrating $A(\omega)$, so it can disagree with the reconstruction; a moment read back off $A(\omega)$ cannot, since compared with itself it gives $\Delta_k \equiv 0$. The reconstruction is rejected when $\Delta_k = |\hat{m}_k - \bar{m}_k|$ exceeds a threshold τ_k fixed in advance from shot noise and a bias budget. These are necessary conditions: failing one rejects the reconstruction at the pre-registered level, while passing does not show that the reconstruction is correct; it rules out only errors that move a tested moment past its threshold. The battery also accepts any exactly known operator identity; Appendix D uses the U(1) Gauss law as a state-level example, estimable from computational-basis samples in principle and here evaluated exactly on the state vector [39, 41].

The design takes both branches from one set of samples; Sec. II A states how far this is realized (not on hardware, where the device supplies only the sampled support; only in part, in simulation; and not in the blinded test).

Checking a computed spectrum against its moments is long-standing. In dynamical mean-field theory and quantum Monte Carlo, low-order moments of Green's functions and self-energies serve as consistency tests of con-

tinued spectra, most systematically in the sum-rule program of Turkowski and Freericks [61, 62], and analytic-continuation pipelines report sum-rule agreement routinely [63, 64]. We claim no novelty for testing a spectrum against sum rules. In the optical case $m_0 = \langle J^2 \rangle$ is the total current-current weight, while the f -sum rule $\int_0^\infty \sigma_1 d\omega = \frac{\pi}{2} \langle -\hat{T} \rangle$ is the separate moment m_{-1} [2, 36, 37]. Which moment sequences a positive measure can carry, and how tightly finitely many moments constrain it, is the classical moment problem [65–68]. Its link to orthogonal polynomials and Gauss quadrature [69–73] also underlies the kernel-polynomial and recursion methods [74–77]. The same machinery now gives rigorous bounds on smeared spectral functions from Euclidean lattice correlators [78–80]; our moments come instead from ground-state operators, and our use is a rejection test on a real-frequency reconstruction. The semidefinite certificates of Wang *et al.* and Mortimer *et al.* return two-sided intervals on ground-state expectations, the latter at finite measurement statistics [81, 82]. Requiring the read-off moments $\bar{m}_k$ to lie in such intervals is a second necessary condition on the reconstruction. Combining the two layers therefore gives two necessary conditions, at least as sharp as either alone, not a necessary condition paired with a sufficient one. We have not applied those programs to our moment operators; Sec. IX A shows the two layers side by side at the exact-moment level only, with a classical interval (Hausdorff) condition in place of the semidefinite program. Finally, some pipelines conserve the zeroth moment by construction. The Green's-function solver of Patel *et al.* augments its response subspace so that the probe acting on the sampled ground state is held exactly, and the spectral weight is then never lost [21]. When both branches use the same ground-state vector, our coverage channel car-

ries no signal for such pipelines; only the higher moments test them.

This paper is the methodological complement of a companion study that reconstructs these spectra from sampled subspaces and bounds their error [20]. Version 3 of the companion contains results that this screen uses, and we adopt its terms. The operator leak of Eq. (B3), with $|\varphi\rangle = J|0\rangle$, contains two of them. At $k = 0$ it is the coverage residual $\Delta_0 = m_0(1 - w)$, with w the companion's captured weight. At $k = 2$, on a subspace of unit captured weight, it is the companion's identity $\mu_2 - \mu_2^S = \|Q_S H P_S \varphi\|^2$, where μ_j is the companion's name for m_j . The companion also proves that a subspace containing the order- K Krylov space reproduces the moments through order $2K + 1$. This is why Krylov reconstructions with two or more Lanczos nodes reproduce m_0 – m_2 exactly and pass our battery, while the one-node case misses m_2 by 2–4% and passes only inside the 2% bias budget. Finally, the companion shows that the captured weight alone cannot certify a reconstruction, and our $L = 8$ device run gives an instance on device data. What this paper adds is not an error bound. It is a rejection test with thresholds that include shot noise and a bias budget, applied to single-particle, current and collective response functions; a blinded, sealed calibration of its false-positive and detection rates; the exact limits of what a pass implies; and a device evaluation of the coverage channel against reference distributions. In detail:

(i) *Screens across observables* (Figs. 3–6). In the doped Hubbard current response a spurious low-frequency peak that preserves $m_0 = \langle J^2 \rangle$ is rejected by m_1 , the mean frequency once normalized by m_0 [36, 37]. The negative-order moment m_{-1} moves most in relative terms (154% against 38% for m_1). Normalized by noise at a 2% bias, however, it fires later than m_1 and m_2 for this error model, on the ring and on the open chain; among the peak positions we scanned it fires first only at $0.5t$ and below. On the ring it also needs the Kohn stiffness from a separate calculation. The same construction applies to $A(k, \omega)$ and to the collective $S(q, \omega)$ and $S^{zz}(q, \omega)$, each a positive $T = 0$ Lehmann measure with its own sum rules; these are exact simulations at $L \leq 12$.

(ii) *Independence under truncation* (Sec. IV). Under determinant-subspace truncation the independent estimator crosses its threshold while a control read back off $A(\omega)$ stays identically zero [15, 16]. The same holds when both branches are evaluated on one subspace state.

(iii) *A blinded, pre-registered calibration* (Sec. V). Before any instance existed we sealed the battery, thresholds, error catalog and seed by a SHA-256 hash; the seal is self-attested, with no third-party timestamp. The sealed primary endpoint was a detection rate of 53.7% (101/188, Wilson 95% interval [46.6, 60.7]%) at a false-positive rate of 0.9% (1/112, [0.2, 4.9]%). By class, the battery caught 56/56 determinant truncations, 45/61 spurious features and 0/71 under-converged Krylov reconstructions; rescoring the 56 sealed truncations post hoc

with the shared-state estimator also rejects all of them. The sealed secondary predictions held for truncations, for weight-changing spurious features (25/27 caught) and for Krylov reconstructions with two or more nodes (0/56 caught); two failed. The one-node Krylov case, expected to be caught, was caught in 0 of 15 instances. The weight-preserving spurious features, expected to be rarely caught, were caught in 20 of 34 (59%), an outcome the pre-registration did not anticipate. The low false-positive rate is largely built in: the simulated estimator is unbiased while the threshold budgets a 2% bias, and a realized bias of +2% (−2%) would raise the modeled rate from 1.3% to 8.0% (11.5%). The 0.9% therefore confirms the implementation, not robustness to a real mitigation bias, which is untested. The positivity branch, applied to the reconstruction's own moments, cannot fire on a catalog of positive measures. The catalog is one doped $L = 6$ ring at $U/t \in \{3, 4, 5, 6\}$ with the current probe; its 56 truncation depths were drawn at random, and by ground-state weight the truncations are mild (they keep 75–99% of it). An exhaustive scan of every depth for the $U/t = 4$ member, with the interval battery of Appendix B, finds 12 of 185 depths at which the full battery, with the sealed thresholds, passes the truncated reconstruction, three of them (46–51% of the support) inside the depth range of the catalog. The 56/56 therefore describes the sealed draw, not every truncation.

(iv) *Exact limits* (Secs. VII and VIII). Classical Christoffel–Markov–Stieltjes and Markov–Krein brackets, re-derived here, bound how far the cumulative weight of a reconstruction with exact moments can deviate from the true one [69, 83]. At the deployed order the bracket is vacuous at the mean ($\max_t W_1 = m_0$), and finite tolerances widen it further. On N distinct poles the first N moments are complete for fixed-pole weight redistributions (Theorem 2), but in exact arithmetic only. At the deployed order (m_0, m_1, m_2) and $N > 3$ the guaranteed power is zero: the redistributions that preserve these moments form an $(N - 3)$ -dimensional space, whatever the pole separation. On the $L = 6$ current spectrum ($N = 5$) an explicit redistribution that keeps m_0, m_1, m_2 exact moves 72% of the weight and empties two of the five poles; one that moves 86% keeps m_0, m_1 exact, shifts m_2 by exactly τ_2 and passes. Closing this gap needs moments beyond m_2 . For weight moved within a cluster of s poles of diameter D , the shots needed to see it grow at least as $D^{-2(s-1)}$.

(v) *A device evaluation of the coverage channel* (Sec. VI). On IBM Heron (`ibm_fez`) we ran seven circuits for an $L = 8$ electron-addition sector at four shot budgets (5×10^4 to 4×10^3 shots per circuit) and kept the raw counts. The circuits prepare a computational-basis determinant and apply on-site R_{zz} phases and number- and spin-conserving $R_{xx}R_{yy}$ hopping without Jordan–Wigner strings, so they sample the addition sector but are not the Hubbard propagator of the probe state; the sampler used measurement twirling, Pauli gate twirling and dynamical decoupling, and the counts were post-selected

to the physical sector without configuration recovery or readout-error mitigation. The device supplies only the support $\mathcal{S}$. The missed weight Δ_0 grows from 0.0044 to 0.37 as coverage falls from 94% to 35%, but it measures support, not fidelity. At 5×10^4 shots per circuit the device value lies below all 2000 noiseless multinomial replicas of the same circuits at the same raw shot count, and at 4×10^3 above all of them. A uniform sampler over the sector, drawing as many in-sector samples as the device retained, reaches a lower Δ_0 and a more accurate reconstruction than the device at every budget. The device reconstruction with $\Delta_0 = 0.0044$ has a relative L_1 error of 0.35; that is still better than all 80 noiseless replicas we reconstructed at the same budget, while at 4×10^3 shots the device reconstruction is worse than all 80. The discriminating moments m_1, m_2 were not measured on the device: at the companion’s circuit depth (292 CZ) the forecast depolarizing bias of the bond-basis estimate of m_1 is 37–116% of the signal (local to global depolarizing at a two-qubit error of 2.5×10^{-3}). A sealed simulated noise gate for an on-device momentum-distribution test failed at the relaxed noise floors of its single permitted amendment ($T_1 = 100 \mu\text{s}$, $T_2 = 70 \mu\text{s}$, two-qubit error parameter 3.5×10^{-3}), and that run was not made.

We state the scope at the outset. The method can reject a reconstruction but cannot certify one; its conditions are necessary, not sufficient. The moment screen targets determinant-subspace truncation and spurious features in sample-based-diagonalization (SQD/QSCI) spectra, that is, weight and subspace misassignment; the Gauss-law identity targets gauge-symmetry violation in the prepared state. The screen is out of scope for Krylov-convergence errors: a subspace containing the order- K Krylov space reproduces the moments through order $2K + 1$ [20], so an under-converged Krylov reconstruction passes the battery (Sec. V). Detecting that error class calls for a separate instrument, most naturally a Ritz-residual convergence check on the Krylov basis, which we leave to future work. Neither check is a general device-noise diagnostic, which remains the province of noise-model and observable-estimation validation [52]. The screen tests integrated moments, not pointwise line shapes; its positivity structure is that of a diagonal $T = 0$ Lehmann spectrum [38]; and every demonstration is at a classically reproducible size (Hubbard chains and rings with $L \leq 12$, a small lattice-Schwinger model, and a DMRG scalability check at $L = 24$). On hardware, moments carry statistical and systematic error, so the exact-moment tests must be replaced by interval tests before they can be trusted there (Appendix B); Table II separates what is measured on hardware, what is simulated and what is deferred.

II. METHOD: RECONSTRUCTION-INDEPENDENT MOMENT TESTS

We formalize the screen in two steps: first the object it tests—a quantum-computed spectrum and the samples that produced it—and then the checks we attach to it.

A. The quantum processor and the origin of the samples

The object the screen tests is the output of a sample-based-diagonalization pipeline, and what the screen can say depends on that pipeline. The pipeline and the $L = 6$ device spectrum of Fig. 11 are those of the companion study [20]. This paper contributes the test described below, its blinded calibration in simulation, its exact limits, and one new $L = 8$ `ibm_fez` job that supplies a device-sampled support (Sec. VI). In sample-based diagonalization [15–17] the processor does not evaluate $A(\omega)$. It prepares a state, and each computational-basis shot returns one bitstring, a candidate Slater determinant. The distinct bitstrings that survive post-selection to the physical particle-number and spin sector form the subspace $\mathcal{S}$. The Hamiltonian is projected onto $\mathcal{S}$ and diagonalized classically, and the spectrum is assembled as the $T = 0$ Lehmann sum $A(\omega) = \sum_n |\langle n | J | 0 \rangle|^2 \delta_\eta(\omega - (E_n - E_0))$ over the resulting eigenpairs, with δ_η a narrow broadening [38]. In the companion’s formulation the prepared state is a Trotterized real-time evolution of the $(N+1)$ -particle seed $c_p^\dagger |\psi_0\rangle$ [84]. The device circuits actually run—the companion’s $L = 6$ job and our $L = 8$ job—are different. Each of their seven circuits prepares a computational-basis $(N+1)$ determinant with X gates and applies Trotter layers of on-site R_{zz} and $R_{xx}R_{yy}$ hopping gates without Jordan–Wigner strings. They are therefore number- and S_z -conserving samplers of the addition sector, not the Hubbard propagator of the seed, as the companion’s v3 discloses for its run [20]. Neither run used configuration recovery [29]; both used plain post-selection. The companion’s $L = 6$ run read the bitstrings in reversed qubit order during post-selection, so every determinant it retained arose from a device error [20]. Our $L = 8$ job ($5\uparrow, 4\downarrow$ determinant) uses the correct bit order and post-selects to $N = 9$ with $N_\uparrow - N_\downarrow = 1$. It ran on SamplerV2 with measurement twirling, Pauli gate twirling and dynamical decoupling; no readout-error mitigation was applied to the counts.

The intended role of the processor appears only at scale. The many-body sector grows exponentially with system size, so the configurations that carry the weight of the probe state $|\varphi\rangle = J|0\rangle$ cannot in general be enumerated classically. An ideal sampler of the targeted state would concentrate shots on them and fix $\mathcal{S}$ by measurement rather than by a classical selection heuristic [16]. We do not show that any device sampler does this better than a classical heuristic, and the companion claims

TABLE I. **Contribution relative to the closest prior art.** “recon.-indep.” states whether the check compares a reconstruction with something the reconstruction did not produce, and with what; “same samples” states whether that comparison quantity comes from the samples that produced the spectrum; “—” marks an axis that does not apply to the method’s object. The semidefinite programs bound the true moments; requiring the read-off moments to lie inside their intervals is a second necessary condition, not a sufficient one (Sec. IX A). ^aDesigned; realized only in part, and only in simulation (Sec. II A); same-sample estimation, blinded pre-registration and on-device execution have not been realized together in one experiment. ^bThe device supplies the sampled support only; the discriminating moments m_1, m_2 are simulation-only (Table II). ^cBlinded simulation with an oracle-centred estimator (Sec. V). ^dThe companion’s $L = 6$ device run covers the whole sector, so its bound is not exercised on device data.

method	recon.-indep.	same samples	on device	blinded	role
Turkowski–Freericks sum-rule checks [61, 62]	yes (vs classical)	no	no	no	consistency check
maxent / stochastic continuation [63, 64]	yes (vs classical)	no	no	no	consistency check
Buchs <i>et al.</i> validation framework [40]	yes (vs experiment, classical, sum-rule targets)	no	no	no	validation metrics, incl. moments
Lee <i>et al.</i> neutron benchmark [14]	yes (vs experiment)	no	yes	no	benchmark
Greene-Diniz <i>et al.</i> cumulant Lanczos [12]	— (moments build G)	—	yes	no	constructive
Wang–Acín / Mortimer SDP [81, 82]	— (bounds moments)	no	no	no	two-sided moment bounds
Abbott <i>et al.</i> / causal bootstrap [78–80]	yes (Euclidean)	no	no	no	two-sided bound
Kokail <i>et al.</i> self-verification [41]	— (certifies state)	—	yes	no	state verification
reference-free observable estimation [52]	— (certifies number)	—	yes	no	validity
companion: captured weight and leakage [20]	— (bounds the error)	—	no ^d	no	a posteriori error bound
this work	yes	designed^a	support only^b	yes^c	necessary condition

no quantum advantage [20]. The circuits run here are not targeted at φ . For them, the coverage residual Δ_0 of Sec. VI measures how much of φ ’s weight lies outside the configurations the sampler happened to reach. At $L = 8$ a uniformly random in-sector sampler, given the device’s retained shot counts, leaves less of that weight uncovered than the device does at every shot budget (Sec. VI).

The design goal is that the moment estimator and the reconstruction share the sampled support, so that $\hat{m}_k$ inherits the shot noise and mitigation choices of the spectrum it tests. The independent estimator $\hat{m}_k$ defined below applies the full operator to the subspace ground state built on $\mathcal{S}$. This is a classical post-processing step that reaches configurations outside $\mathcal{S}$ (Appendix B). A device-native alternative measures $\langle O_k \rangle$ on a separately prepared ground state in qubit-wise-commuting Pauli groups; it is simulated only (Appendix B). We state where the shared-sample design is realized. On hardware it is not: the device supplies only the support $\mathcal{S}$, and every moment in Sec. VI is computed classically. In simulation it is realized in three places: the shared-state test of Fig. 8, where both moments are evaluated on one subspace state (there the subspace is a deterministic truncation by $|\psi_0|^2$, not a set of shots); the joint-sampling simulations of Sec. IX, where one set of shots yields both $\hat{m}_1$ and $\bar{m}_1$ (correlation $\rho \approx 0.43$ at 84% coverage and no gain in detection power; their local estimator uses exact amplitudes, so it is not device-realizable); and the $L = 24$ DMRG test of Sec. IX, where both branches use

one sampled subspace. The blinded test of Sec. V does not use it: there $\hat{m}_k$ is the exact moment plus Gaussian shot noise. Section V also reports a post hoc rescoring of its 56 truncations with the shared-state estimator.

B. The independent moment estimator and the falsifier

A quantum-computed dynamical spectrum is thus delivered as a non-negative function $A(\omega)$ together with the sample set $\mathcal{S}$. For a zero-temperature diagonal Lehmann spectral function of an operator J , $A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \delta(\omega - (E_n - E_0))$ is a positive measure on the real line. On an exact eigenstate its power moments $m_k = \int \omega^k A(\omega) d\omega$ are ground-state expectation values, $m_k = \langle 0|J(H - E_0)^k J|0\rangle = \langle 0|J \text{ad}_H^k(J)|0\rangle$, so that $m_0 = \langle J^2 \rangle$ and $m_1 = \frac{1}{2}\langle [J, [H, J]] \rangle$ (Appendix A) [35, 38]. In the optical case m_0 is the total integrated current–current weight and m_1/m_0 the mean frequency, while the optical f -sum is the separate moment m_{-1} [36, 37]. These operator forms coincide only on an exact eigenstate. On a subspace state they differ, by enough to change a verdict. For the current probe of the doped $L = 6$, $U/t = 4$ ring, truncated to the top 100 of its 225 sector configurations by $|\psi_0|^2$ (44% coverage), $\langle J(H - e_0)J \rangle = 4.56$ while $\frac{1}{2}\langle [J, [H, J]] \rangle = 3.72$, against a reconstruction moment $\bar{m}_1 = 3.74$ and an exact $m_1 = 4.23$. At the deployed threshold $\tau_1 = 0.229$ (defined below) the first

form rejects, as the exact moment would, and the second does not. The cut falls inside a 24-fold degenerate $|\psi_0|^2$ shell, so these values depend on how ties are broken; the verdicts are the same in both tie orders we computed. We therefore fix the deployed estimator as $\hat{m}_k = \langle J^\dagger (H - e_0)^k J \rangle$ with $e_0 = \langle H \rangle$ on the prepared state, the form used in the shared-state tests below.

The screen attaches to the spectrum an estimator $\hat{m}_k$ of each moment that is independent of the reconstruction: it is never obtained by integrating $\omega^k A(\omega)$. We denote by $\bar{m}_k = \int \omega^k A(\omega) d\omega$ the reconstruction moment, the quantity a circular check would use. The test statistic is the discrepancy $\Delta_k = |\hat{m}_k - \bar{m}_k|$. The true spectrum gives $\Delta_k = 0$ up to estimator error, so the reconstruction is rejected when $\Delta_k > \tau_k$, with $\tau_k = z \delta_k$, $z = 1.96$, and $\delta_k^2 = \text{Var}(O_k)/N_s + b_k^2$ the shot variance plus a bias budget b_k (Appendix B); the deployed rule uses $N_s = 5 \times 10^4$ and $b_k = 0.02 |\hat{m}_k|$. A circular check compares $\bar{m}_k$ with itself, returns $\Delta_k \equiv 0$ and can never reject; Results II shows this contrast on controlled truncation. At $k = 0$ the discrepancy is the weight the support misses, $\Delta_0 = \langle \varphi | (\mathbb{I} - \Pi_S) | \varphi \rangle$, which in the companion's notation is $m_0(1 - w)$ with w the captured weight [20]. The companion also proves two results that bound what this test can see. A subspace containing the order- K Krylov space of φ reproduces the moments through order $2K+1$, so a Krylov-adequate reconstruction passes any moment test up to that order by construction. For a subspace of unit captured weight, m_0 and m_1 are reproduced exactly and the first moment that can fail is m_2 , with deficit $\|Q_S H P_S \varphi\|^2$; this is Eq. (B3) at $k = 2$ with $\Delta_0 = 0$. The companion further shows that captured weight alone cannot certify a reconstruction [20]. What the present paper adds is the use of these residuals as a test with fixed thresholds: its sealed blinded calibration in simulation (Sec. V), the shared-state estimator (Sec. IV), exact statements of what passing means (Secs. VII and VIII), the moment test on the negative-order and collective channels (Results I), and an analysis of the coverage residual on a device-sampled support (Sec. VI).

The screen is a battery, not a single test. It applies Δ_k for $k = 0, 1, \dots, K$ (deployed: $K = 2$) and, separately, checks that the estimated sequence $\{\hat{m}_k\}$ is itself a moment sequence of a positive measure. That requires the Hankel matrix $H_n = [\hat{m}_{i+j}] \succeq 0$ and, because a diagonal $T = 0$ Lehmann measure is supported on $\omega = E_n - E_0 \geq 0$, also the shifted Stieltjes matrix $H_n^{(1)} = [\hat{m}_{i+j+1}] \succeq 0$ [65–67] (Fig. 2). A violation rejects the estimate—device, mitigation or estimator—independently of $A(\omega)$. It cannot reject a non-negative reconstruction, whose own moments satisfy these conditions by construction; in the blinded test the branch was applied to $\bar{m}_k$ and could not fire (Sec. V). The battery also accepts any exactly known operator identity: a conserved charge $[H, G] = 0$ with a known physical eigenvalue gives a constraint on the prepared state whose violation rejects that state, as in the gauge-theory exam-

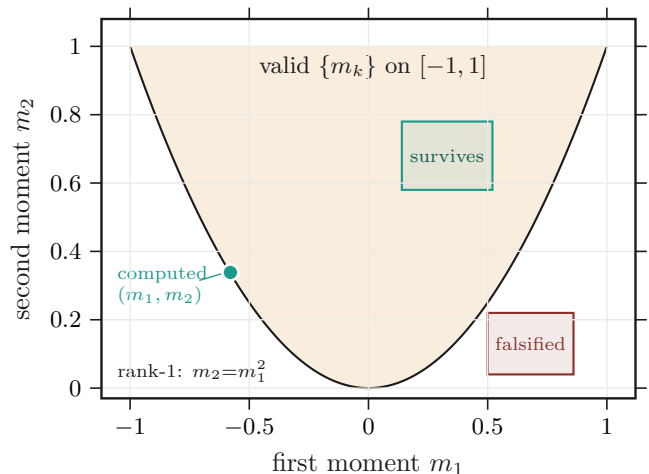


FIG. 2. **Hankel-positivity region in the (m_1, m_2) plane.** For a normalized measure ($m_0 = 1$) with frequencies rescaled to $[-1, 1]$, the moments must lie in the body $m_1^2 \leq m_2 \leq 1$. The lower edge $m_2 = m_1^2$ is the Hankel (rank-one) boundary, reached by a single δ ; the upper edge $m_2 = 1$ is the interval (Hausdorff) localizing condition of the $[-1, 1]$ support. The two boxes are schematic shot-budget intervals on an estimated moment vector. A box that lies wholly in the Hankel-negative region $m_2 < m_1^2$ (oxblood, “falsified”) rejects the estimate with no reference to $A(\omega)$; a box inside the body (teal, “survives”) passes. The check acts on the estimate: a non-negative reconstruction always has its own moments inside the body. The teal dot is the exact (m_1, m_2) of the doped $L = 6$, $U/t = 8$ Hubbard current measure. The rescaling maps onto $[-1, 1]$ the full range of excitation energies of the Fock-space Hamiltonian, $[0.148, 52.21] t$, which includes states of zero weight; the weighted support $[2.49, 13.92] t$ maps to $[-0.91, -0.47]$. In these units the measure is narrow, and one pole carries 78% of its weight, so the dot lies just above the rank-one edge [65, 66].

ple below [39]. The construction transports across models and observables because it needs only (a) a positive spectral measure and (b) operators whose ground-state expectations can be estimated without the reconstruction [69, 74].

III. RESULTS I: WEIGHT-MISASSIGNMENT ERRORS ACROSS OBSERVABLES

We first show the physical spectra on which the screen operates. Figure 3 shows the single-particle spectral function $A(k, \omega)$ of the half-filled Hubbard ring: lower and upper Hubbard bands separated by the Mott gap, with the dispersion of a one-dimensional correlated insulator [3, 85–88]. A truncated or otherwise corrupted reconstruction misplaces spectral weight. To illustrate what the first-moment check measures, we inject a corruption that flattens the band-centroid dispersion by moving weight between the Hubbard bands at each momentum, leaving the Mott gap and the per- k total weight intact. It

is the single-particle analogue of the optical weight misplacement of Fig. 5 and of the truncation errors of Fig. 7. The first moment at each momentum, given by the anticommutator sum rule for the non-Hermitian probe $c_{k\sigma}$ (Appendix A), exposes the flattening (right panel). The corruption is built to move the centroid by a fixed fraction of ϵ_k , and the moment is evaluated in closed form, so the residual is set by construction: the panel illustrates the check rather than testing it. The same construction applies to collective response. Figure 4 shows the charge dynamical structure factor $S(q, \omega)$, gapped by a particle-hole excitation onset Δ_c , and the spin structure factor $S^{zz}(q, \omega)$, whose gapless low-energy weight is the spinon continuum of the gapless spin sector [89], a signature of spin-charge separation [3]. Each channel is a positive $T = 0$ Lehmann measure with its own moment sum rules, so the screen applies to each; Fig. 6 runs it on both. Such spectra are standard output of classical many-body methods—density-matrix renormalization group and tensor-network dynamics, dynamical mean-field theory, and determinant quantum Monte Carlo with analytic continuation [63, 90–93]; at the sizes used here we compute them exactly.

We now apply the screen to a wrong distribution of correct total weight. In the doped one-dimensional Hubbard model the regular ($\omega > 0$) current spectral function $A_J(\omega)$ is dominated by a mid-infrared/Mott band, and its total weight is the ground-state quantity $m_0 = \langle J^2 \rangle$ [2, 36, 37, 96]. On the finite ring this object is entirely regular: the charge stiffness is a separate zero-frequency Drude term that does not appear in $A_J(\omega)$. We construct a wrong reconstruction that moves a fraction f of the weight above $6t$ into a spurious pole at $2.5t$ —a Drude-like low-frequency peak—keeping the total fixed. Figure 5 shows the outcome, with every moment evaluated exactly on the ground state (no sampling). The zeroth moment is unchanged, so a total-weight check passes the wrong spectrum. The first moment $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$ [35, 36] differs from the reconstruction’s by up to 38% of m_1 at $f = 0.5$; under the deployed threshold rule (2% bias budget at $N_s = 5 \times 10^4$) it rejects from $f \approx 0.057$. Integrated weight is a weak check for this error, and the first moment—the mean frequency of the weight—is what a misplaced-weight reconstruction corrupts.

The same misplaced weight also moves the negative-order moment. The positive moments $m_k = \int \omega^k A_J d\omega$ carry the kernel ω^k and weight high frequencies, whereas $m_{-1} = \int_0^\infty A_J(\omega)/\omega d\omega = \sum_n |\langle n|J|0 \rangle|^2/\omega_n$ carries $1/\omega$ and weights low frequencies (Fig. 5). This is the optical f -sum, and it too is a ground-state expectation: since the current is the polarization velocity $J = i[H, P]$, $m_{-1} = \frac{1}{2} \langle [P, [H, P]] \rangle = \frac{1}{2} \langle -\hat{T} \rangle$, a kinetic energy (equivalently $\int_0^\infty \text{Re } \sigma_{\text{reg}} d\omega = \pi m_{-1}$). On open chains we verify this identity against the Lehmann sum over the computed poles, to a relative 2×10^{-15} at $L = 6$ and 8 and 2×10^{-14} at $L = 12$ [35–37]. On the ring the polarization is multivalued and the identity acquires Kohn’s stiffness

correction, $m_{-1} = \frac{1}{2} \langle -\hat{T} \rangle - \mathcal{D}$ with $\mathcal{D} = \frac{1}{2} \partial_\theta^2 E_0$ the finite-size Drude weight (verified at $L = 12$ to a relative 2×10^{-6} , with $\mathcal{D}$ from the flux curvature). On this doped ring only 1.4% of m_0 lies below $6t$, and $m_{-1} = 0.324$ is the small difference of $\frac{1}{2} \langle -\hat{T} \rangle = 5.025$ and $\mathcal{D} = 4.701$: the stiffness cancels 94% of the kinetic term. In relative terms the spurious peak therefore moves m_{-1} by up to 154% at $f = 0.5$, $4.0 \times$ the shift of m_1 (38%) and $3.3 \times$ that of m_2 (47%).

A larger relative shift does not mean earlier detection. Because of the cancellation, a 2% bias on $\frac{1}{2} \langle -\hat{T} \rangle$ sets $\tau_{-1} = 0.198$, which is 61% of m_{-1} , whereas τ_1 and τ_2 are 4.3% of m_1 and m_2 . Under this rule m_{-1} is the last to reject: m_2 rejects first ($f \approx 0.046$), then m_1 (0.057), and m_{-1} only at $f \approx 0.20$. The open chain ($\mathcal{D} = 0$) does not restore an advantage for this corruption, because 41% of its regular weight already lies below $6t$ and $m_{-1} = 4.81$ is large: there the thresholds are crossed at $f \approx 0.17$ for m_{-1} , 0.059 for m_1 and 0.045 for m_2 . When the spurious pole is moved across 0.25 – $4t$, m_{-1} rejects first only for poles at $0.25t$ and $0.5t$, on both geometries (at $0.25t$ on the ring, $f \approx 0.015$ against 0.044 for m_1 and 0.043 for m_2). The negative-order moment is thus a complementary check for weight placed very close to $\omega = 0$, not the most sensitive check for the error model of Fig. 5. On open chains it is also cheap: $\langle -\hat{T} \rangle$ is a one-body operator, measured in two qubit-wise-commuting settings that are a subset of those the m_0, m_1 estimator already requires (Appendix B). On the ring, $\mathcal{D}$ requires ground states at twisted flux and is not a same-sample quantity. We claim no novelty for the identity, which is the textbook f -sum [36, 37]; we use it as an independent negative-order check.

We next run the same screen on the collective channels (Fig. 6). For each probe $A \in \{\rho_q, S_q^z\}$, with $\rho_q = \sum_j e^{-iqj} n_j$ and $S_q^z = \sum_j e^{-iqj} S_j^z$, the two moments are ground-state operator expectations as for the current: $m_0(q) = \langle A^\dagger A \rangle_c$ and $m_1(q) = \langle 0|A^\dagger (H - E_0) A|0 \rangle = \frac{1}{2} \langle [A^\dagger, [H, A]] \rangle$. For this non-Hermitian probe the double commutator is the symmetrized form $\frac{1}{2} [m_1(q) + m_1(-q)]$, equal to $m_1(q)$ by the inversion symmetry of the ring; both forms are evaluated on the exact ground state, where they coincide (App. A). Only the kinetic term contributes to m_1 , since the Hubbard interaction is diagonal in the occupation basis and commutes with ρ_q and S_q^z . On a half-filled $L = 6$ ring at $U/t = 8$ the operator moments reproduce the exact Lehmann moments of $S(q, \omega)$ and $S^{zz}(q, \omega)$ to better than 10^{-14} at every momentum, and the two forms of m_1 agree to the same precision. A reconstruction that moves band weight into a spurious low-frequency peak—in the gap for the charge channel, near zero for the spin channel—while conserving the total weight passes the m_0 check (residual $\sim 10^{-15}$) but its first moment is off by 29–36% once 40% of the band weight is moved, for both observables at each momentum tested ($q/\pi = 1/3, 2/3, 1$). The moments also register the two energy scales: the mean excitation frequency m_1/m_0 is set by U in the charge channel ($\approx 8.5t$ at $q = 2\pi/3$)

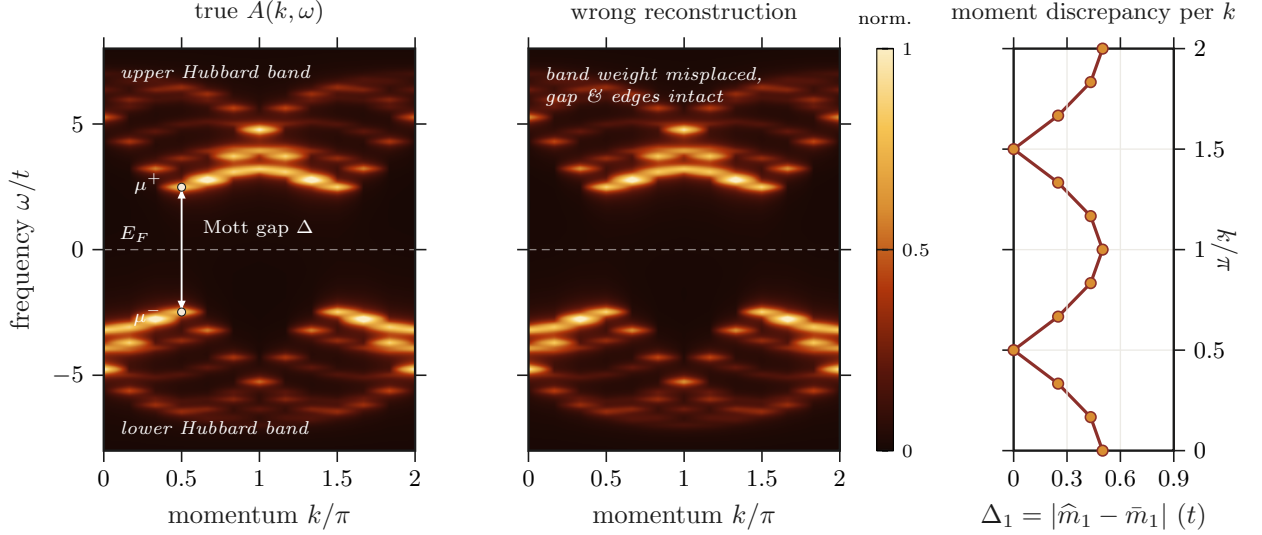


FIG. 3. **The first-moment check on the single-particle spectral function $A(k, \omega)$ of the Hubbard model.** Left: exact $A(k, \omega)$ of the half-filled ring. Frequencies are measured from the mid-gap Fermi level $E_F = U/2$ (dashed). The band edges $\mu^\pm = \pm 2.484 t$ at $k_F = \pi/2$ (dots, arrow) give the single-particle Mott gap $\Delta = \mu^+ - \mu^- = 4.97 t$, which approaches U only in the atomic limit $t \rightarrow 0$. Center: an injected corruption that rebalances intensity between the upper and lower Hubbard bands at each momentum so that the band centroid becomes $(1 - \alpha)\epsilon_k$, with $\alpha = 0.25$. It leaves the gap free of poles (only the Lorentzian tail of the broadening appears at E_F), the edges $\mu^\pm$ unchanged, and the per- k weight $m_0 = \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1$ preserved, so a total-weight check accepts it. Right: $\Delta_1 = |\hat{m}_1 - \bar{m}_1|$ per momentum, with $\hat{m}_1(k) = \langle \{c_{k\sigma}, H\}, c_{k\sigma}^\dagger \rangle$ the Hartree centroid; measured from $U/2$ it is $\epsilon_k + U(\langle n_{-\sigma} \rangle - \frac{1}{2}) = \epsilon_k$ at half filling (Appendix A). Here $\hat{m}_1$ is this closed-form value, not a sampled estimate, and $\Delta_1 = \alpha|\epsilon_k|$ by construction: $0.50 t$ at $k = 0, \pi, 2\pi$ and zero at $k = \pi/2, 3\pi/2$, where $\epsilon_k = 0$. For this probe m_0 is pinned to 1 and cannot discriminate, so m_1 is the first informative moment [38, 62]. Sector-basis Lanczos ground state with Haydock continued-fraction Green's functions [73, 75]; $L = 12$ (12 momenta), $U/t = 8$, broadening $\eta = 0.16 t$. Spinon and holon branches are only partly resolved at this size and broadening; the collective response of Fig. 4 separates the two energy scales more clearly.

and by the exchange $J = 4t^2/U$ in the spin channel ($\approx 1.0 t$). These are exact-simulation demonstrations on a half-filled $L = 6$ ring, not on the $L = 12$ spectra of Fig. 4, and not a hardware result; their purpose is to show that the screen transports across observables.

The battery also admits checks of a different kind: an exactly known operator identity, evaluated on the prepared state, screens an error class the moment test does not reach. An example is the U(1) lattice-Schwinger Gauss law $\langle \sum_x G_x^2 \rangle = 0$, whose violation detects gauge-symmetry breaking, that is, leakage out of the physical sector. This error is distinct from the weight and subspace errors the moment screen targets. The operator $\sum_x G_x^2$ is diagonal in the computational basis, so it is estimable from computational-basis samples in principle; here it is evaluated exactly on the state vector, with no sampling. Because it checks the state rather than a reconstruction, we treat it as a complementary check and develop it in Appendix D [39, 41].

IV. RESULTS II: INDEPENDENCE SUPPLIES THE DISCRIMINATING POWER

The central claim of the method is that *independence* is what gives the screen its discriminating power. We test it on the native failure mode of sample-based diagonalization: determinant-subspace truncation, in which the sampled subspace omits configurations that carry spectral weight, so the reconstructed spectrum is internally consistent with an incomplete basis [15, 16]. We run two checks side by side on the same truncated data (Fig. 7). The independent estimator computes $\hat{m}_1 = \langle O_1 \rangle$ with the full operators J and H , not only their restriction to the retained determinants (here on the exact full-sector ground state; the idealization is removed below). The back-computed control computes $\bar{m}_1$ by integrating the reconstructed $A(\omega)$, which lives entirely inside the truncated subspace. As truncation deepens, the independent discrepancy Δ_1 grows and crosses an illustrative fixed 5% threshold. The circular control compares $\bar{m}_1$ with itself, so it is zero by definition and can only confirm that the reconstruction agrees with itself [16]. This is the concrete reason why every moment in the screen must be estimated independently of $A(\omega)$ and never read back off it [15].

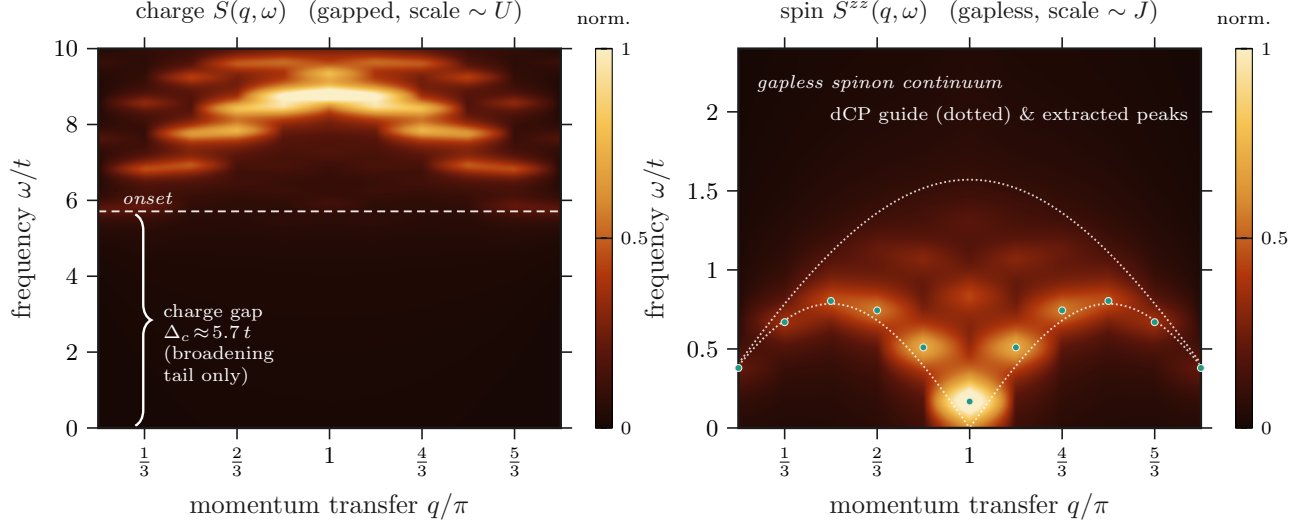


FIG. 4. **Charge and spin structure factors of the half-filled Hubbard ring.** Left: the charge dynamical structure factor $S(q, \omega)$ is gapped by a particle-hole onset $\Delta_c \approx 5.7t$ (dashed line: the lowest frequency carrying more than 5% of the peak intensity); below the line only the Lorentzian tail of the broadening is visible. Δ_c is a finite-size quantity distinct from the single-particle Mott gap $\mu^+ - \mu^- \approx 4.97t$ of Fig. 3; both are set by U and coincide only in the thermodynamic limit. Right: the spin structure factor $S^{zz}(q, \omega)$ carries gapless spinon weight. White dotted curves are the des Cloiseaux–Pearson (dCP) two-spinon boundaries $\omega_L = (\pi J/2)|\sin q|$ [94] and $\omega_U = \pi J|\sin(q/2)|$ [95], with $J = 4t^2/U = 0.5t$; filled teal points are the dominant spinon peak at each of the eleven momenta $q/\pi \in \{1/6, \dots, 11/6\}$ and lie close to the lower edge. The dCP curves are an ideal large- U Heisenberg guide, not a strict bound: finite U , finite size and broadening put some weight below ω_L . The field is linearly interpolated in q for display. The panels use different frequency windows because the two channels live on separated energy scales, $\sim U$ and $\sim J$; the charge data are computed to $11t$ and the spin data to $2.6t$, and the windows clip them at $10t$ and $2.4t$. Both are positive $T = 0$ Lehmann measures with their own moment sum rules [35]. No moment test is applied in this figure; Fig. 6 applies it to both channels on a smaller ($L = 6$) ring. Sector-basis Lanczos ground state with Haydock continued-fraction evaluation [73, 75], $L = 12$ at half filling (11 momentum transfers), $U/t = 8$; broadening $\eta = 0.20t$ (charge) and $0.12t$ (spin).

One boundary must be stated precisely. The independent estimator applies the exact operator O_k to the prepared state, so Δ_k detects spectral weight that leaks *outside* the retained determinant support. A corruption that only redistributes weight among the retained poles, while preserving the moments used, is not caught by those moments. In exact arithmetic this blind spot closes at a finite order: on N distinct poles the weight-to-moment map is injective, so a fixed-pole redistribution that preserves $m_0, \dots, m_K$ changes some m_j with $K < j \leq N - 1$ (Theorem 2(a), proved in Appendix E; part (b) gives the shot cost of each further order). At the deployed order, however, the guaranteed power against such redistributions is zero: every redistribution in the $(N - K - 1)$ -dimensional null space of the moment rows $0, \dots, K$ passes, whatever its size. This follows from the dimension count, not from the conditioning of the poles. For the $U/t = 4$ current measure of the blinded test of Sec. V ($N = 5$ weighted poles, $K = 2$) this null space is two-dimensional. One redistribution in it preserves m_0, m_1, m_2 exactly while moving 72% of the weight and emptying two poles; at the deployed budget it also stays below the m_3 and m_4 thresholds ($0.23\tau_3$ and $0.70\tau_4$). The moment screen is therefore a necessary-condition detector of support leakage and of weight misassignment

that shifts the low moments. Redistributions that preserve the deployed moments are a disclosed blind spot, not an error class we claim to catch.

Figure 7 evaluates the independent estimator on the exact full-sector ground state, while only the reconstruction lives in the truncated subspace. This is an idealization: a sample-based pipeline has only the subspace state built from the sampled support, and both branches must be computed from it. We therefore repeat the test with both moments evaluated on the *same* subspace ground state $|0_S\rangle$ of the truncated Hamiltonian $H_S = \Pi_S H \Pi_S$, with the shared reference $e_0 = \langle 0_S | H | 0_S \rangle$ (Fig. 8). The independent estimator applies the *full* sparse operators, $\hat{m}_1 = \langle 0_S | J(H - e_0)J | 0_S \rangle$, whose action on $|0_S\rangle$ reaches determinants outside $\mathcal{S}$. The reconstruction moment $\bar{m}_1 = \langle 0_S | J_S(H_S - e_0)J_S | 0_S \rangle = \int \omega A(\omega) d\omega$ stays within $\mathcal{S}$. This form, $\langle J(H - e_0)^k J \rangle$ with e_0 the energy of the prepared state, is the estimator we fix throughout; off an eigenstate it differs from the double-commutator form (Appendix A). The shared-state estimator needs only the sampled support and is evaluated classically; it is not a moment measured on a device. Its difference from $\bar{m}_1$ is the operator leak of Eq. (B3) at $k = 1$. At two orders the identity takes the form of results of the companion study [20], which states them for the probe applied to

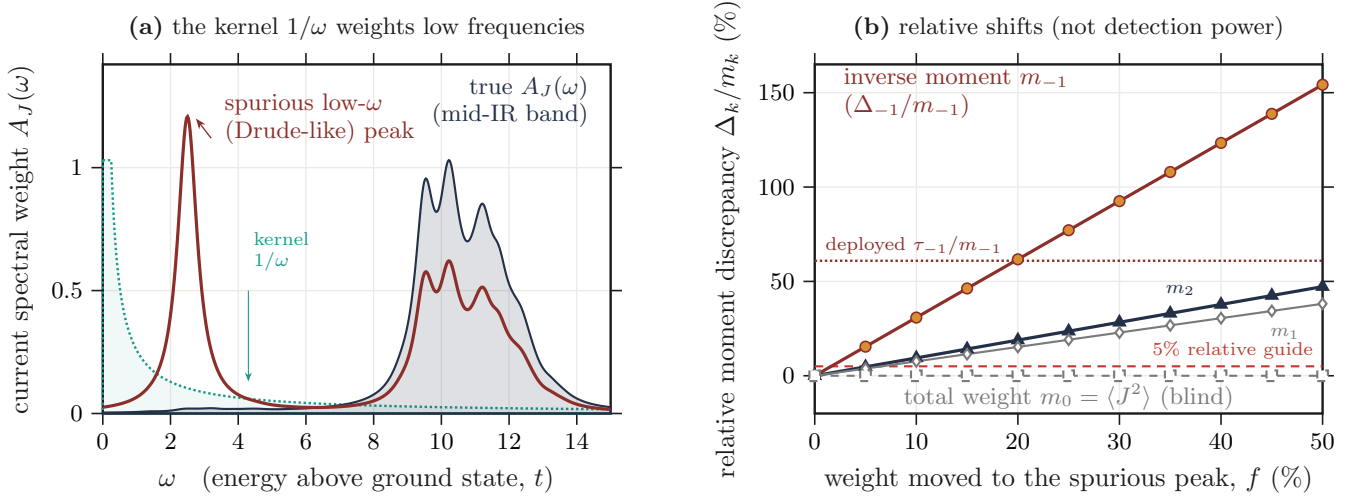


FIG. 5. **The current-response test: a wrong distribution of correct total weight.** Doped one-dimensional Hubbard ring ($L = 12$, $U/t = 8$, filling $2/3$, periodic boundaries; optical conductivity $\text{Re } \sigma \propto A_J/\omega$). A wrong reconstruction moves a fraction f of the weight above $6t$ into a spurious pole at $2.5t$ (a Drude-like peak), keeping $m_0 = \langle J^2 \rangle$ fixed. **(a)** The true regular spectrum (indigo) is a mid-infrared/Mott band with only 1.4% of m_0 below $6t$; the spurious peak (oxblood) sits where the kernel $1/\omega$ (teal) is large. **(b)** Relative discrepancies Δ_k/m_k against f , with every moment evaluated exactly on the ground state. The total weight is unchanged ($\Delta_0/m_0 < 10^{-13}$); $m_1 = \frac{1}{2}\langle [J, [H, J]] \rangle$ moves by up to 38%, m_2 by up to 47%, and $m_{-1} = \frac{1}{2}\langle -\hat{T} \rangle - \mathcal{D} = 0.324$ by up to 154%. The 5% line is a common relative reference, not the deployed threshold. Under the deployed rule (2% bias budget, $N_s = 5 \times 10^4$) the thresholds are 4.3% of m_1 and of m_2 but 61% of m_{-1} (dotted line), because m_{-1} is a 94% cancellation between $\frac{1}{2}\langle -\hat{T} \rangle$ and the Kohn stiffness $\mathcal{D}$; m_2 then rejects first ($f \approx 0.046$), m_1 next (0.057) and m_{-1} last (0.20). $\mathcal{D}$ is obtained from the flux curvature, not from the same samples. The identity for m_{-1} is verified against the Lehmann sum over the computed poles to a relative 2×10^{-15} on open chains of $L = 6$ and 8 , 2×10^{-14} on the open $L = 12$ chain and, with the stiffness correction, 2×10^{-6} on this ring. Sector-basis Lanczos ground state with Haydock/Ritz current poles [36, 37, 73, 75].

the exact ground state rather than to $|0_S\rangle$. At $k = 0$ the leak is $\hat{m}_0(1 - w)$, where w is the captured weight of $J|0_S\rangle$ on $\mathcal{S}$. When $w = 1$ the first nonzero order is $k = 2$, and the leak equals the boundary coupling $\|(\mathbb{1} - \Pi_S)H\Pi_S J|0_S\rangle\|^2$. What this paper adds is the use of the leak as a test statistic against a threshold, its contrast with the circular control, and the blinded calibration of Sec. V.

The shared-state discrepancy is nonzero and rises by about five orders of magnitude across the sweep, from 5×10^{-6} at full support to $\approx 100\%$ at deep truncation (non-monotone near both ends: below 10^{-4} near full support, and at saturation). On the $L = 12$ doped ring at $U/t = 8$ it crosses the 5% line at $d \approx 4.4 \times 10^4$ retained determinants (18% coverage). The idealized curve, which reproduces Fig. 7, crosses at $d \approx 1.6 \times 10^4$ (7%) and dips at moderate truncation, where the full-state reference and the truncated reconstruction happen to track each other. With the shared state the screen therefore fires at milder truncation, so at this threshold the idealization of Fig. 7 does not inflate the result. Neither branch stays at the exact value once the state is truncated: $\hat{m}_1$ rises from 35.1 at full support to 60.1 at the deepest cut, while $\bar{m}_1$ falls toward zero. The discrepancy measures the leak, not the error of either branch against the exact moment.

V. A PRE-REGISTERED TEST PROTOCOL

Every test shown so far targets an error we injected and knew in advance. To probe the screen where the outcome was not known to the classifier, we ran a blinded, pre-registered simulation. Before any instance existed we sealed by SHA-256 hash (0e4c3368...) the battery (m_0, m_1, m_2 plus a Hankel-positivity condition), the decision rule and its thresholds, the error catalog, the primary endpoint, the secondary predictions and the random seed. The rule rejects when $|\hat{m}_k - \bar{m}_k| > \tau_k$ for any $k \leq 2$ or when the Hankel condition fails, with $\tau_k = z\delta_k$, $\delta_k^2 = \text{Var}(O_k)/N_s + (b\hat{m}_k)^2$, $z = 1.96$, $N_s = 5 \times 10^4$ shots per moment (the largest per-circuit budget of the `ibm_fez` runs) and an assumed mitigation bias $b = 2\%$; $\text{Var}(O_k)$ is the local-estimator variance on the prepared state (Appendix B). The generator and the classifier re-verify the hash and refuse to run if it is broken. The seal fixes the protocol but is self-attested: it carries no third-party timestamp, and the blinding is enforced in code. Because the simulation is deterministic given the sealed seed, the seal cannot by itself exclude iteration before sealing.

The generator produced $G = 300$ instances: 112 faithful controls and three *algorithmic* error causes, namely determinant-subspace (SQD) truncation (56), under-

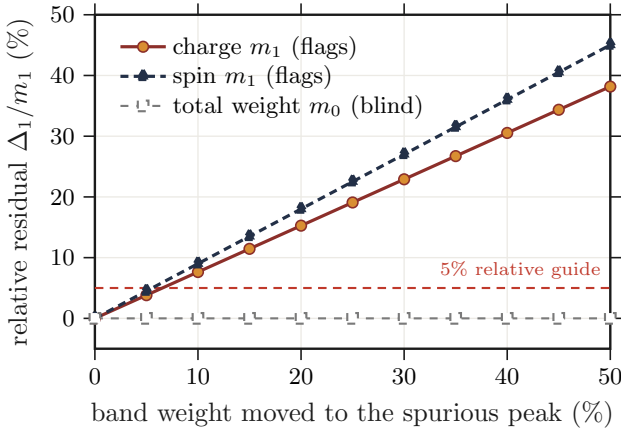


FIG. 6. **The moment test on the collective (charge and spin) channels.** The screen of Fig. 5, applied without change to the charge $S(q, \omega)$ and spin $S^{zz}(q, \omega)$ structure factors of a half-filled Hubbard ring ($L = 6$, $U/t = 8$, periodic boundaries; exact simulation). For each channel a wrong reconstruction moves band weight into a spurious low-frequency peak—in the gap ($\omega = 2.0t$) for the gapped charge channel, near zero ($\omega = 0.10t$) for the gapless spin channel—conserving the total weight m_0 exactly. At $q/\pi = 2/3$ the total weight m_0 (grey) stays blind for both channels, while the relative first-moment residual of the charge ($\rho_q = \sum_j e^{-iqj} n_j$, oxblood) and spin ($S_q^z = \sum_j e^{-iqj} S_j^z$, indigo) probes grows linearly and crosses a fixed 5% relative threshold, a guide rather than the deployed rule τ_1 . The operator moments reproduce the exact Lehmann moments to better than 10^{-14} at every momentum $q/\pi \in \{1/3, 2/3, 1\}$. The $L = 12$ spectra of the same channels are shown in Fig. 4.

converged Krylov reconstruction (71), and an analytic-continuation artifact that adds a spurious spectral feature (61). The classifier applied the frozen battery to each instance without access to its label. The calibrated scope is narrow. Every instance uses the current probe J of the doped $L = 6$ Hubbard ring ($N_\uparrow = N_\downarrow = 2$, sector dimension 225) at $U/t \in \{3, 4, 5, 6\}$. Truncations keep the top $d = 29$ –138 of the 186 ground-state configurations by $|\psi_0|^2$ (16–74% of the support); by weight they are mild, retaining 75.1–98.7% of $|\psi_0|^2$. In four of them $J|0_S\rangle$ has no component inside $\mathcal{S}$, so the reconstruction is empty. Krylov instances are n_l -node Lanczos representations with $n_l \in \{1, 2, 3, 4\}$. A spurious feature is a point atom of weight 2–25% of m_0 at a frequency drawn uniformly between the lowest and highest excitation energies of the sector. (The pre-registration text calls it a Gaussian atom; the generator adds a δ atom.) With probability 1/2 the measure is then rescaled to restore m_0 ; in this m_0 -preserving subclass the atom carries 2–20% of the reconstructed weight. The independent estimate was simulated as the exact moment plus Gaussian shot noise of variance $\text{Var}(O_k)/N_s$. It is therefore unbiased and centered on the exact full-sector value, the idealized estimator of Fig. 7, and it is drawn independently of $\bar{m}_k$, so the test carries no same-sample covariance. (A sep-

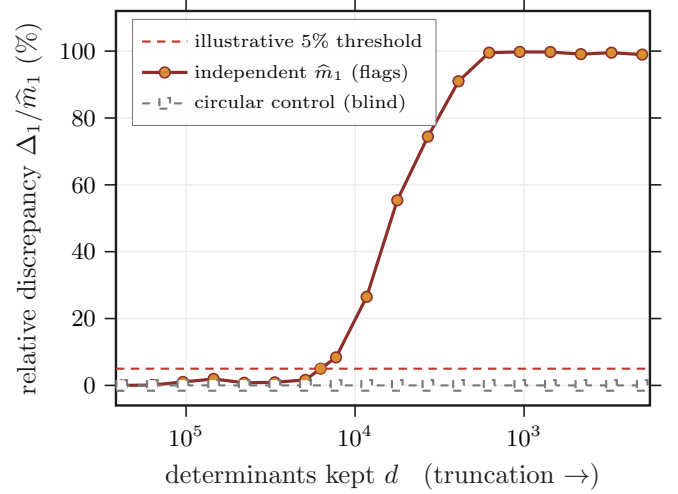


FIG. 7. **Independent vs. back-computed moment discrepancy under determinant-subspace truncation (idealized estimator).** The determinant subspace of the doped $L = 12$ Hubbard ring ($N_\uparrow = N_\downarrow = 4$, $U/t = 8$) is truncated to its top d configurations by $|\psi_0|^2$ (the sample-based-diagonalization ordering; full support $n_{\text{sup}} = 2.4 \times 10^5$), and the current first moment is recomputed within it. The reconstruction-independent discrepancy $\Delta_1^{\text{rel}} = |\hat{m}_1 - \bar{m}_1|/\hat{m}_1$, with $\hat{m}_1$ evaluated on the exact full-sector ground state, stays below 2% down to 8% coverage. It then grows sharply, crosses an illustrative fixed 5% threshold at $d \approx 1.6 \times 10^4$ (7% coverage) and saturates near 100%. The back-computed (circular) control compares $\bar{m}_1$ with itself and is zero by definition [15, 16]. The screen catches severe truncation, not mild undersampling. Figure 8 removes the full-sector idealization. Sector-basis Lanczos: the truncated first moment is $\langle J\psi_0^d | (H_d - E_0^d) | J\psi_0^d \rangle$ from one subspace ground state, with no full diagonalization [73].

arate simulation that draws both from one set of shots finds a same-sample correlation but no gain in power; Sec. IX.)

The sealed primary endpoint was the pooled detection rate, $\text{TPR} = 101/188 = 53.7\%$ (Wilson 95% interval [46.6, 60.7%]), at a false-positive rate $\text{FPR} = 1/112 = 0.9\%$ ([0.2, 4.9%]). The pooled rate is low because 71 of the 188 corrupted instances are under-converged Krylov reconstructions, which the battery misses (0/71); the per-class rates decompose it. All 56 truncations were caught ([93.6, 100%]). Spurious features were caught in 74% of cases (45/61): 93% (25/27) of the weight-changing ones, which the diagonal m_0 channel can reach, and 59% (20/34, [42, 74%]) of the m_0 -preserving ones, which only the off-diagonal moments can reach and which the pre-registration expected to be caught rarely (see below). The truncation result is a property of the sampled cuts, not a guarantee. An exhaustive scan over every cut of the $U/t = 4$ ring, with exact moments, a deterministic tie order and the same z , N_s and bias budget (Appendix B), finds 12 truncated reconstructions that pass

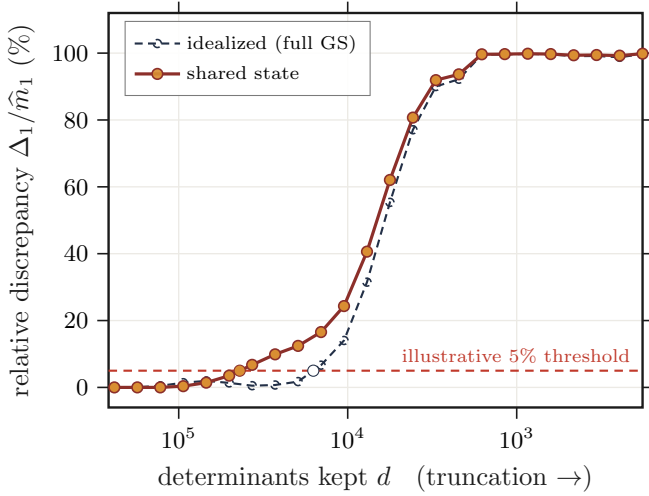


FIG. 8. **Shared-state discrimination: the operator leak on a single subspace state.** Both first-moment branches are evaluated on the *same* subspace ground state $|0_S\rangle$ of $H_S = \Pi_S H \Pi_S$, with shared reference $e_0 = \langle 0_S | H | 0_S \rangle$, using only what a sample-based-diagonalization pipeline has (Appendix B); the evaluation is classical. Solid (oxblood): the shared-state discrepancy $\Delta_1^{\text{shared}} = |\hat{m}_1 - \bar{m}_1|/\hat{m}_1$, with $\hat{m}_1 = \langle 0_S | J(H - e_0) J | 0_S \rangle$ formed with the *full* operators (reaching determinants outside $\mathcal{S}$) and $\bar{m}_1 = \int \omega A(\omega) d\omega$ the within- $\mathcal{S}$ reconstruction moment; their difference is the $k=1$ operator leak of Eq. (B3). Dashed (indigo): the idealized curve of Fig. 7, with $\hat{m}_1$ evaluated on the exact full-sector ground state. As the retained determinant count d falls (truncation $\rightarrow$), the shared-state discrepancy rises by about five orders of magnitude (non-monotone near both ends: below 10^{-4} near full support, and at saturation) and crosses the illustrative 5% threshold at $d \approx 4.4 \times 10^4$ (18% coverage). The idealized curve crosses at $d \approx 1.6 \times 10^4$ (7%), after an accidental dip near $d \sim 4 \times 10^4$ where the full-state reference and the truncated reconstruction track each other. Both saturate near 100%. $L = 12$ doped Hubbard ring ($N_\uparrow = N_\downarrow = 4$, $2/3$ filling), $U/t = 8$, sector-basis Lanczos.

the full battery: three at mid coverage ($d = 86, 87, 95$ of 186) and nine near-complete ones ($d \geq 160$). A Monte-Carlo power scan over the same cuts finds three narrow windows, near 46%, 51% and 87% coverage, in which the first moment alone, at the shot-only threshold, has power only at the false-positive level. The near-complete cuts lie outside the catalog’s range ($d \leq 138$), no $U/t = 4$ instance of the catalog fell on the mid-coverage ones, and the other U values were not scanned.

The Krylov misses follow from Gauss quadrature. An n_l -node Lanczos representation is a Gauss quadrature of the spectral measure and reproduces $m_0, \dots, m_{2n_l-1}$ exactly [69, 71]. The companion study proves the corresponding statement for subspace reconstructions: a subspace that contains the order- K Krylov space reproduces the moments through order $2K + 1$ [20]. For $n_l \geq 2$ the reconstruction matches m_0 – m_2 to below 10^{-15} relative, so at this order it cannot be told apart from a faithful

one. For $n_l = 1$ it misses m_2 by 2.4–4.2% and passes only because the threshold budgets a 2% bias (largest residual $0.99 \tau_2$). Against the shot-only threshold, without the bias term, 6 of the 15 one-node instances would fire on m_2 , but 5 of the 56 exact $n_l \geq 2$ instances would also fire, from noise alone.

Two sealed secondary predictions failed. The one-node Krylov case, which the pre-registration expected to be caught, was caught 0/15. The m_0 -preserving spurious-feature subclass, which the pre-registration described as designed-blind and expected to have low detection, was caught 20/34: the rescaling restores only m_0 and still shifts m_1 and m_2 . We report this 59% as an outcome the pre-registration did not anticipate, not as a designed capability, and the register was not revised. The other secondary predictions held: truncations (56/56) and weight-changing artifacts (25/27) were caught at high rates, and $n_l \geq 2$ Krylov reconstructions at a low one (0/56).

The false-positive rate is low by construction. The simulated estimator is unbiased, while the frozen threshold budgets a 2% bias comparable to the shot term. Redrawing the noise for the 112 faithful instances under the frozen rule gives a modeled FPR of 1.3%, far below the pre-registered nominal target $1 - 0.95^3 = 14.3\%$; one or fewer false positives occur in more than half of the redraws, so the sealed 1/112 is typical. With a realized bias of +2% on all three moments the modeled FPR is 8.0%, and with –2% it is 11.5% (the threshold scales with $|\hat{m}_k|$). The 0.9% therefore confirms the implementation, not robustness to real mitigation bias, which remains untested.

Every rejection came from the moment comparisons. The sealed rule describes an interval-Hankel condition, but the classifier implements it as a positivity check of the 2×2 Hankel matrix H_1 built from the reconstruction moments $\bar{m}_0, \bar{m}_1, \bar{m}_2$. Every reconstruction in the catalog is a positive measure, so this check cannot fire (smallest eigenvalue over the 300 instances -3.6×10^{-15} , rounding error). Its 0/300 holds by construction, and in this test the battery reduces to the three moment comparisons.

Post hoc, we rebuilt the 56 sealed truncation instances and rescored them with the shared-state estimator of Fig. 8. The rebuild reproduces every sealed reconstruction moment, variance and noise draw to rounding error; it depends on the eigensolver’s tie order inside degenerate $|\psi_0|^2$ shells, which we store. With $\hat{m}_k = \langle 0_S | J(H - e_0)^k J | 0_S \rangle$ plus the sealed noise draw, and the sealed variances in the thresholds, the frozen rule rejects all 56, every one on m_0 (smallest $r_0 = 5.9$, where $r_k = |\hat{m}_k - \bar{m}_k|/\tau_k$). Before noise, $\hat{m}_0 - \bar{m}_0 = \hat{m}_0(1 - w)$ is exactly the weight of $J|0_S\rangle$ outside $\mathcal{S}$. With the idealized sealed estimator m_0 fired on 55 of the 56. The exception, id 162 ($r_0 = 0.62$, caught only off-diagonally), is an artifact of the oracle-centered estimate; under the shared-state estimator it has $r_0 = 15.8$ and leaves the separating region of Fig. 9. The shared-state estimate is itself far from the exact moment (for id 162, $\hat{m}_0 = 1.04$

against $m_0 = 0.37$): the test registers the leak, not the distance to the truth.

Within this catalog the screen detects support truncation and most added spectral features, and it does not detect reconstructions that reproduce the tested moments within the thresholds. The scope is limited: the ground truth is classically known here and merely withheld from the classifier, so this is a blinded test, not a field catch of an error whose correctness is unknown, and no device is in the loop; both remain future work.

These aggregates separate the two channels by the errors they catch, and the per-instance residuals show the separation directly (Fig. 9). Determinant truncation moves the total-weight sum rule $m_0 = \langle J^2 \rangle$, so the diagonal channel catches it (55/56 with the idealized estimator, 56/56 with the shared-state one). The spurious-feature class splits in two. The weight-changing artifacts move m_0 , and the diagonal channel catches most of them (21 of the 25 caught). The m_0 -preserving subclass keeps the total weight and the determinant support fixed by construction, so neither the m_0 residual nor a support-coverage check can see it; there the off-diagonal moments carry every rejection (20/34; the pre-registration predicted low detection for this subclass, so this reading is post hoc). In all, 26 sealed instances have $r_0 < 1$ but an off-diagonal residual above threshold (m_1 on 16, m_2 alone on 10): the 20 caught m_0 -preserving artifacts, 4 weight-changing ones whose weight shift fell below τ_0 , truncation id 162 and false positive id 184. This is consistent with Fig. 5: at fixed total weight, misplaced weight is visible only to the off-diagonal moments. Theorem 2 does not apply here, because the artifacts add a pole and lie outside its fixed-pole class. The off-diagonal estimator therefore reaches a corruption class that the diagonal channel cannot see, but so far only in simulation; by the noise analysis of Sec. VI it is deferred to lower-error hardware. On `ibm_fez` only the sampled support came from the device, and the coverage residual was evaluated on it with exact classical amplitudes (Sec. VI).

VI. COVERAGE RESIDUAL ON A DEVICE-SAMPLED SUPPORT (IBM HERON)

No moment in this paper is estimated from device counts. On hardware the device supplies only the sampled support $\mathcal{S}$. The moments $\bar{m}_0, \bar{m}_1$ read off the reconstruction are computed classically on $\mathcal{S}$, and the reference values $\hat{m}_0, \hat{m}_1$ used below are exact classical numbers. Same-sample estimation, in which $\hat{m}_k$ and $\bar{m}_k$ come from one set of shots, has been realized only in simulation (Sec. IX). This section reports what a device-sampled support does and does not show about the coverage channel.

The companion study ran a sample-based-diagonalization pipeline on `ibm_fez` at $L = 6$ (12 qubits; 3.5×10^5 shots, 5×10^4 on each of seven circuits; job `d9s16avpemt573ct6g8g`) [20]. Its retained

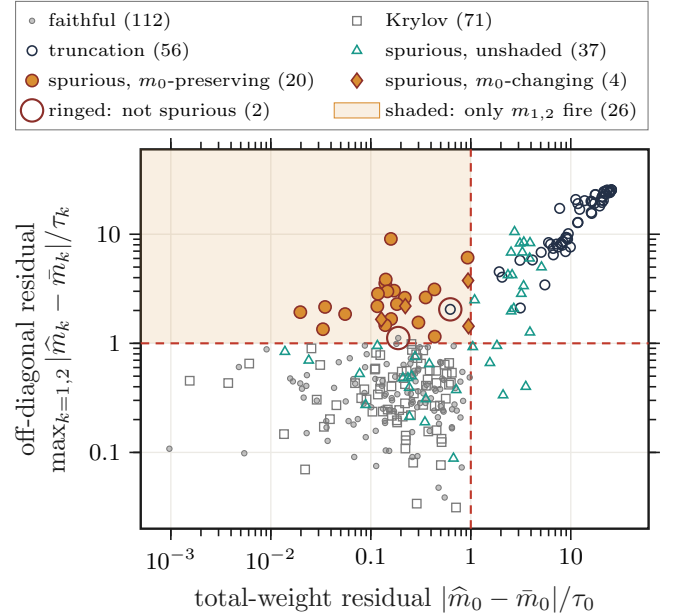


FIG. 9. **The diagonal and off-diagonal moments have complementary reach** (sealed, pre-registered blinded battery, SHA-256 0e4c3368...). For each of the 300 instances every residual is normalized to its frozen threshold, $r_k = |\hat{m}_k - \bar{m}_k|/\tau_k$ ($z = 1.96$ at $N_s = 5 \times 10^4$, including the assumed 2% bias), so $r_k > 1$ is exactly a fire. The horizontal axis is the total-weight residual r_0 ; the vertical axis is $\max(r_1, r_2)$. All 300 points are drawn at their true values. Determinant truncations (56/56 caught) have $r_0 > 1$ in 55 of 56 cases; the exception (id 162, ringed) is caught only off-diagonally with the idealized estimator of the sealed test and has $r_0 = 15.8$ with the shared-state estimator (Sec. V). The shaded region ($r_0 < 1$, $\max(r_1, r_2) > 1$) holds 26 instances rejected by an off-diagonal moment alone (m_1 on 16, m_2 alone on 10): 20 m_0 -preserving spurious features, 4 weight-changing ones whose weight shift stayed below τ_0 , truncation id 162 and the single false positive (id 184, ringed). An example is id 222, m_0 -preserving, with $(r_0, r_1, r_2) = (0.16, 3.3, 9.1)$. Legend counts are numbers of plotted points: the entries for m_0 -preserving (20) and m_0 -changing (4) spurious features count only points in the shaded region, and the other 14 preserving and 23 changing ones are in “spurious, unshaded” (37). The pre-registration predicted low detection for the m_0 -preserving artifacts, so reading the 20 caught (of 34) as a capability is post hoc. Under-converged Krylov reconstructions (0/71) match m_0 - m_2 exactly ($n_l \geq 2$) or within the bias budget ($n_l = 1$) and are caught by neither channel. The thresholds include shot noise and the assumed 2% bias but no device gate-error bias. The off-diagonal moments are simulated (exact value plus Gaussian noise); on current hardware they would carry the depolarizing bias of Sec. VI (~ 37 –116% of the signal). This is a structural separation in simulation, not device-realizable firing.

support is the whole addition sector ($|\mathcal{S}| = 300/300$), so the reconstruction equals exact diagonalization by coverage (Fig. 11). The bitstrings were post-selected after being read in reversed qubit order, which swaps

the spin labels and moves the circuits' own output into the discarded $S_z = -1$ sector, so all 300 retained determinants came from device errors [20]. The raw counts are not deposited. The run shows only that the pipeline executes; no moment residual can be nonzero on a complete support.

To evaluate the coverage residual on an incomplete support we ran our own job at $L = 8$ (16 qubits, periodic chain, $U/t = 4$) and deposit its raw counts. At $L = 6$ even noiseless sampling of the companion's circuits covers the whole sector at 5×10^4 shots per circuit [20], which is why the size was raised. The reference state is $|\varphi\rangle = c_{0\uparrow}^\dagger|0\rangle$, with $|0\rangle$ the exact half-filled ground state ($E_0 = -4.6035t$); $|\varphi\rangle$ lies in the addition sector $N_\uparrow = 5$, $N_\downarrow = 4$, of dimension 3920. Its weight is $m_0 = \langle c^\dagger c \rangle = 1 - \langle n_{0\uparrow} \rangle = 1/2$ exactly, because translation invariance fixes $\langle n_{j\uparrow} \rangle = 4/8$ on every site (four up electrons on eight sites; operator and Lehmann values agree to 6×10^{-17}). The seven circuits belong to the companion's family [20]. Each prepares a computational-basis determinant with X gates (five up and four down electrons) and applies three Trotter steps of on-site R_{zz} and nearest-neighbor $R_{xx}R_{yy}$ hopping, without Jordan-Wigner strings, up to times 0, 0.5, ..., 3.0 in units of $1/t$. The circuits conserve particle number and S_z , so without noise they sample only the addition sector. They do not apply the Hubbard propagator to $c_p^\dagger|0\rangle$, and they are not targeted at $|\varphi\rangle$. We ran them at 50, 30, 16 and 4×10^3 shots per circuit (3.5×10^5 shots in total at the largest budget) with SamplerV2, using measurement twirling, Pauli gate twirling and XpXm dynamical decoupling. No readout-error mitigation was applied; SamplerV2 returns twirled raw counts. Each pooled record is read in the correct bit order and post-selected to the sector, with no configuration recovery. Post-selection keeps 23.5–24.1% of the shots (82,697 of 350,000 at 50k). At 50k it keeps 92% of the shots of the $t = 0$ circuit and 11–13% of each of the six evolved circuits.

The retained bitstrings define $\mathcal{S}$, and the coverage residual $\Delta_0 = |m_0 - \bar{m}_0| = \langle \varphi | (\mathbb{1} - \Pi_{\mathcal{S}}) | \varphi \rangle$ [Eq. (B3) at $k = 0$] is evaluated exactly on it. As the budget falls from 50k to 4k, the configuration coverage $|\mathcal{S}|/3920$ falls from 94% to 35% and the residual grows: $\Delta_0 = 0.0044$, 0.0323, 0.1232 and 0.3726 (Fig. 10). In the companion's notation $\Delta_0 = m_0(1 - w)$, with w the captured weight of $\mathcal{S}$ [20]. Configuration coverage and captured weight differ: at 50k, 94% of the configurations carry 99.1% of the weight ($1 - w = 0.009$). The first-moment residual $\Delta_1 = |\bar{m}_1 - \bar{m}_1|$ (0.019, 0.057, 0.024, 0.869) is not monotone in the budget; like Δ_0 , it combines a device-sampled support with exact classical amplitudes.

Reference distributions. A graded residual needs a reference. We compare the device with four reference samplers and report empirical percentile ranks, that is, the fraction of replicas at or below the device value. We do not use Gaussian z -scores: the weight of $|\varphi\rangle$ is concentrated (1% of the configurations carry 58% of m_0), and the distributions of Δ_0 are skewed.

(i) *Noiseless sampling at the device's raw shots.* Two thousand multinomial replicas of the exact output distributions of the same seven circuits give $\Delta_0 = 0.036 \pm 0.018$, 0.066 ± 0.027 , 0.121 ± 0.035 and 0.269 ± 0.032 (mean $\pm$ sd) at 50/30/16/4k. The device lies below all 2000 replicas at 50k, at the 1.7th percentile at 30k, at the 57.9th percentile at 16k and above all 2000 replicas at 4k. Its support is larger than in every replica at 50k and 30k ($|\mathcal{S}| = 3682$ against 3379 ± 14 ; 3330 against 3183 ± 16) and smaller than in every replica at 16k and 4k (2798 against 2877 ± 18 ; 1379 against 2027 ± 21). Noiseless sampling cannot reach $\Delta_0 = 0$: the seven output distributions together cover 3864 of the 3920 configurations, which leaves 8.8×10^{-5} at infinite shots. The device's Δ_1 lies at the 35th, 82nd and 27.5th percentiles at 50/30/16k and above all replicas at 4k.

(ii) *Noiseless sampling at the device's retained counts.* This comparison depends on how the retained counts are allocated. Matched circuit by circuit, the comparator puts 46,216 of its 82,697 shots at 50k on the $t = 0$ circuit, which is a single determinant without noise; its mean Δ_0 is 0.224, 0.279, 0.343 and 0.443 at 50/30/16/4k. With the same total split equally over the seven circuits it is 0.150, 0.205, 0.273 and 0.404. Under both allocations the device lies below every replica at 50, 30 and 16k; at 4k it lies at the 0.6th (per circuit) or 5.8th (equal split) percentile. The margin shrinks with the budget: the reference mean exceeds the device value 51-, 8.6-, 2.8- and 1.19-fold per circuit, and 34-, 6.3-, 2.2- and 1.08-fold with the equal split.

(iii) *Uniform noise.* Mixing each circuit's output distribution P_k with a fraction p of uniformly random bitstrings, $(1-p)P_k + p/2^{16}$, before post-selection lowers the expected Δ_0 at 50k monotonically, from 0.036 at $p = 0$ to 0.0056 at $p = 0.5$ and 0.0024 at $p = 1$. At 30k and 16k it is lowest near $p = 0.75$ –0.81 and rises again toward $p = 1$. At 4k it increases with p ($0.270 \rightarrow 0.326$), because post-selection keeps only 6% of uniform shots. At the p that reproduces the device's retention ($p \approx 0.81$), the mixture's mean (0.0023, 0.017, 0.075, 0.295) is below the device at every budget. The device nevertheless lies inside the mixture distribution at 50–16k (85.7th, 91.6th and 88.7th percentiles of 1000 replicas) and at its far upper edge only at 4k (99.9th; one replica lies higher).

(iv) *Uniform in-sector sampler.* Drawing configurations uniformly from the sector at the device's retained count N , with no device and no circuit, gives $\mathbb{E}[\Delta_0] = m_0(1 - 1/3920)^N = 3.4 \times 10^{-10}$, 1.5×10^{-6} , 5.2×10^{-4} and 0.093, below the device at every budget. At 50k and 30k this sampler covers essentially the whole sector, and in all ten replicas checked its reconstruction is exact to the Lanczos floor (relative L_1 error below 4×10^{-9}).

What the device run shows. Δ_0 rewards support spreading, not fidelity. The noisy device leaves less weight uncovered than all 2000 noiseless replicas of its own circuits at 50k and than 98% of them at 30k, where its support is the larger one; at 16k and 4k, where its support is smaller than in every replica, it is typical

or worse. Mixing uniform noise into the noiseless output lowers Δ_0 at 50k, and a sampler with no quantum content does best, because the 3920-configuration sector is small enough for uniform sampling to cover all or most of it at these budgets (81% of the configurations at 4k). A related effect has been reported for ground-state quantum-selected configuration interaction, where sampling noise can enlarge the configuration space beyond what the ideal ansatz supports [58]. The growth of Δ_0 as the budget falls holds for every sampler considered. The run therefore shows that the coverage residual can be evaluated on a support sampled by a device. It cannot distinguish a good device from a noisy one, and it does not certify the reconstruction. Relative to the companion, which introduces the captured weight, proves that it bounds the reconstruction error from below and shows that on its own it cannot certify a reconstruction [20], this section adds the evaluation on incomplete, device-sampled supports at four budgets with deposited counts, and the reference distributions above.

A device-data false negative. The records also give the relative L_1 error of each reconstruction, $\int |A_S - A| d\omega / \int A d\omega$ on the job's 600-point frequency grid at broadening $\eta = 0.15t$: 0.35, 0.56, 0.90 and 1.12 at 50/30/16/4k. At 50k both residuals, $\Delta_0 = 0.0044$ and $\Delta_1 = 0.019$, lie below the bias terms that the pre-registered rule of Sec. V would set, $\tau_k \geq z b m_k$ with $z = 1.96$ and $b = 2\%$, that is $\tau_0 \geq 0.020$ and $\tau_1 \geq 0.082$. The reconstruction nonetheless has a relative L_1 error of 0.35. These thresholds are counterfactual: the rule was sealed for the $L = 6$ current probe, and here $\hat{m}_0, \hat{m}_1$ are exact. This is consistent with the companion's result that captured weight alone cannot certify a reconstruction: its bound places the error above $1 - w = 0.009$ and needs the boundary coupling for an upper bound [20]. On the line shape, too, noiseless sampling does worse at the two higher budgets. Its reconstructions give 0.566 ± 0.032 at 50k (all 80 replicas above the device) and 0.630 ± 0.043 at 30k (device at the 1.25th percentile). At 16k and 4k the device is worse (96th percentile, and above all 80 replicas). The uniform in-sector sampler gives 0.014 at 16k and 0.90 at 4k, below the device at every budget.

A superseded noise model. A simpler model draws shots from $|\varphi|^2$ at the pooled retained count; it is kept in the repository as a superseded record. It underestimates the spread of Δ_0 by factors of 34 to 195 relative to the raw-matched noiseless replicas, and under its own threshold a sampler drawing exactly from $|\varphi|^2$ would fire at every budget (modeled false-positive rate 100%). We use neither that model nor any z -score derived from it. A nonparametric bootstrap of the retained counts is biased upward (its 50k interval [0.015, 0.101] excludes the point estimate 0.0044) and is not used as a test.

The discriminating off-diagonal estimator (m_1, m_2 from rotated, qubit-wise-commuting bond-basis settings, Fig. 12) is verified only in simulation (Appendix B). On present hardware, at the depth of the companion's circuits, its forecast bias is a large fraction of the signal. Un-

der global depolarizing noise of strength p , the estimate of $m_1 = \langle 0|O|0 \rangle$, with $O = c_{0\uparrow}(H - E_0)c_{0\uparrow}^\dagger$, becomes $(1 - p)m_1 + p \text{Tr}(O) / \dim \mathcal{H}$, a bias of $p [\text{Tr}(O) / \dim \mathcal{H} - m_1]$. For the $L = 8$ device Hamiltonian $\text{Tr}(O) / \dim \mathcal{H} = 6.80$ against $m_1 = 2.10$ (ratio 3.24), so the bias is upward. An exact density-matrix simulation of a bond-basis probe with nearly the same ratio (the $q = \pi$ density probe of the open $L = 6$ chain at $U/t = 4$: $m_1 = 10.6$, $\text{Tr} / \dim = 34.3$, ratio 3.23), with local two-qubit depolarizing noise at $\varepsilon = 2.5 \times 10^{-3}$, gives a first-moment bias of 13% of the signal at 100 CZ and 37% at 292 CZ, the depth of the companion's circuits. This static, local noise model omits the operator spreading of a real circuit and is a lower bound. Global depolarizing at the same depths gives 49% and 116%, and a coherent- ZZ model gives 47% at 292 CZ. The particle-number post-selection that protects the diagonal channel does not survive the bond-basis rotations. A device-side off-diagonal catch therefore awaits lower gate error (see the n_k gate below), and no device datum here bears on off-diagonal content. Table II summarizes what is demonstrated on hardware, in simulation and analytically, and what is deferred.

The off-diagonal limitation concerns the bond-basis estimator of m_1 . It does not rule out every discriminating on-device channel. A diagonal one exists: the momentum distribution $n_k = \int_{\omega < 0} A(k, \omega) d\omega = \langle c_{k\sigma}^\dagger c_{k\sigma} \rangle$, the occupied spectral weight at momentum k . The total weight $\int A(k, \omega) d\omega = \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1$ is pinned at every k and is blind to how the weight splits between the occupied and empty sides; n_k is not. For the $L = 6$ doped ground state at filling 1/3 (committed script `nk_falsifier.py`) n_k has a finite-size step $\max_k n_k - \min_k n_k = 0.91$ at $U/t = 4$ and 0.80 at $U/t = 8$; in the thermodynamic limit the one-dimensional step softens to a Luttinger power-law cusp. A reconstruction that flattens the dispersion moves $\hat{n}_k$ toward the filling value, while the true n_k keeps the step. (At half filling the chain is a Mott insulator and n_k is analytic, so this discriminant targets doped sectors, not the half-filled spectrum of Fig. 3.) n_k is a number operator in the momentum basis, read out diagonally after a number-conserving fermionic Fourier transform (a Givens network of $L(L-1)/2$ rotations per spin sector). The particle-number and spin post-selection that protects the coverage channel therefore survives the basis change. The loss of post-selection and the full-space trace-to-value bias that affect the bond-basis rotations do not arise; depolarization acts within the sector instead, as described next. In an in-sector depolarizing model with post-selection at $\varepsilon = 2.5 \times 10^{-3}$, the ~ 60 -CZ transform adds a flattening bias of $\approx 9\%$ of the step (a connectivity-free lower bound, before fermionic-SWAP routing). $\hat{n}_k$ is not immune to noise. Depolarization flattens it toward the same filling value that a dispersion corruption does, so it discriminates only corruptions whose flattening exceeds the device's own, and it inherits the depolarization of the state preparation. Readout error and coherent or leakage noise, which need not flatten toward filling, are not modeled, and post-selection costs shots.

A pre-registered on-device n_k test was sealed on 2026-09-02 (manifest SHA-256 4962...ae80; doped $L = 6$ ring, $U/t = 8$, 12 qubits), after a simulated Stage-0 gate passed at the manifest's noise floors, $T_1 \geq 150 \mu\text{s}$ and $T_2 \geq 100 \mu\text{s}$ on every qubit of the chain. That pass had little margin: at those floors the sealed post-mitigation residual check ran at 74–100% of its threshold. The single permitted amendment would have relaxed these floors to $T_1 = 100 \mu\text{s}$ and $T_2 = 70 \mu\text{s}$, provided that three pre-committed simulated verdict rows all passed. The worst-admissible row ($T_1 = 100 \mu\text{s}$, $T_2 = 70 \mu\text{s}$, two-qubit error parameter $\varepsilon = 3.5 \times 10^{-3}$) failed on 2026-09-05 at the tightest per-qubit readout cap of a pre-committed ladder (1.8%): the upper confidence bound of the sealed post-mitigation residual check was 0.0287, against the sealed threshold 0.0267. A row identical except $\varepsilon = 2.85 \times 10^{-3}$ passed (0.0258). In this harness ε is a compound knob that moves the depolarizing and coherent- ZZ components together, so the pair of rows does not locate a physical two-qubit error rate. The amendment was denied, the original floors stand, and the experiment has not been run. The forecasts above assume $\varepsilon = 2.5 \times 10^{-3}$; the failed gate row used 3.5×10^{-3} . An on-device n_k test, like the off-diagonal catch, awaits lower-error hardware.

VII. A CLOSED-FORM BOUND: THE CHRISTOFFEL MISS-DISTANCE

Having shown empirically what the screen catches, we now state what a *passing* screen implies. The bound below is classical, from the theory of moments and orthogonal polynomials. Our contribution is its evaluation as a diagnostic for a reconstructed spectrum, not the mathematics; the Christoffel–Darboux kernel is by now an established data-analysis tool in its own right [69, 70, 98, 99]. Consider the cumulative spectral weight $F(t) = \int_{-\infty}^t A(\omega) d\omega$ of a diagonal $T = 0$ Lehmann measure, and suppose the first $2n+1$ moments $m_0, \dots, m_{2n}$ are known exactly. They fix the orthonormal polynomials $p_0, \dots, p_n$ via the Hankel system [65]. Let $W_n(t) = 1/\sum_{j=0}^n p_j(t)^2$ be the Christoffel function, also written $\lambda_n(t) \equiv W_n(t) = 1/K_n(t, t)$ with $K_n(x, y) = \sum_{j=0}^n p_j(x)p_j(y)$ the Christoffel–Darboux kernel; its construction from the moments is given in Appendix C [69, 70].

Theorem 1 (Cumulative-weight miss-distance bound (classical; re-derived)). *Let $d\mu = A(\omega) d\omega$ be a positive measure of compact support (a diagonal $T = 0$ Lehmann spectrum) whose Hankel matrix $H_n = [m_{i+j}]_{i,j=0}^n$ is positive definite, and suppose the moments $m_0, \dots, m_{2n}$ are known exactly. For $t \in \mathbb{R}$ define the cumulative-weight miss-distance as the extremal spread $\sup |F_{\nu_1}(t) - F_{\nu_2}(t)|$ over all positive measures ν_1, ν_2 on $\mathbb{R}$ sharing those moments. Then for every $t \in \mathbb{R}$ this spread equals the Christoffel function $W_n(t)$, which is also the supremum of the mass a moment-matching measure can place at t*

TABLE II. **What is demonstrated on hardware, in simulation, analytically, and deferred.** The only device-derived quantity is the support set $\mathcal{S}$; no moment is estimated from device counts. Every discriminating result is classically reproducible simulation, and the beyond-classical field test is deferred.

quantity	source	status
support set $\mathcal{S}$ and coverage residual Δ_0 , $L = 8$ addition sector	ibm_fez ($\mathcal{S}$ only)	Δ_0 computed classically on $\mathcal{S}$; rewards support spreading (a uniform in-sector sampler scores lower); tests neither the device nor $A(\omega)$ (Fig. 10)
full-sector recovery, $L = 6$ (companion)	ibm_fez	executability only; the retained determinants all came from device errors (Fig. 11)
off-diagonal moments m_1, m_2 (weight discrimination)	simulation	deferred: forecast depolarizing bias 37–116% of the signal at 292 CZ (Sec. VI; Fig. 9)
momentum distribution n_k (diagonal discriminant)	simulation (sealed gate)	gate-negative at the relaxed noise floors; not run
blinded detection and false-positive rates	simulation	sealed primary endpoint: TPR 101/188 = 53.7% at FPR 1/112 = 0.9%, unbiased simulated estimator; two sealed secondary predictions failed (Sec. V)
same-sample $\hat{m}_k$ and $\bar{m}_k$ (one set of shots)	simulation	correlation $\rho \approx 0.43$ at 84% coverage, no power gain (Sec. IX)
Christoffel miss-distance bound; moment-interval layer	exact / analytic	classical bound, re-derived; Hausdorff interval shown; composition with the Wang/Mortimer SDP not implemented
field catch of an unknown-truth error (beyond-classical)	—	future work (Sec. IX)

(the Gauss–Radau weight $\lambda_n(t)$, attained except at the n zeros of the orthonormal polynomial p_n). In particular, any reconstruction whose moments $m_0, \dots, m_{2n}$ equal the true ones obeys $|F_{\text{rec}}(t) - F_{\text{true}}(t)| \leq W_n(t)$ at every $t \in \mathbb{R}$ (Appendix C) [100, 101].

Remarks. The equality needs no convex-hull hypothesis. Once $H_n \succ 0$, a Gauss–Radau rule with positive weights and a node prescribed at t exists for every real t other than the n zeros of p_n , and its atom at t is $W_n(t)$; at

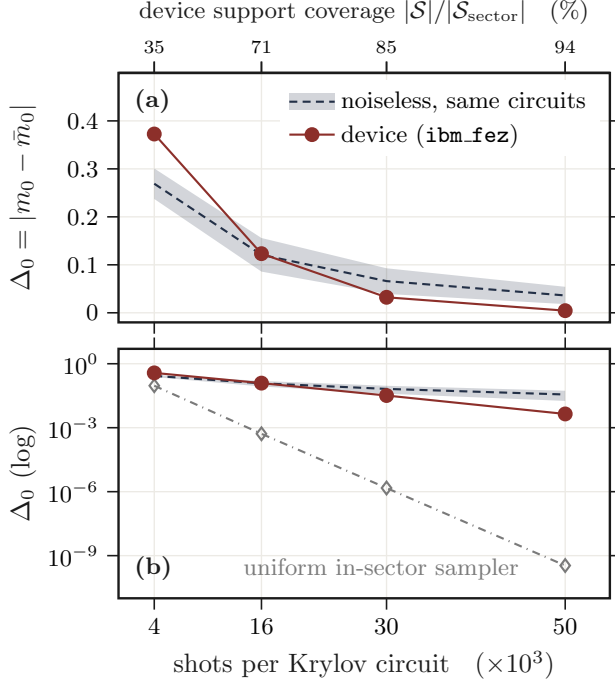


FIG. 10. **Coverage residual on device-sampled supports** (`ibm_fez`, $L = 8$, 16 qubits, $U/t = 4$, addition sector of dimension 3920; seven circuits; measurement twirling, Pauli gate twirling and dynamical decoupling, no readout-error mitigation). Oxblood: $\Delta_0 = |m_0 - \bar{m}_0|$, computed exactly on the support $\mathcal{S}$ retained from the device counts; the top axis of (a) gives the device coverage $|\mathcal{S}|/3920$. Indigo: 2000 multinomial replicas of the exact output distributions of the same circuits at the device’s raw shots per circuit (dashed, mean; shading, mean ± 1 sd; the distribution is right-skewed, so the shading is not a 68% interval). Grey diamonds, drawn in (b) only: a uniform in-sector sampler at the device’s retained count N , $\mathbb{E}[\Delta_0] = m_0(1 - 1/3920)^N$. (a) Linear scale; (b) logarithmic scale. The device lies below all noiseless replicas at 50k, at the 1.7th percentile at 30k, at the 57.9th at 16k and above all of them at 4k. The uniform sampler, which uses no device, scores lower than the device at every budget. Δ_0 measures support coverage and rewards support spreading; it is not a fidelity diagnostic (main text).

those zeros the same value is approached as a limit (Appendix C). We checked this on the measure of Fig. 13: all 24 rules tested ($n = 1, \dots, 4$; 20 of them with t outside the hull of the weighted atoms) have positive weights, reproduce $m_0, \dots, m_{2n}$ to 1.4×10^{-17} , and carry an atom equal to $W_n(t)$ to 2×10^{-16} . The bound is monotone, $W_{n+1}(t) \leq W_n(t)$, since $K_{n+1} \geq K_n$. For a spectrum of N distinct poles $H_n > 0$ only for $n \leq N - 1$, and at a pole ω_i one has $W_n(\omega_i) \geq w_i > 0$ (the residue), with equality at $n = N - 1$; W_n does not vanish on the support. Pointwise vanishing $W_n(t) \rightarrow 0$ is a thermodynamic-limit statement. If the discrete spectrum densifies to a measure that is Stahl–Totik regular with compact support S and absolutely continuous with a density A obeying the local Szegő condition $\int_I \log A > -\infty$ on an interval

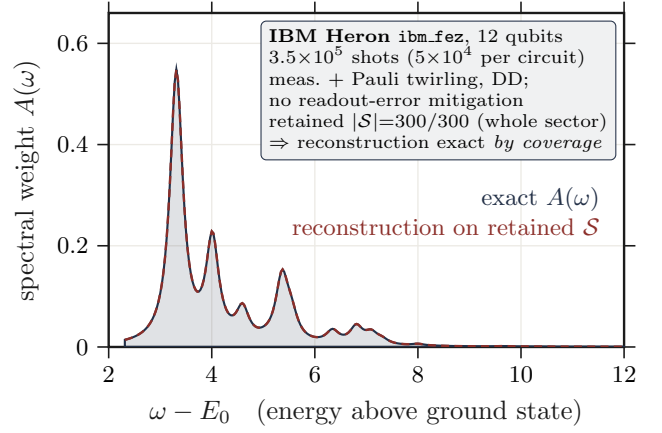


FIG. 11. **Executability run of the companion study** ($L = 6$, full-sector support). Exact $A(\omega)$ (indigo) and the reconstruction on the device-retained support (dashed), for the $L = 6$ Hubbard model at $U/t = 4$ in the addition sector $4 \uparrow, 3 \downarrow$ (300 configurations), $\eta = 0.15t$. `ibm_fez`, 12 qubits, 3.5×10^5 shots (5×10^4 on each of seven circuits; job `d9s16avpems73ct6g8g`), with measurement twirling, Pauli gate twirling and dynamical decoupling and no readout-error mitigation [20]. The retained support is the whole sector ($|\mathcal{S}| = 300/300$), so the reconstruction equals exact diagonalization by coverage. The bitstrings were post-selected after being read in reversed qubit order, so all 300 retained determinants came from device errors. The run shows only that the pipeline executes; it says nothing about accuracy at incomplete coverage. The coverage residual on an incomplete, device-sampled support is Fig. 10.

$I \subset S$, then $n W_n(x) \rightarrow A(x)/\omega_S(x)$ for almost every $x \in I$, with ω_S the equilibrium density of S [83]; on a single interval this reduces to Máté–Nevai–Totik [97].

The equality is the classical content of the Chebyshev–Markov–Stieltjes inequalities and of Gauss–Christoffel moment theory, re-derived in Appendix C [65, 66, 69]. We say “bound”, not “certify”: the screen never certifies that $A(\omega)$ is correct. The bounded object is cumulative weight, not the pointwise value of $A(\omega)$ or the position of a peak [69], and the contraction *rate* holds only under the regularity hypothesis above [83, 97]. The top panel of Fig. 13 shows the mechanism: the true $F(t)$ lies between the Gauss–Radau extremal envelopes of all measures sharing the moments, and the width of that bracket is $W_n(t)$.

The theorem assumes exact moments, and the deployed screen does not supply them. If a reconstruction’s moments $m_0, \dots, m_{2n}$ equal the true ones, its $F(t)$ is pinned within $W_n(t)$. The screen instead passes a reconstruction whose residuals lie within tolerances, $|\Delta_k| \leq \tau_k$, and only through order $2n = 2$. Two consequences follow for the measure of Fig. 13. First, at the deployed order $n = 1$ the exact-moment bracket is the one-sided Chebyshev (Cantelli) bound, $W_1(\omega) = m_0 \sigma^2 / [\sigma^2 + (\omega - \bar{\omega})^2]$ with mean $\bar{\omega} = 11.10t$ and variance $\sigma^2 = 2.10t^2$ (this closed form and the orthogonal-polynomial evaluation

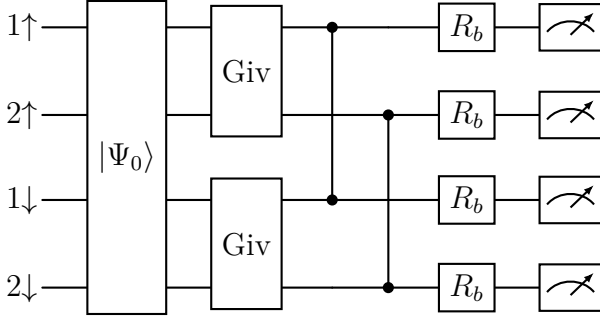


FIG. 12. **Measurement circuit of the bond-basis moment estimator (schematic; not run on hardware).** One circuit of the estimator of Appendix B, drawn for a two-site unit cell (4 qubits: two spin-up and two spin-down orbitals). The first blocks stand for a state preparation: a reference state $|\Psi_0\rangle$, number-conserving hopping within each spin species (Giv), and the on-site interaction $U\hat{n}_\uparrow\hat{n}_\downarrow$ (the controlled pairs, coupling each up orbital to its down partner). A per-qubit basis rotation R_b follows (H or $S^\dagger H$ for the X or Y Pauli components, the identity for Z), then a computational-basis measurement. One circuit is needed per qubit-wise-commuting Pauli group. The diagonal density probe gives m_0 from the single all- Z setting; the hopping content of m_1, m_2 needs the rotated settings (357 groups for the full battery at $L = 6$; 83 for the (m_0, m_1) falsifier at $L = 12$). The verification of Appendix B samples the exact ground state in each basis instead of running a preparation circuit. On present hardware the off-diagonal estimate would carry the bias of Sec. VI.

agree to 1.6×10^{-15}). Its maximum is m_0 , at the mean, which here lies $0.01t$ from the dominant atom ($11.09t$), so the bracket is vacuous at the peak; just above the top pole, at $\omega = 14t$, it is still 0.20. Second, tolerances loosen it further. The largest atom that a positive measure with $|m_k - m_k^{\text{true}}| \leq \tau_k$ ($k \leq 2$, the deployed threshold rule evaluated on this state) can place at ω is 3.2–7.7 times W_1 at $\omega = 14\text{--}20t$ (0.63 against $W_1 = 0.20$ at $14t$) and 10–11 times W_1 far outside the band ($\omega \approx 34\text{--}50t$); at the mean it exceeds m_0 ($1.045m_0$). At exact-moment orders $n \leq 4$ ($m_0, \dots, m_8$) the global maximum falls only to $\max_t W_4 = 0.882$, and at $n = N - 1 = 4$ the value of W_4 at the dominant atom already equals its residue 0.778 and stays there (panel c). No positive-order moment beyond m_2 is estimated in this work. At the deployed order, passing the screen therefore places no constraint on the cumulative weight at the peak. This is the moment-level counterpart of the companion’s finding that captured weight alone cannot certify a sample-based spectral function [20].

Refutation and bracketing are separate statements. A reconstruction is rejected when some $|\Delta_k|$ exceeds τ_k ; that test does not involve W_n , and its reach is set by the thresholds (Sec. VIII). A Hankel/Stieltjes violation of the *estimated* moments rejects the estimate, not a non-negative reconstruction, whose own moments satisfy pos-

itivity by construction. W_n governs only the complementary statement, how tightly a *passing* reconstruction’s cumulative weight is bracketed. The looseness above therefore limits what passing implies; it does not limit the power to reject.

The bracket is not uniformly loose. At a frequency the measure leaves empty, $W_n(t)$ bounds the weight that any moment-matching reconstruction can place there. The rescaled frame of Fig. 13 maps the full-Fock range $[0.148, 52.21]t$ onto $[-1, 1]$; the weighted atoms occupy $t \in [-0.91, -0.47]$, and 73.5% of the axis lies above the top pole ($13.9t$). At the rescaled points $t = 0.3, 0.6$ and 0.9 in that empty region ($\omega = 34.0, 41.8$ and $49.6t$) the 60-digit evaluation of App. C gives $W_1 = 4.0 \times 10^{-3}$, 2.2×10^{-3} and 1.4×10^{-3} , and at $t = 0.6$ the higher orders give $W_2 = 1.4 \times 10^{-5}$, $W_3 = 2.7 \times 10^{-8}$ and $W_4 = 3.6 \times 10^{-10}$. These small values reflect the distance from the band, not resolution: at $n = 1$ they are the Cantelli tail. Nearer the band the bracket is wider ($W_1 = 0.20$ at $\omega = 14t$), and the deployed tolerances raise the far-out-of-band $n = 1$ values about tenfold. This is a computed property of this measure, not a theorem, and we do not extrapolate it to the two-sided single-particle Mott gap, a distinct measure not tested here.

Proposition 1 (Two-sided Markov–Krein window-mass bound). *Let $d\mu = A(\omega) d\omega$ be a positive measure (a diagonal $T = 0$ Lehmann spectrum) with $H_n = [m_{i+j}]_{i,j=0}^n \succ 0$, and suppose $m_0, \dots, m_{2n}$ are known exactly. For any frequency window $[c, d]$ with $c \leq d$ real, the mass $\mu([c, d]) = \int_c^d A(\omega) d\omega$ of every positive measure sharing those moments obeys*

$$\begin{aligned} \max\{0, F_-(d) - F_+(c)\} &\leq \mu([c, d]) \\ &\leq \min\{m_0, F_+(d) - F_-(c)\}, \end{aligned} \quad (1)$$

where $F_\pm(t)$ are the lower/upper principal (Gauss–Radau) cumulative representations of Lemma 3 (App. C): $F_-(t)$ is the mass that the $(n+1)$ -point Gauss–Radau rule with a node fixed at t places strictly left of t , and $F_+(t) = F_-(t) + \lambda_n(t)$ with $\lambda_n(t) = W_n(t)$ its atom at t (at the n zeros of p_n , where that rule degenerates, both are taken as limits). The bracket is two-sided. Its width is at most $W_n(c) + W_n(d)$ (with equality before intersecting with the trivial bounds $[0, m_0]$), so it inherits the monotone contraction $W_{n+1} \leq W_n$ and is tight wherever both endpoints fall in weight-poor frequency. The sharp Krein–Nudelman extremum, attained by a principal representation with prescribed nodes at c and d (equivalently, the optimum of the scalar truncated-moment problem over positive measures), is no wider. Equation (1) follows directly from the Chebyshev–Markov–Stieltjes inequalities and extends the one-sided point bound $|F_{\text{rec}} - F_{\text{true}}| \leq W_n$ to a two-sided bound on integrated weight [67, 69, 100, 101].

We evaluated the two-sided bracket (1) on the same $L = 6$, $U/t = 8$ current measure (App. C; committed script `markov_krein_window.py`), again at exact mo-

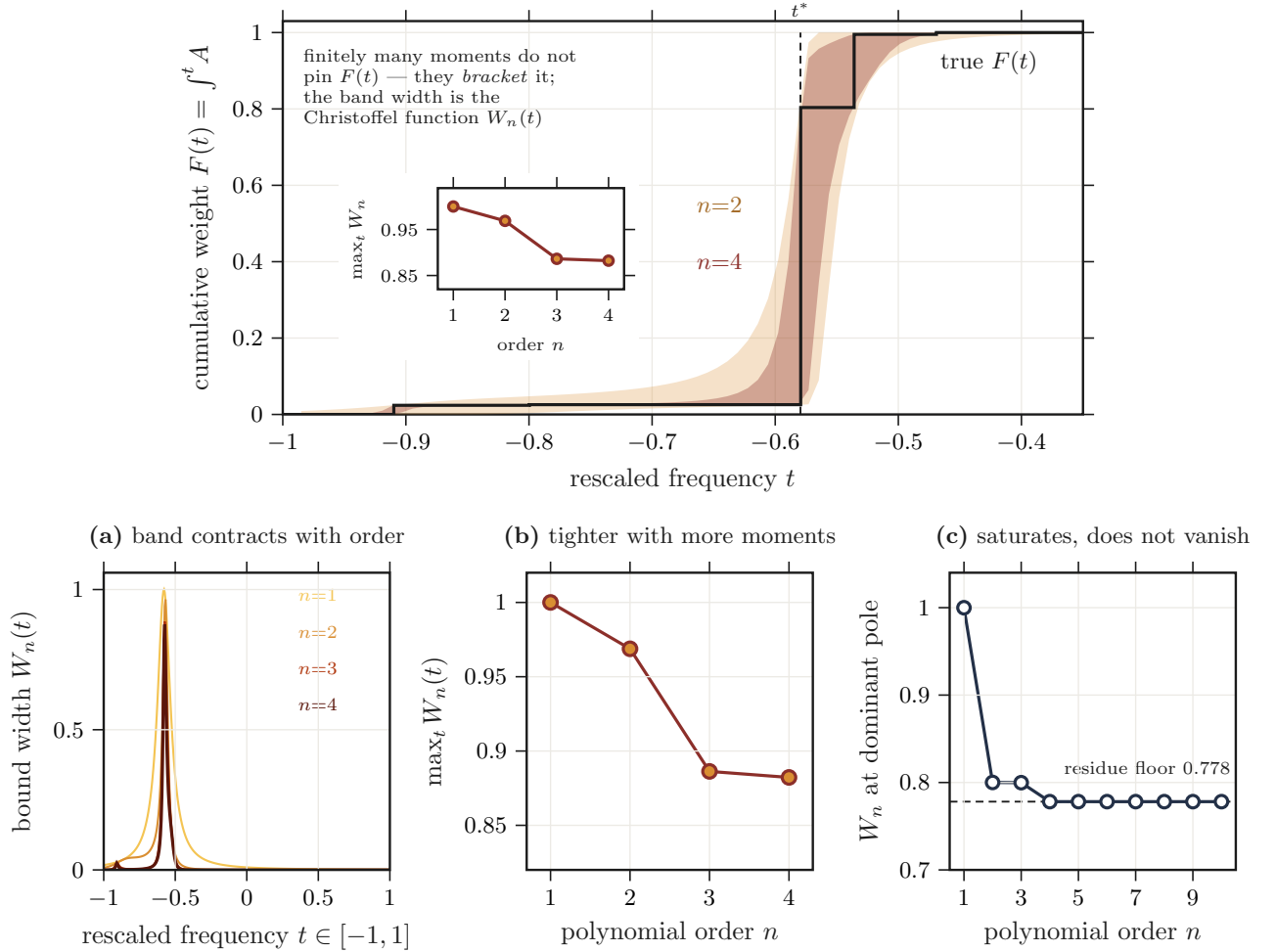


FIG. 13. **The Christoffel miss-distance bound at exact moments.** *Top—the bracketing mechanism.* Finitely many moments do not pin the spectrum; they *bracket* its cumulative weight $F(t)$ (ink staircase) between the extremal Gauss–Radau measures sharing $m_0, \dots, m_{2n}$. The band width is the Christoffel function $W_n(t)$; the $n=4$ band (deep) nests inside the $n=2$ band (light), and t^* marks the dominant atom. *Bottom (a–c)—the bound quantified.* (a) $W_n(t) = 1/\sum_{j=0}^n p_j(t)^2$ contracts as n grows. (b) Its maximum over t decreases with n : 1.000, 0.969, 0.886, 0.882 for $n = 1, \dots, 4$ (exact maxima, by root-finding). At the deployed order $n = 1$ the maximum is $W_1 = m_0$, attained at the mean frequency, so there the bracket is vacuous. (c) At the dominant atom ($\omega = 11.09t$, residue 0.778) W_n reaches the residue at $n = 4$ and stays there: the measure has five weighted atoms, and the bracket at an atom never falls below the jump of F it carries. Pointwise $W_n \rightarrow 0$ is a thermodynamic-limit statement for a continuous density, not a property of a finite atomic spectrum. All curves assume exact moments; a reconstruction that passes with finite tolerances is bracketed less tightly (see text). Doped Hubbard $L = 6$, $U/t = 8$ current Lehmann measure, normalized to $m_0 = 1$ (the ground state is a spin triplet; the measure is the same for each of its S_z components), rescaled to $t \in [-1, 1]$ from the full-Fock frequency range $[0.148, 52.21]t$, zero-weight states included, so the weighted atoms occupy $t \in [-0.91, -0.47]$ [65, 66, 69, 70, 83, 97].

ments. At $n = 4$ ($m_0, \dots, m_8$), a window with one edge at the 400-point-grid maximizer of W_4 , just above the dominant atom, gives $\mu \in [0.070, 0.945]$, width 0.875; the exact maximum of W_4 , at a nearby frequency, is 0.882. A window that contains the peak with both edges in weight-free frequency (rescaled $[-0.75, -0.35]$) gives $\mu \in [0.9732, 0.9758]$ around the exact 0.9744, and the empty window rescaled $[0.3, 0.9]$ ($\omega \approx 34$ – $50t$) gives $\mu \in [0, 3.2 \times 10^{-9}]$. At the deployed order $n = 1$ (m_0, m_1, m_2) the peak-containing window gives $\mu \in [0.848, 1]$ and the empty window $\mu \in [0, 0.004]$, while a window with an

edge on the peak is uninformative (its bracket runs from 0 to within 2×10^{-4} of m_0). These are exact-moment brackets. Under the deployed tolerances the upper end for the empty window is at least 0.040, the largest atom that a measure with m_0, m_1, m_2 within tolerance can place at its lower edge ($\omega \approx 34t$). The two-sided bound is thus tight when both window edges lie where W_n is small, and as loose as W_n when an edge sits on the peak. It is the real-axis, ground-state-moment counterpart of the Euclidean-time two-sided spectral bounds recently obtained from moment-problem semidefinite programs [78–80]; it differs

in that its moments are ground-state operator expectations rather than lattice correlators. As with the one-sided bound, we do not extrapolate it to the two-sided single-particle Mott gap.

VIII. DETECTABILITY AND ITS SAMPLE-COMPLEXITY LIMIT

The Christoffel bound of Sec. VII concerns what a passing screen implies: how tightly a *passing* spectrum's integrated weight is bracketed. Rejection has a matching characterization. It states which reconstruction errors the moment battery can detect in exact arithmetic, and what it can guarantee at finite thresholds, where the Vandermonde structure of the moment map sets the shot cost. The limiting error class is the *within-sector redistribution*: a corruption that keeps the retained determinant support (so the leakage estimator of Sec. IV is blind by construction) and only reweights the reconstructed poles. The statement below is not a new theorem. Part (a) restates the classical Prony/Vandermonde uniqueness of an atomic measure from its moments [65, 102]; part (b) turns it, together with an elementary shifted-power bound, into a detectability-versus-shot-budget statement for the battery. We keep the **theorem** environment for cross-referencing.

Theorem 2 (Detectability and completeness of the moment battery). *Let $A(\omega) = \sum_{n=1}^N w_n \delta(\omega - \omega_n)$ be a diagonal $T = 0$ Lehmann measure on N distinct poles $\{\omega_n\}$ with $w_n = |\langle n|J|0\rangle|^2 \geq 0$, and let a within-sector redistribution $\mathbf{w} \rightarrow \mathbf{w} + \boldsymbol{\delta}$ ($\boldsymbol{\delta} \neq 0$, keeping $w_n + \delta_n \geq 0$) act on the same poles. Write $\mathbf{m} = V\mathbf{w}$ with $V_{kn} = \omega_n^k$, the induced shift $\Delta_k = |(V\boldsymbol{\delta})_k| = |\sum_n \delta_n \omega_n^k|$, and $\rho = \|\boldsymbol{\delta}\|_1$. Then $\|\boldsymbol{\delta}\|_2 \leq \rho \leq \sqrt{N} \|\boldsymbol{\delta}\|_2$, and $\rho/2$ is the weight moved when m_0 is preserved.*

- (a) (Completeness in exact arithmetic.) *If the redistribution preserves $m_0, \dots, m_K$ with $K < N - 1$, then $\Delta_j > 0$ for at least one j with $K < j \leq N - 1$; quantitatively*

$$\max_{K < j \leq N-1} \Delta_j \geq \frac{\sigma_{\min}(V_{0:N-1})}{\sqrt{N-1-K}} \|\boldsymbol{\delta}\|_2, \quad (2)$$

with $V_{0:N-1}$ the $N \times N$ node-Vandermonde on rows $0, \dots, N - 1$ and both sides in one fixed frequency unit. Hence the order- $(N-1)$ battery is complete for fixed-pole redistributions: no nonzero within-sector redistribution preserves all of $m_0, \dots, m_{N-1}$. If the poles may also move (two measures of at most N atoms each), completeness holds at order $2N - 1$ by Prony counting [65, 102]. The only alteration invisible to every moment is weight traded among degenerate eigenstates at one frequency, which leaves $A(\omega)$ unchanged and is not an error.

- (b) (Finite thresholds.) *Let each moment carry the noise floor $\delta_k = \sqrt{\text{Var}(O_k)/N_s + b_k^2}$ at N_s shots, with bias budget b_k , and the two-sided threshold $\tau_k = z_{1-\alpha/2} \delta_k$. We call a given $\boldsymbol{\delta}$ exposed at order j when its noise-free shift exceeds the threshold, $\Delta_j > \tau_j$; whether a finite-shot test then rejects it is a question of power (Corollary below). Some $m_0, \dots, m_K$ -preserving redistribution is exposable at order $K+1$ iff $\Delta_{K+1}^{\max} > \tau_{K+1}$, with $\Delta_{K+1}^{\max}$ the linear program (E1). If rows $0, \dots, K+1$ of V have a nontrivial null space ($N - K - 2 \geq 1$), the smallest Δ_{K+1} over such redistributions is zero for every size ρ up to a limit set by positivity: the guaranteed power at order $K+1$ is zero, however well separated the poles. Two conditions bracket detection.*

- (i) Sufficient. *If $\boldsymbol{\delta}$ preserves $m_0, \dots, m_K$, it is exposed at some order $j \leq N - 1$ whenever the right-hand side of (2) exceeds $\max_{K < j \leq N-1} \tau_j$.*

- (ii) Necessary (cluster bound). *If $\boldsymbol{\delta}$ preserves $m_0, \dots, m_{s-2}$ and is supported on poles within an interval of diameter D and midpoint ω_c , then $(V\boldsymbol{\delta})_{s-1} = \sum_n \delta_n (\omega_n - \omega_c)^{s-1}$, so $\Delta_{s-1} \leq \rho (D/2)^{s-1}$; every higher shift obeys $\Delta_j \leq \rho \sum_{i=s-1}^j \binom{j}{i} |\omega_c|^{j-i} (D/2)^i$, which is also $O(\rho D^{s-1})$. Exposure at order $s - 1$ therefore requires*

$$\rho > \frac{\tau_{s-1}}{(D/2)^{s-1}}, \quad \text{equivalently} \quad N_s > \frac{z_{1-\alpha/2}^2 \text{Var}(O_{s-1})}{\rho^2 (D/2)^{2(s-1)} - z_{1-\alpha/2}^2 b_{s-1}^2}, \quad (3)$$

and no N_s suffices when $\rho (D/2)^{s-1} \leq z_{1-\alpha/2} b_{s-1}$. The shot cost to detect a redistribution of fixed size at its first exposable order grows at least as $D^{-2(s-1)}$ as the cluster contracts.

Proof sketch. Part (a): distinct poles make $V_{0:N-1}$ an invertible Vandermonde, so $V_{0:N-1} \boldsymbol{\delta} = 0$ forces $\boldsymbol{\delta} = 0$. Preserving $m_0, \dots, m_K$ zeroes the first $K+1$ components of $V_{0:N-1} \boldsymbol{\delta}$, whence $\|(\Delta_{K+1}, \dots, \Delta_{N-1})\|_2 = \|V_{0:N-1} \boldsymbol{\delta}\|_2 \geq \sigma_{\min}(V_{0:N-1}) \|\boldsymbol{\delta}\|_2$, and (2) follows from $\ell_2 \rightarrow \ell_\infty$. The moving-pole count is the classical uniqueness of an N -atom measure given $m_0, \dots, m_{2N-1}$. Part (b): a vector in the null space of rows $0, \dots, K+1$, scaled so that $w_n + \delta_n \geq 0$, preserves $m_0, \dots, m_{K+1}$, which gives the zero guaranteed power; (i) compares (2) with $\max_j \tau_j$. For (ii), expand $\omega^j = \sum_i \binom{j}{i} \omega_c^{j-i} (\omega - \omega_c)^i$. The terms with $i \leq s - 2$ are combinations of preserved moments and vanish, and $|\omega_n - \omega_c| \leq D/2$ on the support bounds the rest; inserting $\tau_{s-1} = z_{1-\alpha/2} \delta_{s-1}$ gives (3). The linear program (E1) and the numerical instance are in Appendix E.

Remarks. The clustered-node bounds $\sigma_{\min} \sim \Delta \omega^{s-1}$ of super-resolution theory, with $\Delta \omega$ the node spacing [103–106], are proved for Fourier-type nodes on the unit circle or torus, not for the real nodes of power moments, and

part (b)(ii) does not rely on them. For real nodes the same scaling, $\sigma_{\min} \propto D^{s-1}$ for the $s \times s$ Vandermonde of a cluster at fixed relative geometry, follows from Gautschi's inverse-Vandermonde bound [107, 108] together with the shifted-power identity (Appendix E). Numerically, for s equally spaced nodes at the median pole of the $L = 6$ instance below, $\sigma_{\min}$ falls with the spacing with fitted exponents 0.995, 1.98 and 2.96 for $s = 2, 3, 4$. For an isolated cluster, then, the sufficient condition (i) weakens with the same power of the spacing as the necessary bound (ii). If a cluster contracts uniformly, so that its minimal pole separation falls from $\Delta\omega_{\min}$ to $\Delta\omega$, D shrinks by the same factor and the lower bound (3) on the shot cost grows by at least $(\Delta\omega_{\min}/\Delta\omega)^{2(s-1)}$. A finite separation below which no budget suffices exists only with a bias floor, $b_{s-1} > 0$. Finally, the fixed-pole class excludes under-converged Krylov reconstructions. A reconstruction on a subspace containing the order- K Krylov space reproduces $m_0, \dots, m_{2K+1}$ exactly [20]; a Lanczos representation with $K+1$ nodes is such a reconstruction. With two or more nodes it therefore matches the whole deployed battery by construction, whatever its line-shape error; the one-node case matches only m_0, m_1 , misses m_2 by 2–4%, and passes within a threshold that budgets a 2% bias (largest residual $0.99\tau_2$). This accounts for the Krylov misses of Sec. V; Theorem 2 does not cover them because the poles move.

Corollary (detection power vs. cluster size and shot count). Under the Gaussian estimator validated in Fig. 14 (skewness -0.04 , excess kurtosis -0.05), the test reaches power $1 - \beta$ at level α when $\Delta_j \gtrsim z_{\alpha,\beta} \delta_j$, with $z_{\alpha,\beta} = z_{1-\alpha/2} + z_{1-\beta}$ and the bias budget counted in δ_j as in the deployed rule. By part (b)(ii) this requires $N_s \gtrsim z_{\alpha,\beta}^2 \text{Var}(O_{s-1}) / [\rho^2 (D/2)^{2(s-1)} - z_{\alpha,\beta}^2 b_{s-1}^2]$ at the first exposable order, and no budget suffices below the bias-limited cluster diameter $D_\star = 2(z_{\alpha,\beta} b_{s-1} / \rho)^{1/(s-1)}$. Without bias the cost is finite for every $D > 0$ and diverges only as $D \rightarrow 0$.

For the deployed $L = 6$, $U/t = 4$ battery the $N = 5$ poles are well separated (minimal separation $\Delta\omega_{\min} = 0.87t$; on nodes rescaled to $[-1, 1]$, $\sigma_{\min}(V_{0:4}) = 0.056$ and $\kappa = 50$), yet the guaranteed power of the deployed battery against fixed-pole redistribution is zero. The reason is the null-space dimension, not conditioning: rows 0, 1, 2 of V leave a two-dimensional null space on five poles. Over m_0, m_1 -preserving redistributions the linear program (E1) gives the signed shift of m_2 in $[-0.92, 7.09]$ (Table III). The largest shift is $3.8\tau_2$ ($\tau_2 = 1.848$), but the range contains zero. Three explicit redistributions pass the deployed battery. CE1 lies in that null space: it preserves m_0, m_1, m_2 exactly while moving 72% of the weight ($\rho = 1.43m_0$), emptying two of the five poles and cutting the dominant residue from 0.393 to 0.037. CE2 preserves m_0, m_1 and sets $\Delta_2 = \tau_2$ while moving 86% of the weight (dominant residue $0.393 \rightarrow 0.073$). CE3 holds every residual at or within its threshold and empties three poles, the dominant one included. The random redistribution committed with the original analysis also passes:

it moves 25% of the weight with $\Delta_2 = 0.044\tau_2$. Orders 3 and 4 would catch some of these, but not all. With the same rule ($\tau_3 = 15.0$, $\tau_4 = 122$; the local-estimator variance behind them is a lower bound), CE2 and CE3 give $\Delta_3 \approx 3.3\tau_3$ and $\Delta_4 \approx 7.1\text{--}7.2\tau_4$, whereas CE1 gives $0.23\tau_3$ and $0.70\tau_4$. Over all redistributions that keep m_0, m_1, m_2 exactly, $\Delta_3 \leq 0.35\tau_3$, and Δ_4 reaches $1.19\tau_4$ at the one that maximizes Δ_3 , a marginal case. Neither m_3 nor m_4 is estimated in this work. The sufficient condition (i) is far from met here: for CE1 and CE2 the right-hand side of (2) is below 4×10^{-3} , against $\max_j \tau_j = \tau_4 = 122$ in the same units. The residual blind spot of the deployed screen therefore has two sources. The first is the battery order: with $m_0, \dots, m_K$, every fixed-pole redistribution in the $(N - K - 1)$ -dimensional null space passes, regardless of pole separation. The second is pole clustering, which by (3) raises the shot cost of detection at least as $D^{-2(s-1)}$. Injectivity guarantees that some m_j separates any fixed-pole redistribution; the thresholds decide whether that separation is observable at a given budget.

IX. DISCUSSION AND LIMITATIONS

This paper contributes a method with stated operating limits: a reconstruction-independent moment screen for sample-based quantum spectral functions; estimators that evaluate the moments without reading them off $A(\omega)$, either as full-operator expectation values on the sampled subspace state (classical post-processing) or from separate bond-basis measurements on a prepared ground state (simulated only, Appendix B); a hash-sealed, blinded calibration of the screen's detection and false-positive rates where the truth is known; and classical bounds that state what a passing screen does and does not imply. The screen is a necessary condition. Failing it rejects the reconstruction at the pre-registered level (per-moment $\alpha = 0.05$; blinded false-positive rate 0.9%, $1/112$). Passing it excludes only the tested failure modes. Independence from $A(\omega)$ is what allows the two branches to disagree; it does not turn agreement into a statement about the line shape.

Relation to the companion study. The companion [20] develops an a posteriori error theory for the same class of sample-based spectral functions, and several of its results bear directly on this screen. Its captured weight w gives our coverage residual as $\Delta_0 = m_0(1 - w)$. Its moment-exactness theorem—a subspace that contains the order- K Krylov space of the probe reproduces the exact spectral moments through order $2K+1$ —explains why the under-converged Krylov reconstructions of Sec. V with two or more Lanczos nodes pass our battery by construction (none of 56 caught); the one-node reconstructions reproduce only m_0 and m_1 exactly and pass within the bias budget (none of 15 caught). Its leakage identity $\mu_2 - \mu_2^S = \|Q_S H P_S \varphi\|^2$ is our operator leak, Eq. (B3), at $k = 2$ when the probe state lies in $\text{span } \mathcal{S}$. And its

result that captured weight alone cannot certify the reconstruction has a device-data instance here: the 50k run of Sec. VI has $\Delta_0 = 0.0044$ and a relative L_1 error of 0.35 in $A(\omega)$ at broadening $\eta = 0.15t$. What this paper adds is (i) the use of low-order moments, evaluated independently of the reconstruction, as a test that applies to any reconstructed $A(\omega)$ and not only to Rayleigh–Ritz on a determinant subspace; its first and second moments detect weight moved in frequency at fixed total weight, which the captured weight cannot see; (ii) a sealed, blinded calibration of that test, with its failures reported; (iii) classical brackets on what passing implies, and the result that the deployed order has zero guaranteed power against fixed-pole weight redistributions; and (iv) an evaluation of the coverage channel on a device-sampled support against reference distributions, which shows that this channel cannot distinguish a good device from one that merely spreads its samples.

The method must be positioned by mechanism against two neighboring approaches, because all three aim to verify quantum results without a classical reference. Self-verifying variational simulation certifies a prepared *state* through an energy-variance bound measured on the device [41]. Observable-estimation validation certifies error-mitigated *expectation values* by validating the hardware noise model and the mitigation procedure [52]. Our screen certifies neither. It tests a reconstructed spectral *function* against operator moments evaluated independently of it, and it targets weight misassignment in subspace reconstructions. For sample-based spectral functions this failure mode is documented by the companion [20]; for subspace methods in general it is our extrapolation from the limitations documented for their ground-state energies [54–58]. The other two approaches do not address it. A separate, non-physical layer verifies the numerical linear algebra of the classical diagonalizer, cross-checking eigenpairs across independent LAPACK paths and sealing the result with tamper-evident certificates [109]. It catches software and floating-point errors in the solver and, because it acts only on the classical computation, is blind to sampling incompleteness and weight misassignment. The physical approaches are complementary: a hardware spectrum could be produced under observable-estimation validation [52], prepared with self-verification of its input state [41], and then screened for weight misassignment by our moments. We claim only the third role.

One may ask why a finished reconstruction should be screened at all, rather than building the moment information *into* the reconstruction so that no inconsistent spectrum is produced. This is the strategy of a mature family of analytic-continuation methods. Nevanlinna continuation imposes the interpolation and Pick/positivity structure of a causal response function, returning spectra that are consistent by construction [110]; maximum-entropy and stochastic analytic continuation regularize the ill-posed inversion with moment and sum-rule constraints supplied as prior information [63, 64]; and Prony,

minimal-pole and stochastic-pole-expansion methods fit the data to a compact pole model whose low-order moments can be pinned to their operator values [111, 112]. Where such a moment-constrained reconstruction is available it is preferable, and our screen does not compete with it. A moment-constrained reconstruction is, however, consistent only with the moments *it* was given, only within the method that enforced them, and only for the imaginary-axis or time-domain data it was built from. A downstream consumer of a quantum-generated spectrum typically receives a finished $A(\omega)$ from a pipeline it does not control and cannot re-run. Our screen is built for that setting: a reconstruction-independent necessary condition attachable to any spectrum, however it was produced, that draws its discriminating power from moments evaluated independently of the reconstruction. Building the moments into the reconstruction and screening the result against independently evaluated moments are two layers of one defense, not alternatives.

A related complementarity has appeared on the device side. Irmejs and Santos propose constructing dynamical correlation functions directly on hardware, without real-time evolution, from a continued-fraction representation whose level- k approximant is a weight- $O(k)$ operator—a spectral moment—measured on a prepared eigenstate; they prove exponential accuracy in a region of the complex-frequency plane bounded away from the real axis and demonstrate the method in classical simulations of small lattices [113]. A related circuit-differentiation framework builds the retarded Green’s function from derivatives of real-time-evolution circuits under realistic noise [114]. These are *constructive* programs: they use the same weight- $O(k)$ operator moments to build a response function with a priori convergence guarantees off the real axis. Our screen is their diagnostic counterpart. It assumes no control over how the spectrum was produced and constructs nothing; it takes a finished, real-axis $A(\omega)$ from a sample-based-diagonalization or Krylov pipeline and asks only whether that function is consistent with independently evaluated moments, on the real axis, where the constructive guarantees are weakest.

The Christoffel bound states what passing implies, and its scope is narrow. At exact moments $m_0, \dots, m_{2n}$ the cumulative weight $F(t)$ of any moment-matching measure is pinned to within the Christoffel function $W_n(t) = 1/\sum_{j=0}^n p_j(t)^2$, which shrinks as more moments are enforced [69, 70]. Its asymptotic contraction follows the $n\lambda_n$ behavior established by Totik for Stahl–Totik regular measures [83], reducing on a single interval to Máté–Nevai–Totik [97]; for a finite atomic spectrum W_n instead saturates at the residues (Sec. VII). Three limits apply. The bounded quantity is cumulative weight, not the pointwise value of $A(\omega)$ or the position of a peak [69]. At the deployed order $n = 1$ the bracket is vacuous at the spectral mean, where $\max_t W_1 = m_0$. And a reconstruction that passes with tolerances $|\Delta_k| \leq \tau_k$, rather than with exact equality, is bracketed only by a tolerance-inflated extremum, 3.2–7.7 times W_1 at $\omega = 14\text{--}20t$ on

the $L = 6$, $U/t = 8$ current measure of Fig. 13. A passing screen therefore says that coarse-grained weight is placed consistently at exact moments; it does not resolve a line shape [69, 74].

The limits of the screen are the following.

Necessary, not sufficient; zero guaranteed power at the deployed order. Infinitely many measures share any finite set of moments, so passing can never certify correctness [65, 66]. The deployed battery makes this concrete. On the five weighted poles of the $L = 6$, $U/t = 4$ current measure, m_0, m_1, m_2 leave a two-dimensional null space of fixed-pole redistributions, so the guaranteed power against such redistributions is zero whatever the pole separation; the cause is the null-space dimension $N - K - 1$, not the conditioning of the moment map. A redistribution in that null space moves 72% of the weight with m_0, m_1, m_2 unchanged; another, which keeps only m_0, m_1 exact and has $|\Delta_2| \leq \tau_2$, moves 86%; both pass (Sec. VIII). In exact arithmetic the blind spot closes at order $N - 1$ (Theorem 2); closing it needs moments beyond the deployed order (up to m_4 here), which this work does not estimate. Adding m_3 at the deployed shot budget would not close it: under the same threshold rule, no redistribution that keeps m_0, m_1, m_2 exact shifts m_3 by more than $0.35 \tau_3$. Truncations also escape in places. In the interval-moment analysis of Appendix B ($L = 6$, $U/t = 4$, `ibm_fez` budget), an exhaustive scan over truncation depths finds three narrow windows, near 46%, 51% and 87% configuration coverage, where the first moment alone has Monte-Carlo power only at the false-positive level, and at a 2% bias the full battery passes 12 of the 185 truncated reconstructions, three of them at 46–51% coverage. Device data show the same limitation: at 50k shots the $L = 8$ reconstruction has a coverage residual of 0.0044 while its $A(\omega)$ has a relative L_1 error of 0.35 (Sec. VI).

Error classes. The moment screen addresses weight and subspace misassignment in a reconstruction; the complementary Gauss-law check addresses gauge-symmetry violation at the state level. Neither is a general noise diagnostic, and device-noise trust remains the domain of observable-estimation validation [52]. Reconstructions that preserve the estimated moments pass. Under-converged Krylov reconstructions are the clearest case (0/71 in the blinded test): with two or more Lanczos nodes they match m_0 – m_2 exactly, as the companion’s moment-exactness theorem predicts [20]; the one-node case, whose m_2 is off by 2.4–4.2%, passes only within the 2% bias budget (largest residual 0.99 of its threshold).

Positivity. The positivity that underwrites the moment bounds is that of a diagonal $T = 0$ Lehmann measure. Off-diagonal and finite-temperature spectral functions are not positive measures in general and need separate treatment [38].

Idealized estimator in the calibration. The blinded test simulated the independent estimate as the exact full-sector moment plus unbiased Gaussian shot noise, with no same-sample covariance and no mitigation bias. Its

0.9% false-positive rate is therefore close to what the construction implies: the frozen threshold budgets a 2% bias that the simulated estimator does not have, and redrawing the estimates for the 112 clean instances under the frozen rule gives a modelled rate of 1.3%, far below the sealed nominal joint level $1 - 0.95^3 = 14.3\%$. With a realized bias of +2% (–2%) on all three moments the modelled rate is 8.0% (11.5%). Robustness to a real mitigation bias is untested. A post hoc rescoring of the 56 sealed truncations with the shared-state estimator of Fig. 8 also rejects all 56. The operator forms of the estimator that coincide on an exact eigenstate differ on a subspace state by enough to change a verdict (Appendix A), so the deployed form has to be fixed in advance.

Same-sample statistics. Strict same-shot estimation of both branches has been realized only in simulation. In the joint-sampling analysis of the $L = 6$, $U/t = 4$ current probe (reproduction materials), one set of shots per replica gives both $\hat{m}_k$ and $\bar{m}_k$; there $\hat{m}_k$ is a local estimator that uses the exact ground-state amplitudes, so it is not device-realizable. The falsifier variance is $\text{Var}(\Delta_k) = \text{Var}(\hat{m}_k) + \text{Var}(\bar{m}_k) - 2 \text{Cov}(\hat{m}_k, \bar{m}_k)$, with a same-sample correlation $\rho \approx 0.43$ at 84% coverage that decays towards zero as coverage approaches 100%. Treating the two estimates as independent inputs, as one would when a moment interval comes from separate data, for example from the finite-statistics certificate of Mortimer *et al.* [82], over-estimates this variance, and the test becomes over-conservative in the undersampled regime (realized false-positive rate 1.5% at a nominal 5%). At matched false-positive rate the correlation-aware test gives no detection-power advantage (power differences between –5.6 and +0.05 percentage points across the pre-registered ladder of corruptions and coverages), so the covariance is a calibration correction, not a power gain. This structure is a quantum instance of Hausman-type specification testing [115]: $\hat{m}_k$ is the consistent, model-free estimator and $\bar{m}_k$ the model-implied one. The independent-input variance $\text{Var}(\hat{m}_k) + \text{Var}(\bar{m}_k)$ is exact only at full coverage, where $\rho \rightarrow 0$; Hausman’s efficiency limit, $\text{Cov}(\hat{m}_k, \bar{m}_k) = \text{Var}(\bar{m}_k)$, would instead reduce the falsifier variance to $\text{Var}(\hat{m}_k) - \text{Var}(\bar{m}_k)$, and our correlation lies between the two limits.

Hardware. The device supplies only the sampled support. In the $L = 8$ `ibm_fez` run the amplitudes and moments entering Δ_0 are exact classical values on that support, so no moment has been estimated from device counts. The coverage residual rewards support spreading rather than fidelity: at 50k shots per circuit the device residual lies below all 2000 noiseless-sampling replicas of the same circuits (at 4k it lies above all of them), and a uniformly random sampler over the 3920-configuration sector, at the device’s retained counts, gives a lower residual at every budget (Sec. VI). The discriminating moments m_1, m_2 remain simulation-only: the depolarizing bias of the bond-basis estimate of m_1 is at least 37% of the signal at the companion circuit depth and a two-qubit

error of 2.5×10^{-3} (an exact local-depolarizing surrogate, which gives a lower bound), and the sealed simulated gate for a diagonal n_k device run failed at the relaxed noise floors of its single permitted amendment ($T_1 = 100 \mu\text{s}$, $T_2 = 70 \mu\text{s}$, two-qubit error parameter 3.5×10^{-3}).

Synthetic errors and the seal. Every error rejected here is synthetic: an injected corruption or a controlled truncation of otherwise exact data. The blinded test withholds the labels from the classifier, but the truth is classically known and the calibrated scope is narrow: the current probe on the doped $L = 6$ ring at $U/t \in \{3, 4, 5, 6\}$; truncations that keep 15.6–74.2% of the support configurations but 75.1–98.7% of the ground-state weight; and spurious point atoms carrying 2–25% of m_0 . The seal is self-attested: there is no third-party timestamp and the blinding is implemented in code, and because the simulation is deterministic given the sealed seed, the seal cannot by itself exclude iteration before sealing. The screen has not yet caught an error in a spectrum whose correctness was unknown in advance; that field test is future work.

A final objection concerns scale rather than method: one-dimensional Hubbard rings at $L \leq 12$ are solved outright by density-matrix renormalization group (DMRG) and exact diagonalization, so no quantum resource is needed for their spectra. This is deliberate. These sizes are used because their exact moments are available, which calibration requires; the screen is characterized against these methods, not in competition with them. Its intended regime is the one that motivates the quantum computation—real-time dynamics of two-dimensional and frustrated lattices, and *ab initio* molecular spectra—where no controlled classical spectral reference exists. The reproduction materials also include a code-scalability check at $L = 24$ (open chain at half filling, $U/t = 8$, $S_z=0$ sector of dimension $\binom{24}{12}^2 = 7.31 \times 10^{12}$). Shots drawn by Born-rule sampling of a DMRG ground state computed with TeNPy [116] define $\mathcal{S}$; both $\bar{m}_k$ and $\hat{m}_k$ are evaluated on the same subspace ground state, $\hat{m}_k$ by applying the full operators. As the budget grows from 500 to 15 000 shots, Δ_0 falls from 44.2 to 24.5 (DMRG $m_0 = 55.0$) and tracks the reconstruction error confirmed by DMRG. Even at 15 000 shots the subspace energy $e_0 = -2.90 t$ is far above the DMRG $E_0 = -7.66 t$, so the firing is expected. The test checks that the code runs at this scale; it is not a measure of the screen’s power and not a beyond-classical claim, since one-dimensional Hubbard remains DMRG-solvable.

Several extensions follow. The interval-moment formulation, which propagates shot noise and an assumed bias into a two-sided threshold, is demonstrated on the exact state in Appendix B; a device-side deployment needs the mitigation bias measured and bounded rather than assumed, for example from error-model violation [117] or by validating the noise model and the mitigation procedure [52]. Device estimation of the discriminating moments awaits lower gate error. A pre-registered n_k device run was not executed: its single permitted amendment would have relaxed the sealed noise floors

to $T_1 = 100 \mu\text{s}$ and $T_2 = 70 \mu\text{s}$, and it failed its sealed simulated gate at the worst-admissible row (two-qubit error parameter 3.5×10^{-3} ; upper confidence bound of the sealed post-mitigation residual check 0.0287 against the sealed threshold 0.0267). A row identical except for a two-qubit error parameter of 2.85×10^{-3} passed, although in that harness this parameter is a compound knob rather than a single physical error rate (Sec. VI). The Gauss-law check, evaluated here exactly rather than from samples, shows that the framework accepts any exactly known operator identity as a test, which invites a catalog of domain-specific conditions—conservation laws, symmetry sectors and sum rules—each screening a different error class [39, 74]. Extending positivity-based bounds to off-diagonal and finite-temperature spectra, where Lehmann positivity fails, is the main theoretical opening [38]. Because the moment machinery is that of the kernel-polynomial, recursion and quadrature methods [69, 74, 75], the screen fits the sample-based diagonalization and Krylov pipelines that generate the spectra it tests [15–17, 20, 118, 119].

A. The verification stack: two necessary layers

When the moments are evaluated as full-operator expectation values on the sampled subspace state, the battery is classical post-processing of the pipeline’s output; a device-native estimate of m_1, m_2 instead needs separate bond-basis measurement settings on a prepared ground state (Appendix B; simulated only). The estimator branch is the reconstruction-free counterpart of predicting observables from few measurements with classical shadows [50, 120], whose spectral specialization extracts gaps without forming $A(\omega)$ [121] and whose fermionic variant *constructs* dynamical correlation functions from at-most-two-copy shadows [122]. Those methods construct; our screen takes a finished $A(\omega)$ it did not produce and tests it.

The moment vector of every positive measure lies in the moment body: its Hankel matrices are positive semidefinite, and so are its localizing matrices on any interval known to contain the support [65, 66]. A violated Hankel condition on the estimated moment vector therefore rejects the estimate—device, mitigation or estimator—without reference to $A(\omega)$ (Fig. 2). It cannot reject a non-negative reconstruction, whose own moments satisfy it by construction; for this reason the Hankel branch, applied to the reconstruction moments in the blinded test, fired on none of the 300 instances.

Wang *et al.* turn the same moment/positivity machinery into certified two-sided bounds on ground-state properties of many-body systems [81], and Mortimer *et al.* extend it to finite measurement statistics [82]. Applied to the ground-state operators O_k , these semidefinite programs (SDPs) would bound the true moments, $m_k \in [\ell_k, u_k]$. This is possible in principle; it is not done in the cited works or here. Membership of the read-off

moments $\bar{m}_k$ in such intervals would be a second necessary condition on the reconstruction, of the same logical type as ours. The combined screen is the conjunction of two necessary conditions: at least as strict as either condition alone, and not a composition with a sufficient condition. No finite set of moment conditions certifies $A(\omega)$.

The classical moment-body conditions, which are not the Wang/Mortimer programs, show what such bounds look like. We evaluate them at exact moments on the $L = 6$, $U/t = 8$ current measure, normalized to $m_0 = 1$ (true $m_2 = 125.4$). Positivity bounds m_2 from below, $m_2 \geq m_1^2/m_0 = 123.3$. The Hausdorff (interval-localizing) condition $\int(\omega - a)(b - \omega) d\mu \geq 0$, for a support contained in $[a, b]$, bounds it from above, $m_2 \leq (a+b)m_1 - abm_0$, and its strength depends on what is known about the support. With the weighted support $[2.49, 13.92]t$ the bound is 147.6, but that interval is oracle knowledge: it requires the exact spectrum. Intervals available a priori give 243.3 (the excitation range $[0, 21.91]t$ of the probe's $N = 4$, $S_z = 0$ sector) or 579.8 (the full Fock range $[0, 52.21]t$; 573.7 with the lower edge at the smallest positive Fock excitation, $0.148t$). An under-dispersed moment vector ($m_2 = 122.1$) violates the positivity bound; an over-dispersed one ($m_2 = 148.8$) violates only the oracle Hausdorff bound and passes both a-priori bounds. Neither condition has been applied to device-estimated moments.

The two layers could meet on integrated window weight. The window mass $\mu([c, d]) = \int_c^d A d\omega$ is a linear functional of the measure, and over all positive measures that share given moments it ranges over an interval. Proposition 1 gives that interval in closed form at exact moments (the Markov–Krein envelope, of width at most $W_n(c) + W_n(d)$); an SDP fed with interval-valued moments, as in Ref. [82], would give a finite-statistics counterpart against which a reconstruction's window mass could be checked. We do not implement this. We also do not attempt frequency-resolved spectral weights, and this is a limitation rather than a stylistic choice: a weight $|\langle n|J|0\rangle|^2$ at a resolved frequency is an excited-state matrix element, not a ground-state expectation, so it falls outside the ground-state moment hierarchy that makes these SDPs tractable and would need a separate positivity structure for each excited sector. Integrated window weight, which both layers share through $\{m_k\}$, is the natural bridge; the frequency-resolved line shape almost certainly remains beyond this machinery.

X. CONCLUSIONS

We have developed and calibrated a reconstruction-independent, necessary-condition screen for sample-based quantum spectral functions. It compares the low-order moments read off a reconstructed $A(\omega)$ with ground-state operator expectations evaluated without reference to $A(\omega)$. A failed comparison rejects the re-

construction; a passed one excludes only the tested failure modes. In a hash-sealed, blinded simulation of 300 instances the pre-registered primary endpoint was a detection rate of 53.7% (101/188; Wilson 95% interval $[46.6, 60.7]\%$) at a false-positive rate of 0.9% (1/112; $[0.2, 4.9]\%$), with an unbiased simulated estimator. The screen caught every determinant truncation in the test set (56/56) and 74% of the spurious features (45/61). It caught no under-converged Krylov reconstruction (0/71), and two sealed secondary predictions failed. At the deployed order m_0 – m_2 its guaranteed power against fixed-pole weight redistributions is zero; in exact arithmetic this blind spot closes only at order $N - 1$ for N poles.

The hardware run shows less. On IBM `ibm_fez` the device supplied only the sampled support of an $L = 8$ electron-addition sector. The coverage residual Δ_0 grows from 0.004 to 0.37 as support coverage falls from 94% to 35%, but it rewards support spreading rather than device fidelity: at 50k shots per circuit the device lies below every noiseless-sampling replica of the same circuits, a uniform in-sector sampler does better at every budget, and the 50k reconstruction has a relative L_1 error of 0.35 while its residual is small. The run shows that the diagonal channel executes on device data. It does not validate the device, the reconstruction or the discriminating moments.

Several problems remain open: estimating m_1 and m_2 on hardware, where the depolarizing bias of the bond-basis estimate of m_1 is at least 37% of the signal at the companion circuit depth and a two-qubit error of 2.5×10^{-3} , and the diagonal n_k alternative did not pass its sealed simulated gate; calibrating the screen with a realistic mitigation bias and a device-realizable same-sample estimator; closing the within-sector blind spot with higher moments at an affordable shot cost; applying the semidefinite moment bounds of Refs. [81, 82] to the same moment sequence; and a field test on a spectrum whose correctness is not known in advance.

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COMPETING INTERESTS

The author is also affiliated with Daita AI; the company had no role in this work and the author declares no competing interests.

DATA AVAILABILITY

The code and data that support the findings of this study are archived on Zenodo (DOI to be added at submission) and openly available in the GitHub repository <https://github.com/nicolasbonilla/reconstruction-independent-moment-tests>. The repository contains the exact sector-Lanczos/Haydock engine; the interval-moment, blinded-test, within-sector and hardware-analysis scripts; the pre-registration with its SHA-256 seal and the sealed blinded records; the per-figure data; and Makefile targets and a notebook that run the generators. The file `docs/REPRODUCE.md` maps each figure and the key quoted numbers to their scripts and data files, and lists the items that do not yet regenerate from source. The raw measurement counts of our four $L = 8$ `ibm.fez` jobs are included (`jobs da983qse74ec73ajfr2g`, `da98k16rbfbs73chckvg`, `da98icsjbipc73fej78g` and `da98cksjbipc73fej1tg` at 50, 30, 16 and 4×10^3 shots per circuit), together with the reference-distribution generator `src/delta0_reference_mc.py` (output `data/2026-09-27_delta0_reference_mc.json`) and the exporter `src/export_device_dat.py` of Fig. 10. The raw counts of the companion's $L = 6$ run (Fig. 11) are not deposited; that run was post-selected in reversed bit order, so all of its retained determinants came from device errors [20]. The sealed blinded files are the canonical record: regenerating the truncation instances from the seed depends on tie-breaking inside degenerate $|\psi_0|^2$ shells, and the tie order that reproduces them is stored with the blinded-test addenda (`data/2026-09-27_blind_addenda.json`).

Appendix A: Derivation of the moment identities

We derive the operator form of the spectral moments used throughout. Each m_k is a ground-state expectation value computable without the dynamics, and we state where the derivation needs an exact eigenstate. Let $H = H^\dagger$ have a non-degenerate ground state $|0\rangle$, $H|0\rangle = E_0|0\rangle$, and a complete orthonormal eigenbasis $\{|n\rangle\}$ with $H|n\rangle = E_n|n\rangle$; let $J = J^\dagger$ be a Hermitian probe (a current or density). The diagonal zero-temperature Lehmann spectral function

$$A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \delta(\omega - (E_n - E_0)) \quad (\text{A1})$$

is a non-negative measure supported on $\omega = E_n - E_0 \geq 0$, and its power moments are $m_k = \int \omega^k A(\omega) d\omega = \sum_n (E_n - E_0)^k |\langle n|J|0\rangle|^2$.

Writing $|\langle n|J|0\rangle|^2 = \langle 0|J|n\rangle\langle n|J|0\rangle$, using $\langle n|(H - E_0)^k = (E_n - E_0)^k \langle n|$, and closing with completeness $\sum_n |n\rangle\langle n| = \mathbb{1}$,

$$\begin{aligned} m_k &= \sum_n \langle 0|J|n\rangle (E_n - E_0)^k \langle n|J|0\rangle \\ &= \langle 0|J(H - E_0)^k J|0\rangle. \end{aligned} \quad (\text{A2})$$

The ground state is annihilated by $H - E_0$, so acting to the right $(H - E_0)J|0\rangle = [H, J]|0\rangle$, and iterating,

$$(H - E_0)^k J|0\rangle = \text{ad}_H^k(J)|0\rangle, \quad \text{ad}_H(J) \equiv [H, J], \quad (\text{A3})$$

where ad_H^k denotes the k -fold nested commutator $[H, [H, \dots [H, J]]]$. Substituting Eq. (A3) into Eq. (A2) gives the closed operator form

$$m_k = \langle 0|J \text{ad}_H^k(J)|0\rangle \equiv \langle O_k \rangle, \quad (\text{A4})$$

with $O_k = J \text{ad}_H^k(J)$ built solely from H and J —no reference to the spectrum or to the reconstructed $A(\omega)$.

The lowest moments specialize immediately. For $k = 0$, Eq. (A4) gives the total weight $m_0 = \langle 0|J^2|0\rangle = \langle J^2 \rangle$. For $k = 1$, $m_1 = \langle 0|J[H, J]|0\rangle$; expanding the symmetric double commutator $[J, [H, J]] = 2JHJ - J^2H - HJ^2$ and using $H|0\rangle = E_0|0\rangle$,

$$\langle 0|[J, [H, J]]|0\rangle = 2\langle 0|J(H - E_0)J|0\rangle = 2m_1, \quad (\text{A5})$$

so that

$$m_1 = \frac{1}{2} \langle 0|[J, [H, J]]|0\rangle. \quad (\text{A6})$$

This is the first-moment (double-commutator) sum rule of A_{JJ} ; for the current operator J it fixes the mean frequency of the spectral weight. It is distinct from the optical-conductivity f -sum rule $\int_0^\infty \sigma_1 d\omega = \frac{\pi}{2} \langle -\hat{T} \rangle$, which is the m_{-1} moment of A_{JJ} (a ground-state kinetic energy) [35–37]. On an exact eigenstate the symmetrized form (A6) is manifestly real and independent of the additive constant E_0 , which the one-sided expression $\langle 0|J[H, J]|0\rangle$ obscures.

Off an eigenstate the operator forms differ. Equations (A2), (A4) and (A6) agree because $|0\rangle$ is an exact eigenstate of H . On any other state $|\psi\rangle$, for example the ground state $|0_S\rangle$ of a truncated subspace Hamiltonian (Appendix B), the form $\langle \psi|J(H - e_0)^k J|\psi\rangle$ with $e_0 = \langle \psi|H|\psi\rangle$ and the commutator form $\langle \psi|J \text{ad}_H^k(J)|\psi\rangle$ are no longer equal for $k \geq 1$ (for $k = 1$ the real part of the latter is $\frac{1}{2} \langle \psi|[J, [H, J]]|\psi\rangle$), and in general neither equals the exact moment m_k . The difference can decide a verdict. Take the current probe of the doped $L = 6$, $U/t = 4$ ring and truncate its sector to the $d = 100$ determinants of largest $|\psi_0|^2$ (44% of the 225-dimensional sector, 54% of the 186-determinant ground-state support). With the determinants ordered deterministically (descending $|\psi_0|^2$, ties broken by sector index, NumPy `lexsort`), the subspace ground state gives $\langle J(H - e_0)J \rangle = 4.558$ but $\frac{1}{2} \langle [J, [H, J]] \rangle = 3.717$, against

the reconstruction moment $\bar{m}_1 = 3.743$ and the exact $m_1 = 4.227$. At the deployed threshold $\tau_1 = 0.229$ (Appendix B) the first form rejects the truncated reconstruction, which is the correct verdict since $\bar{m}_1$ is 11% below the exact value, and the second does not. This cut falls inside a 24-fold degenerate shell of $|\psi_0|^2$, so these values depend on how ties are broken: NumPy `argsort`, which does not fix the order of ties, gave 4.605, 3.707 and 3.810 on our machine, with the same two verdicts (`data/2026-09-27_theory_numerics.json`, script `src/estimator_form_offeigenstate.py`). We therefore fix the estimator as $\hat{m}_k = \langle \psi | J(H - e_0)^k J | \psi \rangle$ with $e_0 = \langle \psi | H | \psi \rangle$ on the prepared state, and use the commutator forms only on exact eigenstates.

Two features of Eq. (A4) make it the operational core of the screen. First, O_k is an explicit operator polynomial in H and J . Expanded in the computational (occupation-number) basis, $\langle O_k \rangle$ is a weighted sum of basis observables, so it can be evaluated on a prepared state without forming $A(\omega)$; this is the reconstruction-independent estimator $\hat{m}_k$ of Appendix B. Apart from the idealized evaluation on the exact ground state (Appendix B), it is evaluated in this work in three ways, none of them on hardware: classically, by applying the full operators to the subspace state that also yields the reconstruction (a deterministic truncation by $|\psi_0|^2$ in Fig. 8, a shot-sampled subspace in the $L = 24$ DMRG test of Sec. IX); as a local estimator computed from the same simulated shots that define the subspace, which needs the exact amplitudes (the joint-sampling study of Sec. IX); and from simulated Pauli-grouped measurements of a separately prepared ground state (end of Appendix B). Second, the derivation used only (i) completeness of the eigenbasis and (ii) that $|0\rangle$ is an exact eigenstate, $(H - E_0)|0\rangle = 0$. It therefore holds for any Hermitian probe and any Hamiltonian, which is the algebraic reason the screen transports across models. Positivity of the measure (A1) is not needed for the identities; it enters only in the positivity conditions and in the bounds built on them.

For a non-Hermitian probe the same steps hold with $J^\dagger$ on the bra side, $m_k = \langle 0 | J^\dagger (H - E_0)^k J | 0 \rangle$, so that $m_0 = \langle J^\dagger J \rangle$. The single-particle spectral function $A_{k\sigma}(\omega)$ of Fig. 3 is the two-sided (photoemission plus inverse-photoemission) spectrum of the fermionic probe $c_{k\sigma}$, and its moments close under *anticommutators* rather than the double commutator:

$$\begin{aligned} \int A_{k\sigma} d\omega &= \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1, \\ \int \omega A_{k\sigma} d\omega &= \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle = \epsilon_k + U \langle n_{-\sigma} \rangle, \end{aligned} \quad (\text{A7})$$

the Hartree-shifted band centroid [62]. The Hermitian double-commutator map above is the special case $J = J^\dagger$; Eq. (A7) is the version used for the single-particle screen of Fig. 3, and it is likewise a ground-state expectation of explicit operators.

Derivation of the anticommutator moment identities. Equation (A7) is not a fresh sum rule but the non-

Hermitian specialization of Eqs. (A2)–(A4), derived here for completeness. Write $d \equiv c_{k\sigma}$, $d^\dagger \equiv c_{k\sigma}^\dagger$, with $\{|n\rangle\}$ the complete eigenbasis of H over all particle-number sectors, $H|n\rangle = E_n|n\rangle$, ground state $|0\rangle$, energy E_0 . The two-sided single-particle spectral function is the sum of an electron-addition ($N+1$) and an electron-removal ($N-1$) branch,

$$\begin{aligned} A_{k\sigma}(\omega) &= \sum_n |\langle n | d^\dagger | 0 \rangle|^2 \delta(\omega - (E_n - E_0)) \\ &\quad + \sum_n |\langle n | d | 0 \rangle|^2 \delta(\omega + (E_n - E_0)), \end{aligned} \quad (\text{A8})$$

a non-negative measure whose support *straddles* $\omega = 0$ (addition poles at $\omega = E_n - E_0 > 0$, removal poles at $\omega = -(E_n - E_0) < 0$). Its power moments split accordingly,

$$\begin{aligned} \mu_p \equiv \int \omega^p A_{k\sigma} d\omega &= \sum_n (E_n - E_0)^p |\langle n | d^\dagger | 0 \rangle|^2 \\ &\quad + (-1)^p \sum_n (E_n - E_0)^p |\langle n | d | 0 \rangle|^2. \end{aligned} \quad (\text{A9})$$

Applying to each branch the steps of Eqs. (A2)–(A4) — $(E_n - E_0)^p \langle n | d^\dagger | 0 \rangle = \langle n | \text{ad}_H^p(d^\dagger) | 0 \rangle$ and completeness — gives the closed operator form

$$\mu_p = \langle 0 | d \text{ad}_H^p(d^\dagger) | 0 \rangle + (-1)^p \langle 0 | d^\dagger \text{ad}_H^p(d) | 0 \rangle, \quad (\text{A10})$$

built solely from H , d , and $d^\dagger$. For $p = 0$ this is the canonical anticommutator $\mu_0 = \langle 0 | dd^\dagger + d^\dagger d | 0 \rangle = \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1$. For $p = 1$, using $H|0\rangle = E_0|0\rangle$ and $\langle 0 | H = E_0 \langle 0 |$ on the outer factors one verifies term by term that

$$\mu_1 = \langle 0 | d[H, d^\dagger] | 0 \rangle - \langle 0 | d^\dagger[H, d] | 0 \rangle = \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle, \quad (\text{A11})$$

the anticommutator first-moment identity, manifestly real and independent of the additive constant E_0 . For the Hubbard Hamiltonian $H = \sum_{k'\sigma'} \epsilon_{k'} c_{k'\sigma'}^\dagger c_{k'\sigma'} + U \sum_i n_{i\uparrow} n_{i\downarrow}$ the commutator evaluates in closed form: the kinetic term gives $[c_{k\sigma}, H_0] = \epsilon_k c_{k\sigma}$, while $[c_{k\sigma}, H_U] = \frac{U}{\sqrt{L}} \sum_j e^{-ikj} c_{j\sigma} n_{j,-\sigma}$ (with $c_{k\sigma} = \frac{1}{\sqrt{L}} \sum_j e^{-ikj} c_{j\sigma}$), and since $\{c_{j\sigma} n_{j,-\sigma}, c_{j'\sigma}^\dagger\} = \delta_{jj'} n_{j,-\sigma}$,

$$\begin{aligned} \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle &= \epsilon_k \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle + \frac{U}{L} \sum_j \langle n_{j,-\sigma} \rangle \\ &= \epsilon_k + U \langle n_{-\sigma} \rangle, \end{aligned} \quad (\text{A12})$$

the Hartree-shifted band centroid quoted in Eq. (A7) and evaluated per momentum in Fig. 3 [62].

Scope: which checks transport to the single-particle channel. The zeroth- and first-moment checks carry over intact: μ_0 and μ_1 are ground-state expectation values of explicit operators [Eqs. (A10)–(A11)], so they give reconstruction-independent residuals Δ_0, Δ_1 at each

k . So does the two-sided Hamburger–Hankel positivity $H_n = [\mu_{i+j}] \succeq 0$, since a positive measure on $\mathbb{R}$ has a positive semidefinite Hankel matrix whatever the sign of its support; like every positivity condition, it tests the estimated moments, not a non-negative reconstruction (Appendix B). The shifted-Stieltjes condition $H_n^{(1)} = [\mu_{i+j+1}] \succeq 0$ does *not* transport: it presupposes support on $[0, \infty)$, and $A_{k\sigma}$ is supported on both signs of ω (addition plus removal). The Christoffel bracket of Sec. VII applies to any positive measure on $\mathbb{R}$, but we have evaluated it only for the one-sided current measure and do not deploy it here. In the single-particle channel the screen therefore consists of the residuals Δ_0, Δ_1 and Hamburger–Hankel positivity; the shifted-Stieltjes condition belongs to the one-sided current and density responses.

Appendix B: The reconstruction-independent moment estimator

We give the explicit estimator $\hat{m}_k$ and show why it is independent of the reconstruction and why it, unlike the spectral moment $\bar{m}_k$, detects subspace truncation. Sample-based diagonalization returns a set $\mathcal{S}$ of computational-basis determinants $\{|x\rangle\}$ and diagonalizes the projected Hamiltonian $H_{\mathcal{S}} = \Pi_{\mathcal{S}} H \Pi_{\mathcal{S}}$, with $\Pi_{\mathcal{S}} = \sum_{x \in \mathcal{S}} |x\rangle\langle x|$, on a classical computer [15, 16]. This gives eigenpairs $\{E_n^{\mathcal{S}}, |n_{\mathcal{S}}\rangle\}$, among them the subspace ground state $|0_{\mathcal{S}}\rangle = \sum_{x \in \mathcal{S}} c_x |x\rangle$ with energy $e_0 = \langle 0_{\mathcal{S}} | H | 0_{\mathcal{S}} \rangle$. Unless $\mathcal{S}$ contains the support of the exact ground state, $|0_{\mathcal{S}}\rangle$ is an eigenstate of $H_{\mathcal{S}}$ but in general not of H , so the operator forms of Appendix A differ on it, and we use the one fixed there. The probe J and the Hamiltonian H are sparse in this basis, with known matrix elements $J_{yx} = \langle y | J | x \rangle$ and $H_{yx} = \langle y | H | x \rangle$.

The moment $\langle 0_{\mathcal{S}} | J (H - e_0)^k J | 0_{\mathcal{S}} \rangle$ admits two evaluations that coincide only when $\mathcal{S}$ is complete for the operators involved. Inserting the *subspace* resolution $\Pi_{\mathcal{S}}$ in place of the identity yields the reconstruction (spectral) moment,

$$\begin{aligned} \bar{m}_k &= \sum_{n \in \mathcal{S}} (E_n^{\mathcal{S}} - e_0)^k |\langle n_{\mathcal{S}} | J | 0_{\mathcal{S}} \rangle|^2 \\ &= \langle 0_{\mathcal{S}} | J [\Pi_{\mathcal{S}} (H - e_0) \Pi_{\mathcal{S}}]^k J | 0_{\mathcal{S}} \rangle \\ &= \langle 0_{\mathcal{S}} | J (H_{\mathcal{S}} - e_0)^k J | 0_{\mathcal{S}} \rangle, \end{aligned} \quad (\text{B1})$$

which is exactly $\int \omega^k A(\omega) d\omega$ and lives entirely inside $\mathcal{S}$. Here $H_{\mathcal{S}} - e_0$ acts on $\text{span } \mathcal{S}$, so the $k = 0$ term carries the convention $[\Pi_{\mathcal{S}} (H - e_0) \Pi_{\mathcal{S}}]^0 \equiv (H_{\mathcal{S}} - e_0)^0 \equiv \Pi_{\mathcal{S}}$, and $\bar{m}_0 = \langle 0_{\mathcal{S}} | J \Pi_{\mathcal{S}} J | 0_{\mathcal{S}} \rangle$ is the weight captured by $\mathcal{S}$ rather than the full $\langle J^2 \rangle$. The reconstruction-independent estimator instead applies the sparse operator $O_k = J(H - e_0)^k J$ directly,

$$\begin{aligned} \hat{m}_k &= \langle 0_{\mathcal{S}} | J (H - e_0)^k J | 0_{\mathcal{S}} \rangle \\ &= \sum_{x, x' \in \mathcal{S}} c_x^* c_{x'} \langle x | J (H - e_0)^k J | x' \rangle, \end{aligned} \quad (\text{B2})$$

where each matrix element $\langle x | O_k | x' \rangle$ is a finite product of sparse operators summing over intermediate determinants that need *not* lie in $\mathcal{S}$. This is the operational meaning of independence: $\hat{m}_k$ probes the action of J and H on $|0_{\mathcal{S}}\rangle$ beyond the sampled subspace, whereas $\bar{m}_k$ is confined to it. Evaluated on $|0_{\mathcal{S}}\rangle$, Eq. (B2) is the shared-state estimator of Fig. 8. Evaluated instead on the exact ground state $|0\rangle$ (with $e_0 \rightarrow E_0$), it gives the exact moment m_k ; this idealized version, more independent than a device allows, is used in Fig. 7, in the sealed blinded test (Sec. V, with Gaussian shot noise added), and in the Monte-Carlo and interval-battery demonstrations below.

The two evaluations agree when the states that H generates from $\varphi \equiv J|0_{\mathcal{S}}\rangle$ already lie in $\text{span } \mathcal{S}$. More precisely, if $\text{span } \mathcal{S}$ contains the Krylov space $\text{span}\{\varphi, H\varphi, \dots, H^K \varphi\}$, then $[\Pi_{\mathcal{S}} (H - e_0) \Pi_{\mathcal{S}}]^j \varphi = (H - e_0)^j \varphi$ for $j \leq K$, and $\hat{m}_k = \bar{m}_k$ for every $k \leq 2K + 1$. This moment-exactness statement, and its failure at order $2K + 2$, are proved and measured in the companion study [20]. The same fact explains why the under-converged Krylov reconstructions of Sec. V pass the low-order battery: an n_l -step Lanczos reconstruction started from the probe state projects onto exactly its Krylov space with $K = n_l - 1$ (the argument holds for any orthogonal projector, not only a determinant subspace) and so reproduces $m_0, \dots, m_{2n_l-1}$. The one-step case reproduces only m_0 and m_1 ; its m_2 is 2–4% low, and in the sealed test its residual stayed below τ_2 (at most $0.99 \tau_2$) only because τ_2 includes the 2% bias budget: at the shot-only threshold 6 of these 15 instances would have been rejected. Under determinant-subspace truncation the Krylov space is not contained, and

$$\hat{m}_k - \bar{m}_k = \langle \varphi | (H - e_0)^k | \varphi \rangle - \langle \varphi | [\Pi_{\mathcal{S}} (H - e_0) \Pi_{\mathcal{S}}]^k | \varphi \rangle \quad (\text{B3})$$

measures the spectral weight the truncation leaks outside $\mathcal{S}$.

At $k = 0$, with the convention above, Eq. (B3) reduces to $\Delta_0 = \langle \varphi | (\mathbb{1} - \Pi_{\mathcal{S}}) | \varphi \rangle = \hat{m}_0(1 - w) \geq 0$, where $w = \|\Pi_{\mathcal{S}} \varphi\|^2 / \|\varphi\|^2$ is the captured weight of the companion study ($\Pi_{\mathcal{S}}$ is its $P_{\mathcal{S}}$) [20]. The same expression, with $\varphi = c_p^\dagger |0\rangle$ built from the exact ground state and only $\mathcal{S}$ taken from the device, is the coverage residual of Sec. VI. When $w = 1$ the $k = 1$ term vanishes as well, and the first term that can be nonzero is $\Delta_2 = \|(\mathbb{1} - \Pi_{\mathcal{S}}) H \varphi\|^2$, the boundary-leakage identity of the companion [20]. The reconstruction on the full ground-state support in Fig. 14 is an instance: with $\mathcal{S}$ the whole 186-determinant support of the $L = 6$, $U/t = 4$ ground state, $w = 1$, Δ_0 and Δ_1 vanish to machine precision, and $\Delta_2 = 0.62$ (1.9% of $m_2 = 33.39$). This reconstruction therefore captures all of φ and is still not exact. These identities are the companion's. The companion bounds the error of the reconstructed spectral function from below by $1 - w$ and from above with the boundary coupling, and shows that the captured weight alone cannot certify a reconstruction [20]. What this paper adds is their use as a test: the residuals are formed from the operator algebra on the prepared state, compared with thresholds set in

advance, and used to reject a reconstruction. For the shared-state estimator, $\Delta_k = |\hat{m}_k - \bar{m}_k|$ is nonzero only if $\mathcal{S}$ misses part of φ or of the states $H^j\varphi$ that order k probes. The independent estimator can then fire, while the circular control compares $\bar{m}_k$ to itself and returns $\Delta_k \equiv 0$ (Sec. IV). Weight moved *within* $\mathcal{S}$ without leaking out of it is a separate blind spot (Appendix E).

Equation (B2) is equivalently an importance-weighted average over $\mathcal{S}$,

$$\hat{m}_k = \sum_{x \in \mathcal{S}} |c_x|^2 O_k^{\text{loc}}(x), \quad O_k^{\text{loc}}(x) = \frac{\langle x | O_k | 0_{\mathcal{S}} \rangle}{\langle x | 0_{\mathcal{S}} \rangle}, \quad (\text{B4})$$

the local-operator estimator familiar from variational Monte Carlo [124]; the local moment $O_k^{\text{loc}}(x)$ again sums over determinants connected to x by O_k and so reaches outside $\mathcal{S}$. Drawing N_s configurations from $|c_x|^2$ and averaging O_k^{loc} gives an unbiased Monte-Carlo estimate with variance $\text{Var}(O_k)/N_s$, where $\text{Var}(O_k) = \sum_x |c_x|^2 |O_k^{\text{loc}}(x)|^2 - \hat{m}_k^2$ is the local-estimator variance. This form needs the amplitudes c_x , which a device does not return, so we use it only in simulation (Fig. 14 and the joint-sampling study of Sec. IX). On a device the same expectation would be estimated from Pauli-grouped measurements, described at the end of this appendix, with a variance of its own.

The exact-moment test takes its finite-shot form here. We model the estimator uncertainty as $\delta_k^2 = \text{Var}(O_k)/N_s + b_k^2$, a shot term over N_s samples plus a bias budget b_k for mitigation and device error, and reject the reconstruction at level α when

$$\Delta_k = |\hat{m}_k - \bar{m}_k| > \tau_k, \quad \tau_k \equiv z_{1-\alpha/2} \delta_k.$$

The deployed rule uses $\alpha = 0.05$ per moment ($z_{0.975} = 1.96$), $N_s = 5 \times 10^4$ and $b_k = 0.02 |\hat{m}_k|$. For the current probe of the doped $L = 6$, $U/t = 4$ ring it gives $\tau_0 = 0.029$, $\tau_1 = 0.229$ and $\tau_2 = 1.848$. A second, separate check concerns the estimate alone: whether some moment vector in the box $[\hat{m}_\ell - \tau_\ell, \hat{m}_\ell + \tau_\ell]$ satisfies the positivity conditions that the moments of every positive measure obey, $H_n \succeq 0$ and, for one-sided $T = 0$ support, $H_n^{(1)} \succeq 0$. Infeasibility would show that the estimated moments belong to no spectrum, which points to the device, the mitigation or the estimator. It cannot refute a non-negative reconstruction, whose own moments $\bar{m}_k$ satisfy these conditions by construction. In the implementations reported here (the sealed blinded test and the interval battery below) the positivity condition is evaluated at $\bar{m}_k$ and therefore cannot fire; every rejection comes from a residual Δ_k . The operators, targets and intervals of each demonstration are in the reproduction materials.

The estimator also admits a device-native evaluation. We verify it in simulation because it settles how the off-diagonal (hopping) content of $\hat{m}_1, \hat{m}_2$ would be obtained on hardware. For a probe diagonal in the computational basis, the density probe $\rho_q = \sum_j \cos(qj) n_j$ at $q = \pi$, the moment operator $O_k = \delta \rho_q (H - E_0)^k \delta \rho_q$ is a sum of

Pauli strings, and $\hat{m}_k = \langle 0 | O_k | 0 \rangle = \sum_P c_P \langle P \rangle$ is a linear combination of Pauli expectations estimated from counts. The m_0 term is a pure computational-basis estimator. The hopping terms in m_1, m_2 need rotated-basis measurements, one circuit per qubit-wise-commuting group of Pauli strings, the standard grouping for Pauli observables; classical shadows are a randomized alternative [50]. This path measures a prepared ground state in rotated bases. It is a separate set of circuits from those whose samples define $\mathcal{S}$, so it shares the device with the reconstruction but not the samples.

We verify it by finite-shot sampling of the exact ground state of an open $L = 6$ Hubbard chain at $U/t = 4$ (12 qubits), with 5×10^4 shots in each of the 357 qubit-wise-commuting bases that (m_0, m_1, m_2) require (1.8×10^7 shots per repetition). Over 8 repetitions the mean estimates $(\hat{m}_0, \hat{m}_1, \hat{m}_2) = (2.730, 10.55, 47.01)$ lie 0.87, 0.72 and 0.66 standard errors from the exact $(2.737, 10.61, 47.64)$, with the standard error taken as the spread over repetitions divided by $\sqrt{8}$. For the $L = 4$ chain (8 qubits, 126 bases) every moment is recovered within 0.06 standard errors. The number of bases is set by the Pauli support of O_k , not by the numerical moment values, so it can be counted without the statevector once the probe shift $\langle \rho_\pi \rangle$ is known. That shift vanishes by the reflection symmetry of the open chain: for even L , reflection maps ρ_π to $-\rho_\pi$, so $\langle \rho_\pi \rangle = 0$ in a reflection-symmetric ground state. The ground states are not half filled ($N = 2, 4$ and 5 electrons at $L = 4, 6$ and 8, where $\langle \rho_\pi \rangle = 0$ was checked numerically; it is assumed at $L = 10$ and 12). The committed script `estimator_scaling.py` counts the bases this way and reproduces the count from the full eigen-decomposition where that is affordable ($L \leq 8$). The full (m_0, m_1, m_2) battery needs 126, 357, 721, 1191, 1837 bases at $L = 4, 6, 8, 10, 12$ (8 to 24 qubits); the (m_0, m_1) pair alone needs 19, 35, 51, 67, 83, growing roughly linearly. The number of settings thus grows polynomially in this range, not exponentially. We do not turn this into a deployability claim: the counts-based recovery was validated only up to 12 qubits and the basis count against full diagonalization only up to $L = 8$; the shot budget and accuracy at 24 qubits remain to be characterized; and a device would have to prepare the ground state that the classical check takes as given.

The estimator, including its off-diagonal content, is therefore measurable from a fixed shot budget in principle, but not on present hardware. Under depolarizing noise the estimate of m_1 is pulled upward, toward $\text{Tr}(O_1)/\dim \mathcal{H} = 34.3$ against $m_1 = 10.6$ for this probe. An exact density-matrix simulation with local two-qubit depolarizing noise at $\varepsilon = 2.5 \times 10^{-3}$ on 292 CZ gates puts the bias at 37% of m_1 ; this is exact for that surrogate and a lower bound for a real circuit. A coherent-ZZ error model gives 47% and global depolarizing 116% (Sec. VI). No off-diagonal moment has been measured on a device in this work. A job script that builds the rotated-basis circuits and applies this estima-

tor (`hardware_blind_job.py`) is in the repository; its state preparation is a placeholder, it has not been run on a quantum processor, and an off-diagonal test on hardware awaits lower gate error.

We close with a Monte-Carlo calibration of this test (Fig. 14) at $N_s = 5 \times 10^4$, the largest per-circuit shot budget of the `ibm_fez` jobs. It is a simulation: $M = 4 \times 10^3$ replicas of N_s shots are drawn from the exact ground-state distribution $|\langle x|0\rangle|^2$ of the doped $L = 6$, $U/t = 4$ ring, a system not run on hardware, and $\hat{m}_1$ is the local estimator above for its current probe. Each truncated reconstruction is built from the ground state of the truncated Hamiltonian. The Monte Carlo checks the analytic interval rather than replacing it. The realized estimator is close to Gaussian (skewness -0.04 , excess kurtosis -0.05), its spread matches the analytic shot term to 1% (0.0799 against 0.0806), and its false-positive rate, 4.5% (Wilson 95% interval $[3.9, 5.2]\%$), is consistent with the nominal 5% (Fig. 14a). Two inputs are modeled. The shot term uses the exact-state local-estimator variance, which assumes known amplitudes and omits ratio-estimator and Pauli-grouping overhead, so it likely understates the per-shot uncertainty of a device estimator. The bias term b_k is assumed and swept, not measured. For this probe $\text{Var}(O_1) \approx 3.2 \times 10^2$ gives a shot standard error $\sqrt{\text{Var}(O_1)/N_s} \approx 0.081$ (1.9% of m_1); with the 2% bias budget the threshold is $\tau_1 = 0.229$ (5.4% of m_1), close to the 5% relative line of Figs. 7 and 8.

Power is high but not uniform in the truncation depth. At the shot-only threshold, m_1 alone reaches power ≥ 0.95 at 161 of the 185 truncations, already at $d = 181$ (97% coverage, power 0.98). The exceptions are the four mildest truncations ($d = 182$ –185, one to four determinants dropped) and three narrow windows, $d = 84$ –88, 93–98 and 158–166 (coverage 45–47%, 50–53% and 85–89%). At the window centres power falls to the false-positive level: 5.5%, 6.1% and 5.0% at $d = 86$, 95 and 162. The 22-point grid of Fig. 14b touches the first two windows only at their edges ($d = 88$ and 98, power 0.93 and 0.94) and misses the third. An earlier grid-only version of this analysis, with ties broken by floating-point noise, showed a single low-power point near 53% coverage; the scan at every d supersedes it. The noise-free residuals tell the same story. At the deployed 2% bias budget, $|\hat{m}_1 - \bar{m}_1|$ stays below τ_1 for 17 of the 185 truncations ($d = 183$ –185, 159–165, 94–97 and 85–87), and at a 4% budget for 27.

Where the cut falls matters. The ground-state probabilities $|\psi_0|^2$ of this sector come in exactly degenerate shells of 6, 12 or 24 determinants, and which members of a shell a truncation keeps is a convention. All the window cuts, and 19 of the 21 truncated grid points, fall inside a shell. We use a deterministic convention (descending $|\psi_0|^2$, ties broken by sector index), but verdicts near the windows depend on it. At $d = 98$, for example, the deterministic order gives $|\hat{m}_1 - \bar{m}_1| = 0.284 > \tau_1$, while 300 random choices of the 8 kept members of the 24-fold shell give 0.172–0.295 and rejection by m_1

alone in 57% of cases; an earlier run that broke ties by floating-point noise found 0.225 there, a miss. At the ten truncations whose cut falls on a shell boundary ($d = 6, 18, 30, 42, 66, 90, 114, 138, 150, 174$), and which therefore do not depend on the convention, the noise-free residual of m_1 alone exceeds τ_1 at the 2% budget.

The onset of detection here lies at much higher coverage than in the independence figures, where the idealized full-state curve crosses its 5% line near 7% retained determinants and the shared-state curve near 18% ($d \approx 1.6$ – 4.4×10^4 of 2.4×10^5 ; Figs. 7 and 8). This reflects a different test, not a sharper one. Those curves score the large $L = 12$ sector by a *relative* discrepancy $|\hat{m}_1 - \bar{m}_1|/\hat{m}_1$ against a fixed fractional line, whereas this Monte Carlo scores the small $L = 6$ sector by an *absolute* threshold set by the shot budget. The coverage percentages are therefore set against different rulers on different-sized sectors (and, at $L = 12$, against two reference states); they are not one test giving several answers.

The direction of the residual uncertainty matters for a test that can only reject. Underestimating δ_k shrinks τ_k . That is safe against missed errors (false negatives, the acceptable direction for a necessary condition) but anti-conservative against false rejections (false positives, the costly error for such a test). The false-positive rate above relies on $\bar{m}_1$ being exact at full support ($|\hat{m}_1 - \bar{m}_1| \sim 10^{-15}$), so that the null hypothesis holds exactly there. For m_2 it does not. The 186-determinant support contains all of φ but not $H\varphi$, so the reconstruction on it is wrong at second order: $\Delta_2 = 0.62$ ($0.48\tau_2$ at the shot-only threshold, $0.34\tau_2$ at the deployed one), the boundary-leakage term discussed above. An m_2 rejection of this reconstruction would be a correct detection, not a false positive. The Monte Carlo tests m_1 only. Away from full coverage the threshold should be set on $\text{Var}(\hat{m}_k - \bar{m}_k)$, for which $\text{Var}(O_k)$ is a proxy valid only when the reference is trusted (Sec. IX).

The second moment catches some lone- m_1 misses, not all. In the noise-free scan at the 2% budget, m_2 rejects 3 of the 17 truncations that m_1 alone passes ($d = 97, 96$ and 85). The clearest is $d = 97$ (52% coverage): in the deterministic order $|\hat{m}_1 - \bar{m}_1| = 0.186 < \tau_1 = 0.229$ while $|\hat{m}_2 - \bar{m}_2| = 2.73 > \tau_2 = 1.85$, and this verdict does not depend on the tie convention (m_1 alone rejects in none and the battery in all of 300 random shell choices). The full (m_0, m_1, m_2) battery nonetheless passes 12 truncated reconstructions at the 2% budget: $d = 183$ –185 (98–99.5% coverage), $d = 160$ –165 (86–89%), and $d = 86, 87$ and 95 (46–51%). Some of these passes are tie-convention effects (the battery rejects in 44% of random shell choices at $d = 87$, 40% at $d = 160$ and 11% at $d = 95$), but not those at $d = 86$ (1.3%), $d = 161$ –165 (at most 0.3%) or $d = 183$ –185 (none). Without a bias budget the battery catches the mid-coverage windows but still passes $d = 161$ –164, 184 and 185; with a 4% budget it passes 22 truncations. The joint battery is thus a partial remedy for the single-moment windows. With finite thresholds no battery of necessary conditions is free of blind spots,

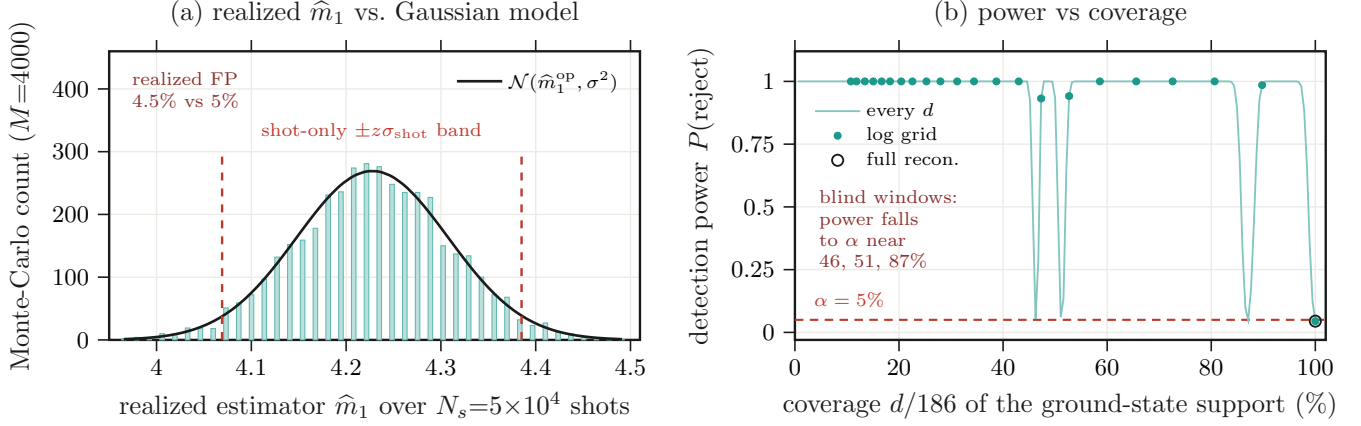


FIG. 14. **Monte-Carlo calibration of the interval-moment test** (simulation only: current probe of the doped $L = 6$ ring, $U/t = 4$; $N_s = 5 \times 10^4$ shots per replica drawn from $|\langle x|0 \rangle|^2$, $M = 4 \times 10^3$ replicas; no device data enter). **(a)** The realized estimator $\hat{m}_1$ is close to Gaussian, and its false-positive rate on the full reconstruction is 4.5% (Wilson 95% interval $[3.9, 5.2]\%$) against the nominal 5%. Dashed: the shot-only band $\pm z\sigma_{\text{shot}} = \pm 0.158$; the deployed threshold with the 2% bias budget, $\tau_1 = 0.229$, is wider. **(b)** Power of m_1 alone against truncated reconstructions at the shot-only threshold, for every retained count d (line) and on a 22-point log grid (filled markers); open marker: the reconstruction on the full support ($d = 186$), whose $\bar{m}_1$ is exact, so its rejection rate is the false-positive rate. Coverage $d/186$ is the retained fraction of the ground state's 186-determinant support. Power is at least 0.95 at 161 of the 185 truncations. It falls to the false-positive level in three narrow windows near 46%, 51% and 87% coverage; the grid touches the first two only at their edges and misses the third. The window cuts lie inside degenerate $|\psi_0|^2$ shells, so their exact location depends on how ties are broken (text). The shot term is the exact-state local-estimator variance, which likely understates the per-shot uncertainty of a device estimator.

and at the deployed order this one has blind spots that we can list.

Two further boundaries frame the demonstration. First, the discrepancy is measured against the exact moments, which serve as an oracle. The demonstration calibrates the interval where classical ground truth exists; a device-side version that propagates measured rather than assumed bias remains to be done. Second, moment-based certification uses the moments differently. Wang *et al.* bound ground-state properties through moment-matrix feasibility, and Mortimer *et al.* carry this to finite measurement statistics [81, 82]. Here two estimates of the same moments are compared, and a discrepancy beyond threshold rejects a reconstruction; the screen never certifies. The spectral-moment sum rules themselves are textbook [35, 74], and sum-rule residuals already serve as validation metrics for quantum-simulated spectra against experimental and classical anchors [40]. What this paper adds is their use as reconstruction-independent, interval-valued rejection criteria for sample-based spectral functions, with thresholds fixed in advance from a shot budget.

Appendix C: The Christoffel miss-distance: proof and evaluation

We re-derive the equality of the extremal cumulative-weight spread with the Christoffel function from three classical lemmas [66, 67, 69]; nothing in this derivation is new. Write $K_n(x, y) = \sum_{j=0}^n p_j(x)p_j(y)$ for the

Christoffel–Darboux kernel and $\lambda_n(t) = 1/K_n(t, t)$ for the Christoffel function, and assume the moment vector is interior to the moment cone, $H_n = [m_{i+j}]_{i,j=0}^n \succ 0$ (for a finite N -atom Lehmann spectrum this holds for $n \leq N - 1$). The measures ν compared below are positive measures on $\mathbb{R}$. Restricting them to the Lehmann half-line $\omega \geq 0$ can only narrow the bracket, so W_n remains an upper bound there.

Lemma 1 (variational identity). $\lambda_n(t) = \min\{\int |P|^2 d\mu : P \in \mathcal{P}_n, P(t) = 1\}$, with minimizer $P^* = K_n(\cdot, t)/K_n(t, t)$. Expanding $P = \sum_{j=0}^n c_j p_j$ gives $\int |P|^2 d\mu = \sum_j c_j^2$ while $1 = P(t)^2 = (\sum_j c_j p_j(t))^2 \leq (\sum_j c_j^2) K_n(t, t)$ by Cauchy–Schwarz, with equality at $c_j = p_j(t)/K_n(t, t)$; the minimum is $1/K_n(t, t) = \lambda_n(t)$.

Lemma 2 (maximal atom). Among positive measures ν with $\int \omega^k d\nu = m_k$ for $k \leq 2n$, the supremum of the mass at a point t is $\sup_\nu \nu(\{t\}) = \lambda_n(t)$. For the extremal P^* of Lemma 1, $\nu(\{t\}) = P^*(t)^2 \nu(\{t\}) \leq \int (P^*)^2 d\nu = \int (P^*)^2 d\mu = \lambda_n(t)$, the middle equality using $\deg(P^*)^2 = 2n$ and that ν shares those moments. If $p_n(t) \neq 0$ the bound is attained by the $(n+1)$ -point Gauss–Radau rule with a node prescribed at t : a positive measure matching $m_0, \dots, m_{2n}$ whose weight at t is $\lambda_n(t)$. At the n zeros of p_n this rule does not exist (one node escapes to infinity), and the supremum is approached but not attained.

Lemma 3 (ordinate gap). Let $p_n(t) \neq 0$, and let $x_0 < \dots < x_n$ be the nodes of that Gauss–Radau rule, one of them equal to t , with weights $\lambda_n(x_i) > 0$. For every positive ν sharing $m_0, \dots, m_{2n}$, with $F_\nu(t) = \nu((-\infty, t])$,

the Chebyshev–Markov–Stieltjes inequalities give

$$\sum_{x_i < t} \lambda_n(x_i) \leq F_\nu(t^-) \leq F_\nu(t) \leq \sum_{x_i \leq t} \lambda_n(x_i)$$

[66, 67, 100, 101]. They follow from polynomials $Q_\pm \in \mathcal{P}_{2n}$ with $Q_- \leq \mathbf{1}_{(-\infty, t)}$ and $\mathbf{1}_{(-\infty, t]} \leq Q_+$ that coincide with these indicators at the nodes, so that $\int Q_\pm d\nu = \int Q_\pm d\mu$ equal the two sums. The bracket has width $\lambda_n(t)$. Its upper end is attained by the Radau rule itself; its lower end is approached by Radau rules whose prescribed node t' decreases to t . Hence $\sup_{\nu_1, \nu_2} |F_{\nu_1}(t) - F_{\nu_2}(t)| = \lambda_n(t)$. At the zeros of p_n the same supremum follows from the continuity of λ_n and the right-continuity of F .

Combining the three, the extremal spread equals $\lambda_n(t) = W_n(t)$ for every real t , and any single reconstruction consistent with the moments obeys $|F_{\text{rec}}(t) - F_{\text{true}}(t)| \leq W_n(t)$. No convex-hull hypothesis is needed: the Radau rule has positive weights for every t with $p_n(t) \neq 0$, inside or outside the convex hull of the support. We checked this on the $U/t = 8$ current measure of Fig. 13: 24 Radau rules ($n = 1, \dots, 4$ at six values of t , 20 of them outside the convex hull of the weighted support) all have positive weights, place mass $W_n(t)$ at t to a relative error of 2×10^{-16} , and match $m_0, \dots, m_{2n}$ to 1.4×10^{-17} . Monotonicity $W_{n+1} \leq W_n$ follows from $K_{n+1} = K_n + p_{n+1}^2 \geq K_n$.

Evaluation of the bound from the moments. The Christoffel function is computed from the moments $\{m_k\}_{k=0}^{2n}$ by a short pipeline [69, 71]. With the inner product $\langle f, g \rangle_\mu = \int f(\omega)g(\omega) d\mu(\omega)$, so that $\langle \omega^a, \omega^b \rangle_\mu = m_{a+b}$, the orthonormal polynomials follow from Gram–Schmidt on $\{1, \omega, \dots, \omega^n\}$ with $p_0 = 1/\sqrt{m_0}$ and obey the three-term recurrence

$$\beta_{j+1} p_{j+1}(\omega) = (\omega - \alpha_j) p_j(\omega) - \beta_j p_{j-1}(\omega), \quad (\text{C1})$$

with $\alpha_j = \langle \omega p_j, p_j \rangle_\mu$ and $\beta_j = \langle \omega p_j, p_{j-1} \rangle_\mu$, all fixed by $\{m_k\}$. The map from power moments to (α_j, β_j) becomes ill-conditioned as n grows, so high orders need high-precision arithmetic. The nodes and weights of the $(n+1)$ -point Gauss rule are the eigenpairs of the Jacobi matrix $\mathbf{J}_{n+1} = \text{tridiag}(\beta_j, \alpha_j, \beta_j)$ (Golub–Welsch), and the kernel on the diagonal, $K_n(t, t) = \sum_{j=0}^n p_j(t)^2$ evaluated by the same recurrence, gives $W_n(t) = 1/K_n(t, t)$. Every quantity in the bound is thus an explicit function of the moments, and the bound can be read at any frequency.

We evaluate it on the $L = 6$, $U/t = 8$ current measure of Fig. 13, normalized to $m_0 = 1$; five of its atoms carry weight. (Its ground level is a spin triplet; the current measure is the same for all three states, because J is a spin scalar.) The numbers below come from `src/export_christoffel.dat.py` and `src/christoffel_tolerance_lp.py` (data/2026-09-27.theory_numerics.json). We run the recurrence in 60-digit arithmetic on the atoms and weights of the measure, which fix the same coefficients as

its moments, and locate the maxima by root-finding: $\max_t W_n = 1, 0.969, 0.886, 0.882$ for $n = 1, \dots, 4$. The $n = 1$ maximum equals m_0 and lies at the mean frequency $\bar{\omega} = 11.10t$, the zero of p_1 . At the deployed order (m_0, m_1, m_2) the bracket is therefore vacuous at the spectral peak. At the dominant atom ($\omega = 11.09t$, residue 0.778), W_n reaches the residue at $n = 4 = N - 1$. It cannot go lower at any order: Lemma 2 with $\nu = \mu$ gives $W_n(t) \geq \mu(\{t\})$. No number of moments therefore narrows the bracket at the peak below 0.778.

The figure’s abscissa $t \in [-1, 1]$ is the affine image of the frequency range $\omega \in [0.148, 52.21]t$, the range of positive excitation energies in the full Fock space, zero-weight states included. The weighted support $[2.49, 13.92]t$ maps to $[-0.910, -0.471]$, and 73.5% of $[-1, 1]$ lies above the last weighted pole. The affine map moves points along the axis but does not change W_n , which is a mass. The rescaled points 0.3, 0.6, 0.9 correspond to $\omega = 34.0, 41.8, 49.6t$, far above the band. There $W_1 = 4.0 \times 10^{-3}, 2.2 \times 10^{-3}, 1.4 \times 10^{-3}$, and at 0.6 the higher orders give $W_2 = 1.4 \times 10^{-5}$, $W_3 = 2.7 \times 10^{-8}$, $W_4 = 3.6 \times 10^{-10}$. For $n = 1$ these small values are Cantelli’s one-sided Chebyshev bound, $W_1(\omega) = m_0 \sigma^2 / [\sigma^2 + (\omega - \bar{\omega})^2]$ with σ^2 the variance of the normalized measure (agreement to 1.6×10^{-15}). They reflect the distance from the mean, not a feature of this spectrum. Just above the top pole, at $\omega = 14t$, $W_1 = 0.20$.

All of this assumes exact moments. If the moments are matched only within the deployed tolerances, $|\Delta_k| \leq \tau_k$ for $k \leq 2$, the largest atom a positive measure can place at ω is a linear program. With thresholds from the deployed rule applied to this state ($\tau_0, \tau_1, \tau_2 = 0.051, 0.57, 6.5$ before normalization), it is $3.2 W_1$ (0.63 of m_0) at $\omega = 14t$, $7.7 W_1$ at $20t$, and 10.1 – $10.7 W_1$ at the rescaled points 0.3–0.9. These τ_k use the lower-bound shot variance of App. B; wider, realistic tolerances would inflate the atom further. At the deployed order a passing reconstruction is therefore bracketed much more loosely than W_1 alone suggests.

Appendix D: Complementary state-level check: the Gauss law

The Gauss-law check mentioned in Results I is not a moment test; we record its construction and scope here. In the $U(1)$ lattice Schwinger model the generators G_x of gauge transformations commute with the Hamiltonian, $[H, G_x] = 0$, and every physical state satisfies $G_x|\psi\rangle = 0$ at every site, so $\langle \sum_x G_x^2 \rangle = 0$ exactly. The operator $\sum_x G_x^2$ is diagonal in the computational basis, so on hardware it could be estimated from computational-basis samples of the prepared state, with no reconstruction and no moment. It tests the prepared state, not a reconstructed spectrum: it responds to leakage out of the physical sector, an error class the moment test does not target.

Figure 15 illustrates the identity on a small instance: the quantum-link form of the model with $N = 4$ staggered matter sites and 3 spin- $\frac{1}{2}$ links, open boundaries, $w = 1$, $m = 0.6$ ($[H, G_x] = 0$ to machine precision). The plotted state is $|\psi(\epsilon)\rangle \propto (1 - \epsilon)|\psi_0\rangle + \epsilon|\phi_u\rangle$, with $|\psi_0\rangle$ the physical ground state and $|\phi_u\rangle$ an unphysical state. The parameter ϵ is an amplitude, not a weight: the unphysical weight is $f_u(\epsilon) = \epsilon^2 / [(1 - \epsilon)^2 + \epsilon^2]$, which is 0.059 at $\epsilon = 0.2$ and 0.155 at $\epsilon = 0.3$. Because $\sum_x G_x^2 |\psi_0\rangle = 0$ and $\langle \psi_0 | \phi_u \rangle = 0$, the violation is $\langle \sum_x G_x^2 \rangle = g_u f_u(\epsilon)$ with $g_u = \langle \phi_u | \sum_x G_x^2 | \phi_u \rangle$. The curve is therefore fixed by construction: it is zero for the physical state, it rises monotonically with ϵ , and its scale is set by g_u alone. It illustrates the identity; it does not measure detection power. The values are computed exactly on the state vector, and no sampling is simulated.

The admixed state is chosen by an explicit rule. The generator `src/export_gauss_state.py` takes $|\phi_u\rangle$ as the lowest-energy eigenstate of H among charge-neutral states with the smallest nonzero violation, $\sum_x G_x^2 = 2$ (one site with $G = +1$ and one with $G = -1$; a 27-dimensional subspace). Every state in that subspace has $g_u = 2$, so the plotted curve is $2f_u(\epsilon)$. The figure was first made with a different rule: the unphysical eigenstate of H with the largest overlap $|\langle k | M | \psi_0 \rangle|$, where $M = \sum_n (-1)^n n_n$ is the staggered occupation. That rule is ill-posed. M is gauge-invariant, so every such overlap vanishes (the largest is 1.8×10^{-15}), and the pick was set by rounding noise; it happened to select a state with $g_u = 2$. The explicit rule reproduces the plotted values to all five printed decimals (`data/2026-09-27_gauss_state.json`). Over the 124 unphysical eigenstates of this model g_u ranges from about 1 to 10, so the slope of the curve reflects this choice. An earlier sweep under an independent bit-flip channel on every qubit (`data/2026-08-18_gausslaw_falsifier.json`) also gives $\langle \sum_x G_x^2 \rangle > 0$ at every nonzero flip rate (0.25 at $p = 0.05$). That sweep averages only 40 sampled flip patterns per rate and is not monotone in p at that sample size (3.05 at $p = 0.35$, 2.52 at $p = 0.40$), so we do not use it as a graded curve.

Gauss-law violation has been measured in a 71-site cold-atom quantum simulator [39] and analyzed theoretically under gauge-breaking errors [125]. The lattice Schwinger model is also the model of self-verifying variational simulation on a trapped-ion analog quantum simulator [41] and of early digital simulations of Schwinger-model dynamics [126, 127].

Appendix E: Within-sector redistributions: injectivity and the reach of the deployed battery

This appendix proves Theorem 2 and evaluates it on one spectrum. Results II states the boundary: the independent estimator detects weight that leaks *outside* the retained determinant support, but not a redistribution strictly *within* it that preserves the moments used

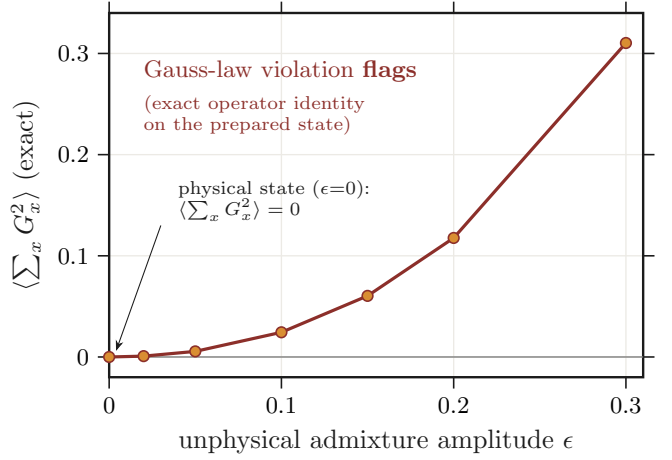


FIG. 15. **A state-level check: the Gauss-law identity.** In the $U(1)$ lattice Schwinger model every G_x annihilates physical states, so they have $\langle \sum_x G_x^2 \rangle = 0$; $[H, G_x] = 0$ keeps the dynamics in that sector. The plotted state is $|\psi(\epsilon)\rangle \propto (1 - \epsilon)|\psi_0\rangle + \epsilon|\phi_u\rangle$, with $|\psi_0\rangle$ the physical ground state and $|\phi_u\rangle$ the lowest-energy charge-neutral state with the smallest violation, $\sum_x G_x^2 = 2$. Here ϵ is an amplitude: the unphysical weight is $f_u = \epsilon^2 / [(1 - \epsilon)^2 + \epsilon^2]$, only 0.155 at $\epsilon = 0.3$, and the curve equals $2f_u$ by construction. It is computed exactly on the state vector; no sampling is simulated. The check tests the prepared state, not a reconstructed spectrum, and targets gauge-symmetry breaking, an error class the moment test does not address. $N = 4$ staggered sites, 3 spin- $\frac{1}{2}$ links, open boundaries, $w = 1$, $m = 0.6$; generator `src/export_gauss_state.py` (see text).

(Sec. IV). We treat that residual in three steps. *Injectivity* (part (a)): in exact arithmetic, $m_0, \dots, m_{N-1}$ fix the weights on N distinct poles. *Conditioning* (part (b) and the Corollary): how large a shift a redistribution can produce, which gives a sufficient and a necessary condition for exposing it at finite shots. *Reach*: a linear program and explicit counterexamples on the doped $L = 6$, $U/t = 4$ current spectrum whose thresholds are set in App. B (Table III). The result at the deployed order is negative: the battery (m_0, m_1, m_2) has no guaranteed power against within-sector redistributions of this spectrum, although its poles are well separated. We characterize this residual; we do not close it. Unless stated otherwise, the numbers below are in `data/2026-09-27_theory_numerics.json` (script `src/within_sector_lp.py`).

Injectivity (proof of Theorem 2(a)). A $T = 0$ Lehmann current spectrum on a finite sector is a discrete measure $A(\omega) = \sum_{n=1}^N w_n \delta(\omega - \omega_n)$ on N distinct poles $\{\omega_n\}$, where $w_n = \sum_a E_a - E_0 = \omega_n |\langle a | J | 0 \rangle|^2 \geq 0$ is the total weight of the eigenstates at ω_n . Its moments are $m_k = \sum_n w_n \omega_n^k$, i.e. $\mathbf{m} = V\mathbf{w}$ with $V_{kn} = \omega_n^k$ a Vandermonde matrix. Because the poles are distinct, the block $V_{0:N-1}$ of rows $0, \dots, N-1$ is invertible, so the weight-to-moment map is injective through order $N-1$:

TABLE III. Within-sector redistributions of the $L = 6$, $U/t = 4$ current spectrum ($N = 5$ distinct poles). Each row keeps $m_0, \dots, m_K$ exactly and asks how far the next moment m_{K+1} can move. “LP range” is the range of the signed shift $\sum_n \delta_n \omega_n^{K+1}$ over all such redistributions with non-negative corrupted weights [Eq. (E1)]; every range contains 0. The threshold $\tau_{K+1} = z_{0.975} \delta_{K+1}$ follows the deployed rule ($N_s = 5 \times 10^4$, 2% bias budget, $z_{0.975} = 1.96$; App. B); for m_3 and m_4 , which the deployed battery does not use, it is the value the same rule would give. When $\dim \ker V_{0:K+1} > 0$, some redistributions of finite size leave m_{K+1} exactly unchanged, so the guaranteed power at that order is zero and the maximum describes only the best case.

preserved exactly	next moment	LP range of shift [Eq. (E1)]	τ_{K+1}	max shift/ τ_{K+1}	$\dim \ker V_{0:K+1}$	in deployed battery?
m_0, m_1	m_2	$[-0.92, 7.09]$	1.848	3.8	2	yes
$m_0, \dots, m_2$	m_3	$[-0.14, 5.26]$	14.98	0.35	1	no
$m_0, \dots, m_3$	m_4	$[-0.30, 0.39]$	122.0	0.003	0	no

a redistribution $\mathbf{w} \rightarrow \mathbf{w} + \boldsymbol{\delta}$ with $\boldsymbol{\delta} \neq 0$ cannot preserve all of $m_0, \dots, m_{N-1}$. A redistribution that preserves $m_0, \dots, m_K$ ($K < N - 1$) zeroes the first $K+1$ entries of $V_{0:N-1}\boldsymbol{\delta}$, so $\|(\Delta_{K+1}, \dots, \Delta_{N-1})\|_2 = \|V_{0:N-1}\boldsymbol{\delta}\|_2 \geq \sigma_{\min}(V_{0:N-1})\|\boldsymbol{\delta}\|_2$; the largest of these $N-1-K$ shifts is at least this norm divided by $\sqrt{N-1-K}$, which is Eq. (2). If the poles may also move, an N -atom measure has $2N$ parameters and is fixed by $m_0, \dots, m_{2N-1}$ (Prony) [65, 102]. Weight traded among degenerate eigenstates at a single frequency changes no moment and leaves $A(\omega)$ unchanged; it is not an error.

For the $L = 6$, $U/t = 4$ spectrum the sixteen eigenstates with nonzero weight fall on $N = 5$ distinct frequencies, $\omega_n/t = 2.73, 5.57, 7.83, 8.70, 10.66$, with weights 0.040, 0.004, 0.393, 0.116, 0.001 ($m_0 = 0.554$; the pole at $7.83t$ carries 71% of it). The map is therefore injective through m_4 : a redistribution among these poles that preserves $m_0, \dots, m_K$ changes some m_j with $K < j \leq 4$ (at most m_9 if the poles also move). In exact arithmetic the residual is thus resolved by m_4 . That is two orders beyond the deployed battery, and it says nothing about detection at finite shots. Equation (2) is valid here but loose: for the redistribution CE1 below, its right-hand side is 3.8×10^{-3} , while the larger of the shifts in m_3 and m_4 is 85.

Conditioning, and when detection is possible (proof of Theorem 2(b)). Invertibility of $V_{0:N-1}$ holds in exact arithmetic. At finite shots a shift counts only if it exceeds the threshold τ_j . Two bounds, in opposite directions, turn detectability into a statement about the size of $\boldsymbol{\delta}$ and the geometry of the poles it touches.

Sufficient condition. By Eq. (2), a redistribution preserving $m_0, \dots, m_K$ is exposed (its noise-free shift exceeds the threshold) at some order $j \leq N - 1$ if $\sigma_{\min}(V_{0:N-1})\|\boldsymbol{\delta}\|_2/\sqrt{N-1-K} > \max_{K < j \leq N-1} \tau_j$. Since $\|\boldsymbol{\delta}\|_1/\sqrt{N} \leq \|\boldsymbol{\delta}\|_2 \leq \|\boldsymbol{\delta}\|_1$, this is a condition on $\rho = \|\boldsymbol{\delta}\|_1$ up to a factor $\sqrt{N}$. It is informative only when $\sigma_{\min}$ is not small, and it uses moments up to order $N - 1$.

Necessary condition. Let $\boldsymbol{\delta}$ preserve $m_0, \dots, m_{s-2}$ and be supported on poles inside an interval of diameter D and midpoint ω_c . (Among s poles, preserving $m_0, \dots, m_{s-2}$ leaves a one-parameter family, and m_{s-1} is the first moment it changes.) Since $\sum_n \delta_n q(\omega_n) = 0$ for every polynomial q of degree at most $s - 2$, the signed

shift is $\sum_n \delta_n \omega_n^{s-1} = \sum_n \delta_n (\omega_n - \omega_c)^{s-1}$, hence

$$|\Delta_{s-1}| \leq \rho (D/2)^{s-1}.$$

Subtracting from ω^j its Taylor polynomial of degree $s - 2$ at ω_c gives, for every $j \geq s - 1$, $|\Delta_j| \leq \rho \binom{j}{s-1} (|\omega_c| + D/2)^{j-s+1} (D/2)^{s-1}$ (Lagrange remainder; Theorem 2(b)(ii) states the term-by-term form), so every shift vanishes as D^{s-1} . Exposure at order $s - 1$ therefore requires $\rho (D/2)^{s-1} > \tau_{s-1}$. With $\tau = z_{1-\alpha/2} \delta$ and $\delta^2 = \text{Var}(O_{s-1})/N_s + b_{s-1}^2$ this reads

$$N_s > \frac{z_{1-\alpha/2}^2 \text{Var}(O_{s-1})}{\rho^2 (D/2)^{2(s-1)} - z_{1-\alpha/2}^2 b_{s-1}^2},$$

and no budget suffices once $\rho (D/2)^{s-1} \leq z_{1-\alpha/2} b_{s-1}$. The shot cost of detecting a redistribution of fixed size inside a collapsing cluster thus grows at least as $D^{-2(s-1)}$. Numerically, the identity holds to a relative error of 1.5×10^{-12} , and the bound holds in 11 structured cases (CE1 and CE2 below, the six optima of Eq. (E1) and three committed random cases) and in 20 000 random null-space redistributions on random pole subsets (seed 20260927, no violation). The largest ratio $|\Delta_{s-1}|/[\rho (D/2)^{s-1}]$ is 1.000, 0.499 and 0.231 at $s = 2, 3, 4$, so the bound is tight at $s = 2$.

Conditioning numbers. The same collapse appears in $\sigma_{\min}$: for s nodes of fixed relative geometry and diameter $D \rightarrow 0$, $\sigma_{\min}$ of the $s \times s$ Vandermonde block scales as D^{s-1} . The sharp super-resolution results for this regime [103–106] are proved for Fourier (unit-circle) nodes, not for the real nodes used here. For real nodes both directions follow from elementary bounds on the $s \times s$ block V of the cluster. Row n of V^{-1} holds the coefficients of the Lagrange polynomial $\prod_{j \neq n} (\omega - \omega_j)/(\omega_n - \omega_j)$, which gives Gautschi’s bound $\|V^{-1}\|_\infty \leq \max_n \prod_{j \neq n} (1 + |\omega_j|)/|\omega_n - \omega_j|$ [107, 108] and hence $\sigma_{\min}(V) \geq [\sqrt{s} \max_n \prod_{j \neq n} (1 + |\omega_j|)/|\omega_n - \omega_j|]^{-1}$, a lower bound that scales as D^{s-1} at fixed relative geometry. Conversely, the redistribution that spans the null space of rows $0, \dots, s - 2$ has $\|V\boldsymbol{\delta}\|_2 = |\Delta_{s-1}| \leq \rho (D/2)^{s-1} \leq \sqrt{s} \|\boldsymbol{\delta}\|_2 (D/2)^{s-1}$, so $\sigma_{\min}(V) \leq \sqrt{s} (D/2)^{s-1}$. We also checked the scaling directly: $s = 2, 3, 4$ equally spaced nodes near $\omega = 7.83t$,

with spacing from 0.02 to $0.2t$, give fitted exponents 0.995 , 1.98 and 2.96 (`data/2026-08-24_within_sector_control.json`). Condition numbers depend on the frequency unit, so we quote them only on nodes mapped affinely onto $[-1, 1]$ (outermost poles at ± 1): for the five poles above, $\sigma_{\min}(V_{0:4}) = 0.056$ and $\kappa = 50$. The minimal pole separation is $0.87t$; these poles are well separated. Good conditioning does not, however, give the deployed battery guaranteed power. That is set by the dimension of a null space, as the next paragraph shows.

Reach of the deployed battery (instantiation of Theorem 2(b) and Corollary). The deployed battery uses m_0, m_1, m_2 with thresholds $\tau_k = z_{0.975} \delta_k$ from App. B: $\tau_0 = 0.0287$, $\tau_1 = 0.229$, $\tau_2 = 1.848$ at $N_s = 5 \times 10^4$ with a 2% bias budget $b_k = 0.02 m_k$ (τ includes $z_{0.975} = 1.96$, unlike the 1σ half-width δ). The shot term uses the exact-state local-estimator variance, a lower bound on the true per-shot variance, so realistic thresholds would be wider. For a redistribution that preserves $m_0, \dots, m_K$, the largest shift it can impose on the next moment is the linear program

$$\begin{aligned} \Delta_{K+1}^{\max} &= \max_{\delta} \left| \sum_n \delta_n \omega_n^{K+1} \right| \quad \text{s.t.} \\ \sum_n \delta_n \omega_n^j &= 0 \quad (j \leq K), \quad \delta_n \geq -w_n, \end{aligned} \quad (\text{E1})$$

the constraint $\delta_n \geq -w_n$ keeping the corrupted weights non-negative; together with $\sum_n \delta_n = 0$ it also bounds $\|\delta\|_1 \leq 2m_0$.

Table III gives the full range of the signed shift. A redistribution that preserves m_0, m_1 can move m_2 by up to $7.09 = 3.8 \tau_2$, so some such redistributions are exposed at m_2 . The guaranteed power, however, is set by the smallest shift a redistribution of a given size can produce, and that is zero. Rows 0, 1, 2 of V have a null space of dimension $N - 3 = 2$: some redistributions preserve m_0, m_1 and m_2 exactly, and scaling one of them down gives an exactly invisible redistribution of every smaller size. Three explicit cases, all positive measures that pass the deployed battery: (i) CE1 preserves m_0, m_1, m_2 exactly and moves 72% of the weight ($\|\delta\|_1/2 = 0.72 m_0$); it empties two of the five poles and lowers the dominant residue from 0.393 to 0.037 . (ii) CE2 preserves m_0, m_1 exactly and shifts m_2 by exactly τ_2 ; it moves 86% of the weight and lowers the dominant residue to 0.073 . (iii) CE3 keeps $|\Delta_k| \leq \tau_k$ for $k \leq 2$, has $\|\delta\|_1 = 2.03 m_0$ (above $2m_0$ because m_0 itself moves by τ_0), and empties three poles, the dominant one included. The random $K = 1$ case committed in `src/within_sector_control.py` (seed 0) moves 25% of the weight ($\|\delta\|_1/2 = 0.25 m_0$; its output

file lists $\|\delta\|_1/m_0 = 0.49$ as the moved fraction) and shifts m_2 by $0.044 \tau_2$; it passes as well. The guaranteed power of the deployed battery against within-sector redistributions of this spectrum is therefore zero. The reason is the dimension of the null space, not conditioning, and the same holds for any spectrum with four or more distinct poles.

Adding m_3 and m_4 would remove the null space (rows $0, \dots, 4$ have $\dim \ker = 0$), but at this shot budget the extended battery would still miss a redistribution that moves 72% of the weight. Under the same threshold rule $\tau_3 = 14.98$ and $\tau_4 = 122.0$. Among redistributions that preserve m_0, m_1, m_2 exactly, the largest m_3 shift is $5.26 = 0.35 \tau_3$ (Table III), and CE1, like every scaled-down copy of it, stays below both thresholds ($0.23 \tau_3$ and $0.70 \tau_4$). Some such redistributions would be caught at m_4 : the one that maximizes the m_3 shift moves m_4 by $145 = 1.19 \tau_4$, a marginal case given the lower-bound variance. CE2 and CE3, which use the tolerance on m_2 , would be caught ($|\Delta_3| = 3.3 \tau_3$; $|\Delta_4| = 7.1$ and $7.2 \tau_4$).

The joint battery does catch some errors that m_1 alone misses, but the example is a truncation, not a fixed-pole redistribution. Truncating the determinant basis of the same system to the $d = 97$ determinants of largest $|\psi_0|^2$ (52% of the 186-determinant support) gives a reconstruction that m_1 alone passes and m_2 rejects. The gaps depend on how ties in $|\psi_0|^2$ are ordered, because the cut falls inside a 24-fold degenerate shell. The committed smoke test `src/verify.py` (NumPy `argsort` order, which does not fix the order of ties; our machine) gives $|\bar{m}_1 - \hat{m}_1| = 0.155 < \tau_1$ and $|\bar{m}_2 - \hat{m}_2| = 2.560 > \tau_2$; the deterministic `lexsort` order of `src/interval_battery.py` gives 0.186 and 2.729 . The verdict is the same for all 300 random tie-breaks within the shell: m_1 alone rejects none and the battery rejects all (`data/2026-09-27_interval_battery.json`). This is one catch, not a guarantee. In an exhaustive scan of $d = 1, \dots, 185$ at the same thresholds (`lexsort` order), the full (m_0, m_1, m_2) +Hankel battery passes 12 truncated reconstructions ($d = 86, 87, 95, 160$ – 165 and 183 – 185 ; App. B). Under random tie-breaks within the degenerate shells, ten of these pass in at least 89% of 300 draws, and $d = 87$ and $d = 160$ in 56% and 60%.

Positivity cannot help against within-sector redistributions: a redistribution with non-negative weights is itself a positive measure, so the Hankel and Stieltjes conditions hold by construction. Only independent moments of high enough order, estimated precisely enough, can expose it, and at the deployed order the guaranteed power is zero. We report this residual; we do not claim to close it.

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